

Endogenous Returns to Scale*

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Abstract

We develop a model in which firms choose the returns to scale of their production technologies. More scalable technologies facilitate output expansion but are less effective at small scale. In equilibrium, more productive firms adopt more scalable technologies and grow disproportionately large, thickening the tail of the size distribution. Reallocation of resources toward these firms raises GDP. When aggregate productivity grows, firms adopt more scalable technologies, lowering their customers' input prices and inducing further adoption downstream. Through this supply-chain amplification, endogenous returns to scale also raise GDP growth. A calibration shows sizable effects, and firm-level data support the model's predictions.

JEL Classifications: E23, D24, D57, O40, L11

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1 Introduction

At the turn of the twentieth century, the automobile was a luxury good, carefully assembled by skilled craftsmen. State-of-the-art models like the 1909 Cadillac Model Thirty were already built with interchangeable parts, yet their production remained artisanal and their price prohibitively high. Just a few years later, Henry Ford’s Highland Park plant was assembling a Model T every 93 minutes. By adopting the moving assembly line, Ford reorganized production to increase its returns to scale. The benefit was a large decline in production costs that transformed the automobile into a mass-market product.

Ford’s story is not unique. In his seminal work, Chandler (1990) argued that the adoption of processes, organizational structures, and technologies designed to achieve greater economies of scale was a key driver of modern economic growth. This historical perspective suggests that scalability is not a fixed constraint but a dimension of technology that firms actively manage. Building on this insight, we explore the idea that returns to scale are endogenous equilibrium objects driven by incentives, and study its implications for the firm size distribution, the response of the economy to shocks, and long-run growth.

To do so, we develop a multi-sector general equilibrium model with endogenous returns to scale. Within each sector, a continuum of firms with heterogeneous productivity produce a common good using labor and intermediate inputs. Importantly, firms in our setup are free to choose their returns to scale subject to a technological trade-off: achieving larger scale economies comes at a cost in terms of productivity. As a result, while more scalable technologies make it easier for firms to expand output, they are less effective at small scale.

The existence of a trade-off between scale and productivity is well documented. Chandler (1977) describes how the historical shift from small, artisanal producers to large, integrated enterprises depended on costly investment in a new “technology of organization.” Achieving and managing large scale required the creation of professional hierarchies and complex administrative systems. While this “visible hand” of management enabled firms to coordinate large-scale production, it introduced administrative overhead and reduced the operational flexibility that smaller firms enjoyed.

Our analysis of the model begins with the individual firm’s decisions in partial equilibrium. Following McKenzie (1959), we interpret decreasing returns to scale as arising from a fixed entrepreneurial factor. A firm’s choice of returns to scale therefore reflects a trade-off between this constrained in-house factor and the variable bundle of labor and intermediate inputs. We show that the optimal degree of scalability is reached when the marginal productivity loss from expanding scalability exactly offsets the cost savings from relying less on the fixed factor. This condition dictates how firms adapt to their environment: any change that pushes the firm to expand puts pressure on the fixed factor and encourages the adoption of a technology with higher returns to scale. Consequently, higher productivity, higher output prices, or cheaper intermediate inputs all induce the firm to adopt a more scalable, input-intensive production function.

This mechanism generates a “double blessing” for the most productive firms. Their intrinsic productivity (the first blessing) naturally leads to larger size and higher profits. This expansion, in

turn, tightens the constraint imposed by the fixed entrepreneurial factor, creating an incentive to adopt more scalable technologies (the second blessing), which leads to a further increase in size. This disproportionate growth creates superstar firms and thickens the tail of the firm-size distribution.

Despite heterogeneity in returns to scale, we show that firms in a sector can be aggregated in a tractable way. Because of free entry, production exhibits *constant* returns to scale at the *sector* level, with firm-level scalability decisions affecting the importance of labor and intermediate inputs in sectoral production. Returns-to-scale decisions also manifest themselves in sectoral productivity. We show that by allowing high-productivity firms to grow larger, endogenous returns to scale increase sectoral productivity and, through that channel, the level of GDP.

The model admits a unique and efficient competitive equilibrium, which can be characterized as the solution to a social planner’s problem. We use this characterization to study the determinants of returns to scale in general equilibrium. We find that any shock that lowers the relative cost of intermediate inputs induces firms to adopt more scalable technologies. For instance, a productivity improvement in an upstream sector reduces input costs for downstream customers, encouraging them to increase their scalability. This shift toward greater scalability implies a heavier reliance on intermediate inputs, which raises the Domar weights of upstream suppliers. Consequently, sectors experiencing productivity gains become more central to the economy, amplifying their impact on GDP. Through this channel, endogenous returns to scale magnify the benefits of positive shocks. Symmetrically, the same mechanism dampens the adverse impact of negative shocks, as firms substitute away from intermediate inputs and reduce the importance of the affected sectors.

Endogenous returns to scale also matter for long-run growth. With recurrent productivity improvements, firms continuously adopt more scalable technologies. This leads to an increase in Domar weights, making subsequent productivity gains even more impactful. As this process unfolds, the economy enters an acceleration phase in which the growth rate of GDP rises over time, eventually converging to a long-run rate strictly higher than in a fixed-technology economy.¹ In our model, growth is therefore driven by the interaction between exogenous innovation and endogenous scaling decisions. A back-of-the-envelope calculation suggests that this mechanism can have a sizable impact on long-run growth.

To study how policy interventions and market frictions affect returns-to-scale decisions, we extend our baseline model to include wedges, such as sales taxes or tariffs on intermediate inputs. We find that such distortions, by incentivizing firms to shrink, lead to the adoption of inefficiently low returns to scale. In this distorted equilibrium, productivity shocks have a first-order effect on welfare by altering the economy’s returns to scale. A positive productivity shock, for instance, not only increases output directly but also acts as a corrective force: by lowering input costs, it encourages firms to increase their scalability, moving the economy’s production technology closer to the efficient benchmark.

We use detailed data covering the near-universe of firms in Spain to test the predictions of the

¹Convergence is slow in the calibrated model, suggesting that the growth-accelerating effect of endogenous returns to scale becomes economically important only over the long run.

model. In both the cross-section and within firms over time, productivity, size, and returns to scale move together, in line with the model’s core mechanism. We then turn to the model’s distinctive supply-chain channel. Exploiting variation in firms’ direct and indirect exposure to energy-producing sectors, we find that more exposed firms reduce their returns to scale by more after increases in global oil prices, in line with the downstream adjustment the model predicts. Finally, cross-country patterns corroborate these mechanisms at the aggregate level. Countries where returns to scale covary more strongly with productivity, indicating a stronger endogenous scalability mechanism, exhibit higher income per capita, suggesting that endogenous scalability decisions may play a role in long-run economic development.

Finally, to quantify the importance of our mechanism, we calibrate the model to the Spanish economy. We find that eliminating the ability of firms to adjust their scalability reduces the level of GDP by nearly 12% and lowers the long-run growth rate of the economy by 0.8 percentage points. Crucially, these gains are driven by the capacity of high-productivity firms to adopt more scalable technologies. To illustrate this, we examine the impact of size-dependent distortions that are particularly detrimental to large firms. Constructing wedges as in Hsieh and Klenow (2009), we confirm that larger firms face higher effective distortions in the data. We find that removing these wedges yields welfare gains that are more than twice as large in our model compared to an economy where returns to scale are fixed. This highlights that policies burdening large firms are particularly costly when they stifle the adoption of highly scalable technologies.

Literature review

Early work emphasizes the importance of changing returns to scale for economic outcomes. Kuznets (1973) argues that the rise of large-scale firms reflected an adaptive process through which firms learned to coordinate production and distribution across expanding markets. Chandler (1977) documents how managerial hierarchies and integrated production systems enabled firms to realize “economies of scale and scope.” These classic accounts emphasized the importance of changes in scalability for growth. Our work formalizes that idea in a general equilibrium framework.

Since we focus on the aggregate impact of endogenous scalability, we adopt a holistic approach and do not take a stance on the underlying margins firms use to adjust their returns to scale (several are likely at work). In contrast, some contemporaneous work focuses on specific mechanisms. Argente et al. (2026) propose a model of multi-product firms in which standardization increases a firm’s returns to scale. Like us, they find that endogenous scalability leads to fat-tailed firm-size distributions. Engbom et al. (2026) build a model in which entrepreneurs can hire white-collar workers to complete administrative tasks, thereby increasing their returns to scale. They find that the scarcity of skilled labor in developing countries limits this reorganization, and that increasing the supply of skills can explain two-thirds of the shift toward large firms observed during development. Also in the development literature, Gottlieb et al. (2026) propose a model in which firms can choose between high- and low-returns-to-scale technologies. They use the model to explain empirical patterns related to the effect of skill endowments on the firm size distribution.

Smirnyagin (2023) proposes a business cycle model with financial frictions in which firms can choose between two returns to scale levels. Focusing on long-run patterns, Lashkari et al. (2024) document that the decline in IT prices led to an increase in returns to scale in France. To explore the implications of this finding, they propose a model with non-homothetic production in which returns to scale can vary with input factors. Hubmer et al. (2026) use administrative data from Canada and the United States to document that larger firms operate technologies with higher returns to scale—a finding that is consistent with our model. They explore the implications of these patterns in an entrepreneurial model with fixed heterogeneous returns to scale and financial frictions.²

Chen (2026) documents that the right tail of the firm size distribution systematically thickens with economic development. Our model can explain this fact through a declining cost of scalability along the development path. When new technologies make higher returns to scale easier to adopt, the firm-size tail thickens. Consistent with this interpretation, we find that the endogenous-scalability mechanism is stronger in richer countries.

In addition to its tractability, a distinguishing feature of our work is that we study the impact of input-output linkages on scalability. In doing so, we identify a novel channel through which adjustments in returns to scale propagate through supply chains, reshaping Domar weights throughout the economy. This channel has important implications for the aggregate impact of endogenous scalability and is essential for our long-run growth results.

Our work also relates to different strands of the production network literature (Long and Plosser, 1983; Acemoglu et al., 2012). As in Baqaee and Farhi (2019b), Hulten’s (1978) theorem only provides a first-order approximation to the economy’s response to shocks in our model. We also build on previous work that studies the role of wedges in network economies (Jones, 2011; Baqaee and Farhi, 2019a; Liu, 2019; Bigio and La’O, 2020). Finally, we relate to a literature on production networks that treats production functions as endogenous (Oberfield, 2018; Acemoglu and Azar, 2020; Kopytov et al., 2024a; Kopytov et al., 2024b). We share with this literature the assumption that firms have control over their production technologies.³ Unlike this literature, we endogenize returns to scale, which yields novel predictions for the firm size distribution and has important aggregate implications.

2 A model of endogenous returns to scale

We introduce endogenous returns to scale into a multisector economy with input–output linkages. Each sector produces a differentiated good that can be used for final consumption and as an intermediate input. Within each sector, there is a continuum of firms with heterogeneous produc-

²In many models, firms must pay a fixed cost to operate, and this fixed cost therefore influences the *average* returns to scale of the firm. In contrast, our setup focuses on *marginal* returns to scale, which capture how a marginal increase in size affects the marginal cost of production.

³Our work is also related to the appropriate-technology literature (Caselli and Coleman, 2006), which, like us, models technology choice along a frontier that trades off the productivity of different inputs. That literature focuses on the choice of factor-augmenting technologies; we instead apply the idea to returns to scale and study its interaction with production networks.

tivity. A representative household supplies labor, owns the firms, and consumes a bundle of sectoral goods. Crucially, firms optimally choose their returns to scale as part of their production decisions. Consequently, changes in the environment affect firm-level scalability and, through that channel, macroeconomic aggregates.

2.1 Production technology

There are N goods, each produced by a different sector. A sector i consists of a continuum of competitive firms whose mass M_i is determined by a free-entry condition. Upon paying $\kappa_i > 0$ units of labor to enter, a firm l draws a random productivity level $\varepsilon_{il} \sim \text{iid } \mathcal{N}(\mu_i, \sigma_i^2)$ from a normal distribution. The firm can then produce using a Cobb–Douglas technology and, crucially, choose how scalable that technology is. Specifically, if it selects returns to scale $0 < \eta_{il} < 1$, firm l 's output is

$$Q_{il} = F_i(L_{il}, X_{il}, \eta_{il}) := e^{\varepsilon_{il}} A_i(\eta_{il}) \zeta_i(\eta_{il}) \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}}, \quad (1)$$

where L_{il} is labor, $X_{il} = (X_{i1,l}, \dots, X_{iN,l})$ is a vector of intermediate inputs, and $\zeta_i(\eta_{il})$ is a normalization term to simplify subsequent expressions.⁴ The parameter $\alpha_{ij} \geq 0$ is the intensity of intermediate input j (with $1 - \sum_j \alpha_{ij} > 0$). We assume that operating technologies with higher returns to scale is costly, so that the productivity shifter $A_i(\eta_{il})$ is strictly decreasing in η_{il} . This captures the idea that achieving greater scalability often requires more complex processes, incurs substantial coordination and communication costs, and demands more managerial attention (Chandler, 1977). We impose that A_i is smooth, strictly log-concave and that $A_i(\eta_{il}) \rightarrow 0$ as $\eta_{il} \rightarrow 1$.⁵

2.2 Firm problem

To explore what drives a firm's returns-to-scale decision, we first analyze its problem in partial equilibrium. A firm l in sector i simultaneously chooses its returns to scale η_{il} and its inputs to maximize profits:

$$\Pi_{il} := \max_{\eta_{il}, L_{il}, X_{il}} P_i F_i(L_{il}, X_{il}, \eta_{il}) - W L_{il} - \sum_{j=1}^N P_j X_{ij,l}, \quad (2)$$

where W is the wage and P_j is the price of good j . Solving this problem, we derive the firm's marginal cost of production as a function of its output Q_{il} and its chosen technology η_{il} .

Lemma 1. *The firm's marginal cost of production λ_{il} is given by*

$$\lambda_{il} := \frac{1}{e^{\varepsilon_{il}} A_i(\eta_{il})} H_i^{\eta_{il}} \Pi_{il}^{1-\eta_{il}}, \quad (3)$$

⁴We set $(\zeta_i(\eta))^{-1} := \left(\left(1 - \sum_j \alpha_{ij} \right) \eta \right)^{\eta(1-\sum_j \alpha_{ij})} \prod_j (\eta \alpha_{ij})^{\eta \alpha_{ij}} (1-\eta)^{1-\eta}$ to simplify the unit cost expression below. Here, this term can be subsumed in $A_i(\eta_{il})$.

⁵Our baseline model is static but we consider a dynamic extension in Appendix E.

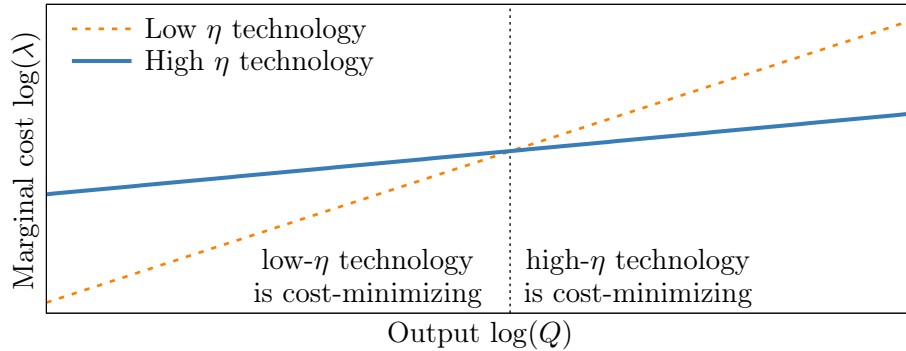
where $H_i := W^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}}$ is the price of the variable input bundle in sector i , and Π_{il} is profits,⁶

$$\Pi_{il} = (1 - \eta_{il}) \lambda_{il} Q_{il}. \quad (4)$$

As usual, the firm's marginal cost λ_{il} decreases with productivity and increases with input prices. The wage and the price of intermediate inputs, in particular, affect λ_{il} through the variable input bundle price H_i . Crucially, the firm's profit Π_{il} also shows up as an input price in (3). To understand why, recall that we can interpret any decreasing-returns production function as a constant-returns technology with an additional fixed entrepreneurial factor in unit supply (McKenzie, 1959). Under this interpretation, profits Π_{il} are simply the payment to that input. As output Q_{il} increases, the pressure on this fixed factor rises, increasing its shadow cost Π_{il} and, in turn, driving up the marginal cost λ_{il} .

Returns to scale η_{il} play a dual role in shaping marginal costs. Higher returns to scale lower baseline productivity through $A_i(\eta_{il})$, but they also increase the weight of the variable bundle (labor and intermediate) relative to the fixed entrepreneurial factor. Through this second channel, η_{il} determines how steeply the marginal cost λ_{il} rises with output Q_{il} . Figure 1 illustrates this trade-off by plotting λ_{il} as a function of Q_{il} for a high and a low value of η_{il} . The high- η technology offers greater scalability and thus a flatter marginal cost curve, allowing the firm to increase its size with only a small increase in its marginal cost. This makes it particularly effective for large firms. However, because it incurs a large productivity penalty through A_i , this technology is ineffective at small scales. In contrast, the low- η technology benefits from high productivity $A_i(\eta_{il})$, making it the preferred choice for small-scale production.

Figure 1: The trade-off between baseline productivity and returns to scale.



This leads to the key sorting mechanism of our model. While adopting a technology with higher returns to scale is costly in terms of baseline productivity, firms that choose these technologies are, in equilibrium, more productive overall. This is because only firms with a sufficiently high idiosyncratic productivity draw ε_{il} find it optimal to operate at the large scale necessary to make a high- η technology worthwhile. In Section 6, we will show that this positive correlation between productivity and returns to scale is supported by the data.

⁶All proofs are in Appendix C.

Finally, note that profit maximization implies that the firm selects output Q_{il} so that its marginal cost λ_{il} equals the price of its good P_i .

2.3 Choosing returns to scale

Building on these insights, we can characterize the optimal returns-to-scale decision of the firm.

Lemma 2. *At an interior solution, the firm chooses its returns to scale $\eta_{il} \in (0, 1)$ according to*

$$\frac{da_i(\eta_{il})}{d\eta_{il}} = \log H_i - \log \Pi_{il}, \quad (5)$$

where $a_i(\eta_{il}) := \log A_i(\eta_{il})$.

Equation (5) describes the key trade-off behind the returns-to-scale choice. It is better understood as the derivative of the log of λ_{il} , given by (3), with respect to η_{il} . When increasing η_{il} at the margin, the firm shifts its input mix away from the fixed entrepreneurial factor, whose shadow cost is Π_{il} , toward the variable input bundle, whose cost is H_i . The right-hand side of (5) captures the change in cost associated with that shift. The firm balances that change in cost with any loss in TFP associated with the higher returns to scale, as reflected by the left-hand side of (5).

From (5), we can determine how η_{il} responds to changes in the economic environment. The top two panels of Figure 2 illustrate the forces involved. Since a_i is concave, its derivative is decreasing. Therefore, any change that increases profits Π_{il} —such as a higher output price P_i or a better productivity draw ε_{il} —makes the fixed factor more expensive, pushing the firm to adopt a higher η_{il} . Conversely, an increase in the variable input cost H_i incentivizes the firm to rely more on its fixed factor, that is, to lower the optimal η_{il} . The following lemma formalizes this intuition.

Lemma 3. *At an interior solution, the returns-to-scale parameter η_{il} satisfies⁷*

$$\frac{d\eta_{il}}{d\varepsilon_{il}} = \frac{d\eta_{il}}{d\log P_i} = - \left[(1 - \eta_{il}) \frac{d^2 a_i}{d\eta_{il}^2} \right]^{-1} > 0, \quad \text{and} \quad \frac{d\eta_{il}}{d\log H_i} = \left[(1 - \eta_{il}) \frac{d^2 a_i}{d\eta_{il}^2} \right]^{-1} < 0.$$

This result highlights that the elasticities of the returns to scale with respect to prices and productivity depend on η_{il} itself and on the concavity of a_i .

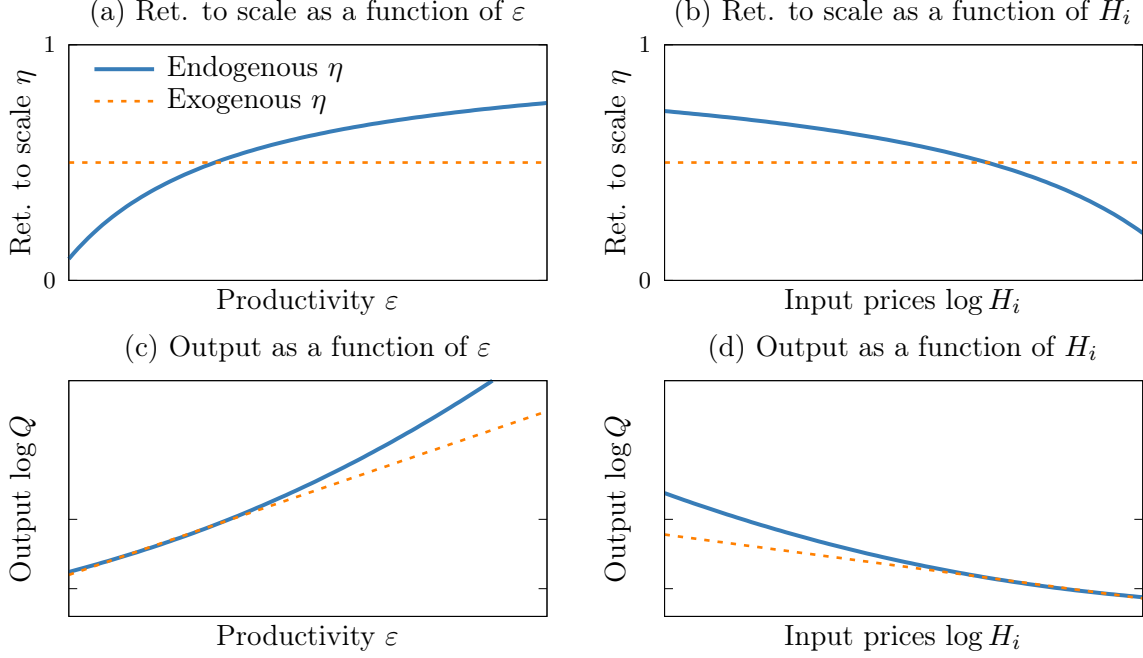
Endogenous scalability also has crucial implications for the output Q_{il} of the firm.

Lemma 4. *At an interior solution, the elasticity of output Q_{il} with respect to productivity ε_{il} is given by*

$$\frac{d\log Q_{il}}{d\varepsilon_{il}} = \underbrace{\frac{1}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d\varepsilon_{il}}}_{\text{Flexible } \eta \text{ effect}} > 0.$$

⁷When increasing P_i , we keep the price of the variable input bundle constant to distinguish the two channels that affect η_{il} .

Figure 2: Impact of productivity ε_{il} and input prices H_i on the firm



In addition, the elasticities of output Q_{il} with respect to prices are given by

$$\frac{d \log Q_{il}}{d \log P_i} = \underbrace{\frac{\eta_{il}}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d \log P_i}}_{\text{Flexible } \eta \text{ effect}} > 0, \text{ and } \frac{d \log Q_{il}}{d \log H_i} = \underbrace{-\frac{\eta_{il}}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d \log H_i}}_{\text{Flexible } \eta \text{ effect}} < 0.$$

With fixed returns to scale, productivity and prices affect output Q_{il} through standard channels, captured by the first terms in the expressions of Lemma 4. Higher productivity ε_{il} , for instance, allows the firm to produce larger quantities before its marginal cost reaches the price P_i of its output. The magnitude of this response depends on returns to scale: a high- η firm is more sensitive to productivity and prices than a low- η firm.

In addition to this fixed- η mechanism, Lemma 4 reveals an additional mechanism at work when returns to scale are endogenous. Following an increase in productivity ε_{il} , the firm not only expands to exploit its lower marginal cost but also increases its returns to scale to better accommodate the higher production volume. This amplification mechanism creates a superstar effect, causing high-productivity firms to grow disproportionately large. A similar mechanism operates in response to price changes.

The bottom two panels of Figure 2 illustrate these forces. With exogenous returns to scale (dashed orange lines), $\log Q_{il}$ varies linearly with ε_{il} and $\log H_i$, as in standard models. In contrast, with endogenous returns to scale (blue lines), the response is convex: productivity and input prices have an outsized impact on output.

This amplification mechanism has important implications for the firm-size distribution. To

explore them transparently, it helps to specialize the returns-to-scale cost function a_i .⁸

Assumption 1. *The TFP shifter function A_i takes the form*

$$a_i(\eta_{il}) = -\frac{\gamma_i}{1 - \eta_{il}}, \quad (6)$$

where the parameter $\gamma_i > \sigma_i^2/2$ governs the productivity cost of raising η_{il} in sector i .

We also define the *effective productivity dispersion* $\varphi_i := \sigma_i^2 / (2\gamma_i)$ as a measure of productivity dispersion in sector i relative to the cost of adjusting returns to scale. Since $\gamma_i > \sigma_i^2/2$ by Assumption 1, we have that $0 < \varphi_i < 1$.⁹ This parameter will play an important role in our analysis. We maintain Assumption 1 throughout the rest of the paper, unless stated otherwise.

With that assumption, we can describe the impact of endogenous returns to scale on the tail of the firm-size distribution.

Proposition 1. *Without endogenous returns to scale, the distribution of Q_{il} in sector i is log-normal. With endogenous returns to scale, the right tail of the distribution of Q_{il} behaves like a Pareto distribution with tail index $1/\varphi_i$, in the sense that*

$$\log(\mathbb{P}(Q_{il} > q)) \sim -\frac{1}{\varphi_i} \log q, \text{ as } q \rightarrow \infty.$$

In the absence of endogenous scalability, all firms within a sector operate with identical returns to scale. Consequently, the distribution of firm output mirrors that of the underlying productivity distribution and is log-normal. In contrast, when returns to scale are endogenous, the most productive firms choose higher returns to scale. This stretches the right tail of the distribution, making it thick and Pareto-like. Proposition 1 shows that the tail's thickness depends on the effective productivity dispersion φ_i . When productivity shocks are highly dispersed (large σ_i^2) or scalability is cheap (low γ_i), the firm-size distribution is thicker. Figure 3 illustrates these mechanisms.

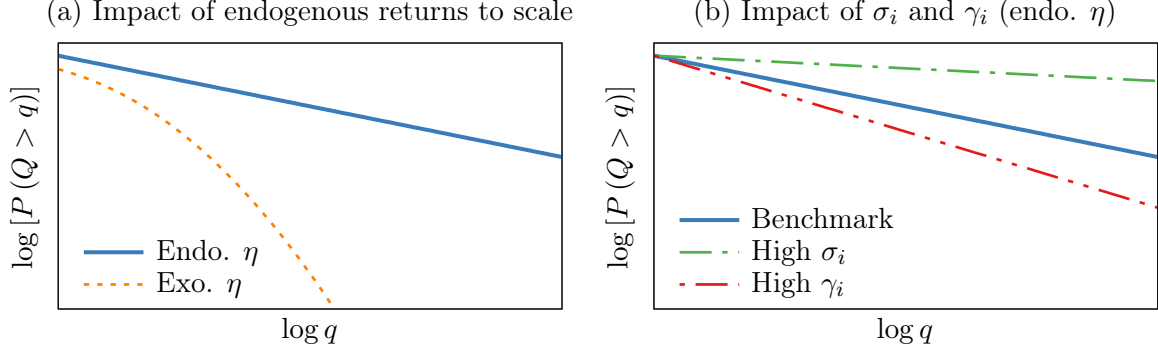
The thick tail of the firm size distribution is well-documented empirically (Axtell, 2001). Our model generates this property endogenously from fundamental productivity shocks that are not themselves fat-tailed, with superstar firms emerging from the decisions of high-productivity producers to become more scalable. Relatedly, Chen (2026) documents that this right tail thickens systematically with economic development. Proposition 1 suggests an interpretation: if technological progress lowers the cost γ_i of adopting more scalable technologies, the effective dispersion φ_i rises and the tail of the firm-size distribution thickens. Consistent with this interpretation, we find in Section 6.6 that the endogenous-scalability mechanism is stronger in richer countries.¹⁰

⁸Assumption 1 imposes that $A_i(\eta_{il})$ satisfies an Inada condition as $\eta_{il} \rightarrow 1$, but not as $\eta_{il} \rightarrow 0$. Since productivity shocks ε_{il} are unbounded, firms with very low ε_{il} might choose $\eta_{il} \notin (0, 1)$. In Appendix D.1, we explore the model in which the distribution of ε_{il} is truncated such that $\eta_{il} \in (0, 1)$ for all firms. We show that aggregate quantities converge to their main-text counterparts as the mass of firms picking $\eta_{il} \notin (0, 1)$ shrinks. The mass of such firms is indeed small under our calibration.

⁹Without this constraint, returns to scale increase so rapidly with ε_{il} that output can be infinite.

¹⁰In Appendix D.2, we generalize Proposition 1 by replacing the parametric TFP shifter imposed in Assumption

Figure 3: Tail of the distribution of firm-level output Q_{il}



2.4 Household

A representative household owns the firms, supplies $\bar{L} > 0$ units of labor inelastically, and consumes a bundle $Y := \prod_{i=1}^N (\beta_i^{-1} C_i)^{\beta_i}$ of the different consumption goods, where $\sum_{i=1}^N \beta_i = 1$. Since Y measures aggregate value added in this economy, we refer to it as (real) GDP.

The household maximizes Y subject to the budget constraint¹¹

$$\sum_{i=1}^N P_i C_i \leq W \bar{L}. \quad (7)$$

Because of the free-entry condition, all profits from the firms are dissipated through entry costs, and the household's only income comes from labor.

Maximization by the household implies that spending on good i amounts to a fraction β_i of total expenditure, so that $P_i C_i = \beta_i \bar{P} Y$, where $\bar{P} := \prod_{i=1}^N P_i^{\beta_i}$ is the ideal price index which we adopt as the numeraire. Consequently, nominal and real GDP are equal, and the household's budget constraint simplifies to $Y = W \bar{L}$.

2.5 Equilibrium conditions

For any firm-level quantity B_{il} , we denote by $B_i = \int_0^{M_i} B_{il} dl$ the sum of that quantity across all firms in sector i . We also use brackets $\{B_{il}\}$ to denote the set of that quantity over all sectors and firms.

We define an equilibrium as an allocation in which the optimality conditions of the firms and the household hold simultaneously, and all markets clear.

¹ with a condition on its asymptotic behavior as $\eta \rightarrow 1$. We show that the thickness of the right tail of the sales distribution depends on how rapidly the productivity cost of scalability increases near $\eta = 1$. When this cost becomes steeper, highly productive firms choose returns to scale that approach one more slowly. This weakens the amplification generated by endogenous returns to scale and produces a thinner sales tail. Nevertheless, the tail remains robustly heavier than the log-normal tail implied by a model with exogenous returns to scale.

¹¹ Because the model is static and there is no aggregate uncertainty, the household could instead maximize a strictly increasing function of Y without affecting the results.

Definition 1. An *equilibrium* is a set of prices (P^*, W^*) , a choice of returns to scale $\{\eta_{il}^*\}$, a tuple of quantities $\{C_i^*, L_{il}^*, X_{il}^*, Q_{il}^*\}$, and a mass of firms M^* in each sector such that

1. (Optimal returns-to-scale choice) For each $i \in \{1, \dots, N\}$ and $l \in [0, M_i^*]$, the returns-to-scale decision η_{il}^* solves (2) given prices (P^*, W^*) .
2. (Optimal input choice) For each $i \in \{1, \dots, N\}$ and $l \in [0, M_i^*]$, factor demands L_{il}^* and X_{il}^* solve (2) given prices (P^*, W^*) .
3. (Consumer optimization) The consumption vector C^* maximizes GDP Y subject to (7) given prices (P^*, W^*) .
4. (Free entry) For each $i \in \{1, \dots, N\}$, the expected profit of a potential entrant in sector i solves

$$E_i [\Pi_i(\varepsilon_{il}, P^*, W^*)] = \kappa_i W^*, \quad (8)$$

where Π_{il} is given by (4), and where the expectation E_i is taken over ε_{il} .

5. (Market clearing) For each $i \in \{1, \dots, N\}$,

$$C_i^* + \sum_{j=1}^N X_{ji}^* = Q_i^* = \int_0^{M_i^*} F_i(L_{il}^*, X_{il}^*, \eta_{il}^*) dl, \text{ and } \sum_{i=1}^N L_i^* + \sum_{i=1}^N M_i^* \kappa_i = \bar{L}. \quad (9)$$

Conditions 2 to 5 are standard and imply that the household and the firms maximize their objective functions, that all markets clear, and that the free-entry condition holds. Condition 1 states that firms pick their returns to scale to maximize profits.

3 Aggregation

In this section, we aggregate the economy and derive equations for equilibrium prices and GDP. Under Assumption 1, we can derive a tractable mapping between a firm's productivity ε_{il} and its returns to scale η_{il} in equilibrium. Indeed, (5) implies that

$$\frac{1}{1 - \eta_{il}} = \frac{1}{2\gamma_i} (\varepsilon_{il} + \log P_i - \log H_i). \quad (10)$$

In the remainder of this section, we take advantage of (10) by first aggregating firms in a sector and then for the whole economy.

3.1 Sectoral aggregation

To aggregate the economy, we define the Domar weight of a production unit (a firm or a sector) as the share of its sales in nominal GDP. For a firm l in sector i and for sector i as a whole, those are given by

$$\omega_{il} := \frac{P_i Q_{il}}{\bar{P} Y} \quad \text{and} \quad \omega_i := \frac{P_i Q_i}{\bar{P} Y}.$$

We also introduce the *effective returns to scale* $\hat{\eta}_i$, defined as the sales-weighted average of firm-level returns to scale in sector i :

$$\hat{\eta}_i := \int_0^{M_i} \frac{P_i Q_{il}}{P_i Q_i} \eta_{il} dl. \quad (11)$$

Using that quantity, we can show that the *sectoral* input cost shares are given by

$$\frac{WL_i}{P_i Q_i} = \hat{\eta}_i \left(1 - \sum_{j=1}^N \alpha_{ij} \right), \quad \frac{P_j X_{ij}}{P_i Q_i} = \hat{\eta}_i \alpha_{ij}, \quad \text{and} \quad \frac{\Pi_i}{P_i Q_i} = 1 - \hat{\eta}_i.$$

In addition, we can characterize the returns to scale of any firm in sector i using $\hat{\eta}_i$.

Lemma 5. *The returns to scale η_{il} of firm l in sector i are given by*

$$\frac{1}{1 - \eta_{il}} = \frac{1 - \varphi_i}{1 - \hat{\eta}_i} + \frac{\varepsilon_{il} - \mu_i}{2\gamma_i}. \quad (12)$$

Furthermore, the moments of the firm-level returns-to-scale distribution are given by

$$E_i \left[\frac{1}{1 - \eta_{il}} \right] = \frac{1 - \varphi_i}{1 - \hat{\eta}_i}, \quad V_i \left[\frac{1}{1 - \eta_{il}} \right] = \frac{\varphi_i}{2\gamma_i}, \quad \text{and} \quad \text{Cov}_i \left[\frac{1}{1 - \eta_{il}}, \varepsilon_{il} \right] = \varphi_i > 0. \quad (13)$$

Equation (12) links a firm's own returns to scale η_{il} to its productivity ε_{il} and the effective sectoral returns to scale $\hat{\eta}_i$. For the firm with the median productivity ($\varepsilon_{il} = \mu_i$), this equation simplifies to

$$\hat{\eta}_i = \eta_i(\mu_i) + \varphi_i(1 - \eta_i(\mu_i)). \quad (14)$$

Since $\varphi_i > 0$, it follows that the effective returns to scale $\hat{\eta}_i$ of a sector is larger than that of its median firm. This is because high-productivity firms, which have higher returns to scale and are larger (Lemmas 3 and 4), are weighted more heavily in the calculation of $\hat{\eta}_i$. The gap between $\hat{\eta}_i$ and $\eta_i(\mu_i)$ increases with $\varphi_i = \sigma_i^2 / (2\gamma_i)$. Intuitively, greater productivity dispersion σ_i^2 places more mass in both tails, and because high-productivity firms receive disproportionate sales weight, effective returns to scale rise. Lower adjustment costs γ_i also allow these high-productivity firms to adopt more scalable technologies and grow more aggressively, leading to a higher sectoral returns to scale $\hat{\eta}_i$.

The second part of Lemma 5 describes the cross-sectional moments of returns to scale within a sector. The first expression shows again that φ_i controls the gap between the median and effective returns to scale. The second expression shows that a higher productivity dispersion σ_i^2 and a lower adjustment cost γ_i both contribute to greater cross-sectional dispersion in η_{il} . The third expression confirms that high ε_{il} firms choose higher η_{il} . As the endogenous returns-to-scale mechanism shuts down ($\varphi_i \rightarrow 0$), the covariance between productivity and returns to scale goes to zero. In the empirical analysis, we will rely on that covariance to measure the strength of the mechanism in the data.

Aggregating firms within a sector using the free-entry condition (8) yields the following results.¹²

Proposition 2. *The marginal cost of sector i is given by*

$$\lambda_i = \frac{1}{Z_i(\hat{\eta}_i)} W^{1-\hat{\eta}_i} \sum_{j=1}^N \alpha_{ij} \prod_{j=1}^N P_j^{\hat{\eta}_i \alpha_{ij}}, \quad (15)$$

where sectoral total factor productivity $Z_i(\hat{\eta}_i)$ is defined as

$$\log Z_i(\hat{\eta}_i) := \underbrace{\mu_i + a_i(\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1-\hat{\eta}_i}}_{\text{Exogenous returns to scale}} + \underbrace{\frac{1}{2} (1-\hat{\eta}_i) \log \left(\frac{1}{1-\varphi_i} \right)}_{\text{Superstar effect}} - \underbrace{(1-\hat{\eta}_i) \log \kappa_i}_{\text{Entry cost}}. \quad (16)$$

Furthermore, the effective returns to scale $\hat{\eta}_i$ is given by

$$\frac{1}{1-\hat{\eta}_i} = \frac{1}{2\gamma_i(1-\varphi_i)} (\mu_i + \log P_i - \log H_i). \quad (17)$$

The sectoral marginal cost λ_i takes the standard form associated with a Cobb–Douglas production function with two factors: labor, which is used for production and entry costs, and intermediate inputs. As the cost shares of those inputs sum to one, the sector as a whole exhibits *constant* returns to scale. Intuitively, free entry acts as an adjustment margin: while individual producers operate under decreasing returns, the entry of new firms allows the sector to expand to achieve constant returns.

The effective returns to scale $\hat{\eta}_i$ is the main driver of the sectoral cost shares. Specifically, the sectoral labor share is the sum of labor used for entry, with share $1-\hat{\eta}_i$, and labor used for production, with share $\hat{\eta}_i(1-\sum_j \alpha_{ij})$. These sum to a total labor share of $1-\hat{\eta}_i \sum_j \alpha_{ij}$, which is decreasing in $\hat{\eta}_i$. Intuitively, a higher $\hat{\eta}_i$ implies that firms are larger and more scalable, so fewer entrants are required to produce a given output. This economizes on fixed entry costs, which are paid in labor, and shifts the input mix toward the variable bundle, which includes intermediate goods. Consequently, as scalability increases, the sector becomes less labor-intensive.

Equation (16) characterizes the sector's total factor productivity $Z_i(\hat{\eta}_i)$. As expected, sectoral productivity depends on the mean firm-level productivity μ_i and the productivity cost $a_i(\hat{\eta}_i)$ associated with the effective returns to scale. These are captured by the first two terms in (16). Firm heterogeneity also plays a role. The third term in (16) captures a standard *dispersion effect*: as more productive firms grow larger, they receive a larger share of input factors, raising sectoral productivity. These first three terms would also appear in an exogenous returns-to-scale model in which all firms share a common $\eta_{il} = \hat{\eta}_i$. The fourth term in (16), however, captures a novel amplification channel from endogenous returns to scale. In our model, high-productivity firms not only

¹²Since all firms in a sector face the same output price, they have the same marginal cost through profit maximization. We therefore define the marginal cost λ_i of sector i as the marginal cost of any firm in that sector. Equivalently, since, as we show later, the economy is efficient, one can write the cost minimization problem of the sector and find the same expression.

produce more but also choose more scalable technologies, which allows them to grow even larger. This *superstar effect* amplifies their contribution to sectoral productivity beyond the standard dispersion effect. Finally, the last term in (16) captures the role of entry costs. A higher entry cost κ_i diverts labor away from production, lowering sectoral productivity, with the magnitude of this loss determined by the importance of the fixed factor, $1 - \hat{\eta}_i$.

Proposition 2 also provides an expression for the sector's effective returns to scale. This expression is analogous to the one determining firm-level returns to scale, given by (10), but it includes the adjustment term φ_i to account for firm heterogeneity. This adjustment reflects that larger, more productive firms have a disproportionate impact on the aggregate measure of returns to scale $\hat{\eta}_i$.

3.2 Prices and GDP

We can now aggregate the economy to derive expressions for prices and GDP. To do so, we define the sectoral Leontief inverse matrix $\mathcal{L} := (I - \text{diag}(\hat{\eta})\alpha)^{-1}$, where $\text{diag}(\hat{\eta})$ is the diagonal matrix with the effective returns-to-scale vector $\hat{\eta}$ on the main diagonal. An element $\mathcal{L}_{ij}$ of this matrix captures the importance of sector j in the production of good i , taking into account direct and indirect connections through the production network. For example, $\mathcal{L}_{ij}$ is large if sector i spends a large share of its costs on inputs from j (i.e., $\hat{\eta}_i\alpha_{ij}$ is large), or if i relies on another sector k that, in turn, relies heavily on j , and so on.

We show in Appendix C.1 that the Leontief inverse can be used to write the sectoral Domar weights as

$$\omega_i := \frac{P_i Q_i}{\bar{P} Y} = \beta^\top \mathcal{L} \mathbf{1}_i, \quad (18)$$

where $\mathbf{1}_i$ is the i th standard basis vector. As usual in network economies, the Domar weight ω_i provides a measure of the importance of sector i . A sector i has a large Domar weight if its output is heavily demanded, either directly by the household (high β_i), or indirectly by other sectors favored by the household (high $\beta_j \mathcal{L}_{ji}$).

Sectoral returns to scale $\hat{\eta}$ play an important role in shaping the production network. Intuitively, when a downstream producer increases its returns to scale, it effectively shifts its input mix toward intermediate goods, thereby increasing its demand for upstream suppliers. This strengthens the input-output linkages and raises the Domar weights of those suppliers. As we will show, this mechanism has important implications for the impact of endogenous returns to scale on the macroeconomy.

We can now characterize equilibrium prices and GDP.

Proposition 3. *The equilibrium price vector $P = (P_1, \dots, P_N)$ satisfies*

$$\log \frac{P}{W} = -\mathcal{L}(\hat{\eta}) z(\hat{\eta}), \quad (19)$$

where $z(\hat{\eta}) = (\log Z_1(\hat{\eta}_1), \dots, \log Z_N(\hat{\eta}_N))$ is the vector of log sectoral productivities (16). Further-

more, equilibrium log GDP $y := \log Y$ is given by

$$y(\hat{\eta}) = \underbrace{[\omega(\hat{\eta})]^\top z(\hat{\eta})}_{\text{Aggregate productivity}} + \underbrace{\log \bar{L}}_{\text{Labor endowment}}. \quad (20)$$

In equilibrium, prices must equal marginal production costs ($P_i = \lambda_i$). This condition, combined with Proposition 2, allows us to solve for the vector of sectoral prices as (19). Intuitively, the price of good i is low if its key suppliers—both direct and indirect, as captured by the i -th row of $\mathcal{L}$ —have high productivity z .

Equation (20) shows that the contribution of a sector's productivity z_i to GDP is proportional to its Domar weight ω_i , as in standard network economies. A key feature of our model, however, is that both ω and z depend on the endogenous effective returns to scale $\hat{\eta}$. We explore below the role played by $\hat{\eta}$ in shaping GDP.

3.3 Equilibrium existence, uniqueness and efficiency

The preceding analysis describes key equilibrium objects, such as prices and GDP, as functions of the vector of effective returns to scale $\hat{\eta}$. To solve for $\hat{\eta}$ itself and characterize how it responds to changes in the environment, it is convenient to rely on the problem of a social planner. Since there is a single representative household in the economy, the planner's problem is to maximize that household's utility (GDP) subject to the physical constraints of the environment. The following result characterizes that problem and its relation to the set of equilibria.

Proposition 4. *There exists a unique equilibrium, and it is efficient. Furthermore, the equilibrium vector of effective returns to scale $\hat{\eta}$ maximizes GDP $y(\hat{\eta})$.*

The proof of this proposition establishes an equivalence result between the set of equilibria and the set of efficient allocations. It further shows that since there exists a unique efficient allocation, there is also a unique equilibrium. We can then use the first-order conditions of the planner to find the equilibrium $\hat{\eta}$. With that object in hand, the returns to scale of all the firms can be recovered using (12).

4 Forces shaping returns to scale

In this section, we study how changes in the environment affect returns to scale in equilibrium. To do so, we rely on the fact that the equilibrium is efficient, and that the effective returns-to-scale vector $\hat{\eta}$ maximizes GDP. The first-order condition associated with that problem is¹³

$$\underbrace{\omega_i \alpha_i^\top \mathcal{L} z}_{\text{Network adjustment } d\omega^\top/d\hat{\eta}_i} + \underbrace{\omega_i \left[\frac{da_i}{d\hat{\eta}_i} + \frac{\sigma_i^2}{2} \frac{1}{(1 - \hat{\eta}_i)^2} - \frac{1}{2} \log \left(\frac{1}{1 - \varphi_i} \right) + \log \kappa_i \right]}_{\text{Productivity adjustment } dz_i/d\hat{\eta}_i} = 0. \quad (21)$$

¹³See the proof of Lemma 6 for a derivation.

A marginal increase in $\hat{\eta}_i$ has two effects on the economy. First, it makes intermediate inputs more important production factors. As a result, the network becomes more connected and Domar weights increase. Through this channel, captured by the first term in (21), the productivities z of the sectors that are upstream of i have a larger impact on GDP.

Second, increasing $\hat{\eta}_i$ affects the productivity z_i of sector i itself, as captured by the term in square brackets in (21). The first term there captures the additional productivity cost a_i of selecting higher returns to scale $\hat{\eta}_i$. In addition, when returns to scale are larger, production can move more easily to the most productive firms within the sector. This amplified dispersion effect is captured by the second term between brackets. Higher returns to scale also affect the importance of returns-to-scale *heterogeneity* within a sector. When $\hat{\eta}_i$ is close to 1, there is less room for the most productive firms to increase their returns to scale and, through that channel, reach a larger size. This superstar effect is captured by the third term between brackets. Finally, a higher $\hat{\eta}_i$ means that firms can scale up more freely and thus produce more. Consequently, fewer firms enter, and sectoral productivity benefits from a reduction in total entry costs. This effectively leads to an increase in sectoral productivity, as the last term in (21) shows.

4.1 Sectoral productivity

We now analyze how changes in the environment affect equilibrium returns to scale. Two prices play a key role in this process. The first is the price H_i of the variable input bundle. Indeed, when inputs are expensive (high W or high P), individual firms move toward technologies with lower returns to scale. But the wage W also plays an additional role at the sector level. When labor is expensive, entry is costly, and few firms enter. This allows incumbents to expand and encourages the adoption of higher returns to scale. Combining these two channels, we find that the ratio H_i/W is key in determining how returns to scale evolve at the sector level.¹⁴

To capture how changes in the environment affect this ratio, we define the input-price sensitivity matrix $\mathcal{K}$ with typical element

$$\mathcal{K}_{ij} := \frac{\partial}{\partial z_j} \log \left(\frac{H_i}{W} \right) \leq 0,$$

where the partial derivative holds $\hat{\eta}$ fixed. The matrix $\mathcal{K}$ summarizes how productivity shocks propagate through the network to affect the input cost ratio. Using the pricing equation (19), we can show that $\mathcal{K} = -\alpha \mathcal{L}$. Since higher productivity z_j lowers the production cost of all of j 's downstream customers, the elements of $\mathcal{K}$ are non-positive, with $\mathcal{K}_{ij} < 0$ whenever sector j is a direct or indirect upstream supplier to sector i (i.e., $\mathcal{L}_{ij} > 0$). The matrix $\mathcal{K}$ plays an important role in our analysis and depends crucially on the input-output structure of the economy. Without network connections, $\mathcal{K} = 0$, and several of the mechanisms that we explore below disappear.

We also introduce a measure of how responsive returns to scale are to changes in the environment. In the planner's first-order condition (21), the term $dz_i/d\hat{\eta}_i$ represents the marginal productivity

¹⁴One can show that H_i/W is the quantity that is raised to the power $\hat{\eta}_i$ in the marginal cost expression (15), which explains its importance for scalability decisions.

benefit of increasing scalability $\hat{\eta}_i$. Consequently, the sensitivity of the optimal $\hat{\eta}_i$ to changes in the environment depends on the curvature of this function. We therefore define

$$\Psi_i := \frac{d^2 z_i}{d\hat{\eta}_i^2} = (1 - \varphi_i) \frac{d^2 a_i}{d\hat{\eta}_i^2} < 0,$$

where the second equality follows directly from (16). The inverse Ψ_i^{-1} captures how elastic returns to scale are in sector i , with a large $|\Psi_i^{-1}|$ implying that returns to scale are flexible and respond strongly to changes in the environment. Both σ_i^2 and γ_i influence this elasticity through φ_i . Specifically, if φ_i is large, which is the case if productivity is dispersed (high σ_i^2) or scalability is cheap (low γ_i), the returns to scale are more elastic. Intuitively, in such sectors, there is a larger mass of high-productivity firms able to aggressively scale up in response to changes in the economic environment.

From these definitions, we can characterize how μ_j and σ_j^2 affect returns to scale.

Lemma 6. *An increase in average productivity μ_j increases returns to scale in all downstream sectors, such that*

$$\frac{d\hat{\eta}_i}{d\mu_j} = \Psi_i^{-1} \mathcal{K}_{ij} \geq 0. \quad (22)$$

Furthermore, the impact of productivity dispersion σ_j^2 on $\hat{\eta}_i$ with $i \neq j$ is given by¹⁵

$$\frac{d\hat{\eta}_i}{d\sigma_j^2} = \Psi_i^{-1} \mathcal{K}_{ij} \frac{\partial z_j}{\partial \sigma_j^2} \geq 0, \quad (23)$$

where $\frac{\partial z_j}{\partial \sigma_j^2} = \frac{1}{2(1-\hat{\eta}_j)} + \frac{1-\hat{\eta}_j}{4\gamma_j(1-\varphi_j)} > 0$. In particular, $d\hat{\eta}_i/d\sigma_j^2 \geq 0$ for $i \neq j$.

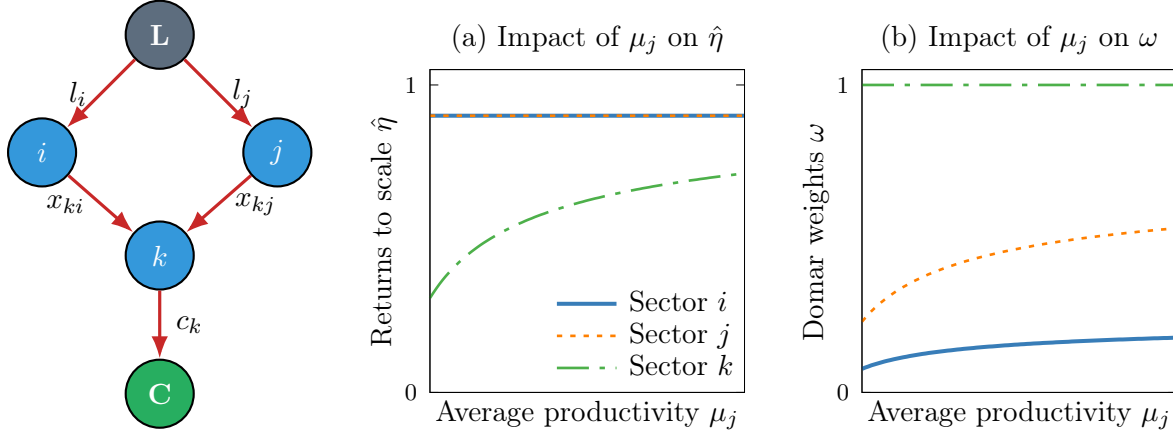
Consider (22) first. An increase in μ_j makes firms in sector j more productive, which lowers the price P_j through competition. If sector i is a downstream customer of j ($\mathcal{L}_{ij} > 0$), this lowers the price of its variable input bundle by an amount proportional to $|\mathcal{K}_{ij}|$. This in turn pushes firms in sector i to increase their returns to scale to take advantage of the cheaper intermediate inputs (Proposition 2). The magnitude of this response depends on how elastic $\hat{\eta}_i$ is, as given by Ψ_i^{-1} .

Example. Consider the economy depicted in the left panel of Figure 4. Since sector k is downstream from j , an increase in μ_j reduces the price of k 's input bundle. In response, firms in sector k increase their returns to scale $\hat{\eta}_k$, as panel (a) shows. In contrast, since sector i is not downstream from j , input prices in sector i are unchanged and so are its returns to scale $\hat{\eta}_i$.

The second part of Lemma 6 shows that an increase in σ_j^2 affects returns to scale similarly to a productivity shock μ_j . Indeed, a higher σ_j^2 raises sectoral productivity z_j through two channels. First, it strengthens the standard dispersion mechanism: even with fixed returns to scale, a higher variance σ_j^2 reallocates market share to more productive firms. Second, it amplifies the superstar effect: with more variation in productivity, the most productive firms adopt even higher returns

¹⁵In Appendix C.11, we provide a richer version of Lemma 6 that also covers the case $i = j$.

Figure 4: Productivity, returns to scale, and Domar weights in a vertical economy



to scale, further boosting sectoral efficiency. This increase in z_j lowers output prices, triggering a downstream increase in returns to scale $\hat{\eta}_i$ analogous to the response to a μ_j shock.

Lemma 6 focuses on changes in the productivity process, but it is straightforward to derive similar results for changes in entry costs κ_j and the cost of scalability γ_j . We provide these results in Appendix D.3. In a nutshell, an increase in the entry cost κ_j in sector j always reduces the effective returns to scale of any other downstream sector $i \neq j$ since it leads to an increase in the price of good j . An increase in the cost of scalability γ_j naturally leads to a reduction in returns to scale in sector j . The price of good j increases as a result, leading to lower returns to scale in other downstream sectors as well.

4.2 Implications for Domar weights

The comparative statics results derived so far describe how the equilibrium returns to scale $\hat{\eta}$ respond to the environment. These movements in $\hat{\eta}$ are important because they directly alter the sectoral Domar weights ω and, therefore, GDP.

Lemma 7. *The impact of a parameter $\chi \in \{\mu_j, \sigma_j^2, \kappa_j, \gamma_j\}$ on Domar weights is*

$$\frac{d\omega_i}{d\chi} = - \sum_{k=1}^N \mathcal{K}_{ki} \omega_k \frac{d\hat{\eta}_k}{d\chi}. \quad (24)$$

Proof. Follows directly from differentiating the Domar weight expression (18). $\square$

This result shows that if a shock to χ leads sector k to increase its returns to scale ($\hat{\eta}_k \uparrow$), the Domar weights of all sectors *upstream* of k increase as well. To illustrate this mechanism, consider again the example in Figure 4.

Example. Recall that an increase in μ_j raises productivity in sector j , lowering P_j through competition. Since k is a downstream customer of j , firms in k respond by increasing their returns to

scale $\hat{\eta}_k$ to take advantage of the cheaper input (Lemma 6). Lemma 7 shows that this shift raises the Domar weights of all of k 's suppliers (panel (b) in Figure 4). Intuitively, as sector k scales up, it becomes more input-intensive, increasing its demand for upstream goods. Consequently, the sales and the Domar weights of sectors i and j rise, reflecting their increased centrality to aggregate production. Notably, sector i 's importance grows even though its own returns to scale are unaffected by the shock.

This example highlights a key feature of our model: while productivity shocks propagate *downstream* to affect returns to scale, adjustments in returns to scale propagate *upstream* to reshape the network structure and alter Domar weights. As we show next, this interaction affects how GDP responds to changes in the environment.

5 Endogenous returns to scale and GDP

Endogenous scalability has important implications for GDP and welfare. It raises the level of GDP, amplifies the impact of beneficial shocks while dampening that of adverse shocks, and increases the long-run growth rate of the economy. We explore these mechanisms in this section.

Throughout our analysis, we compare the equilibrium of our model to a counterfactual economy in which returns to scale are exogenously fixed.

Definition 2 (Fixed returns-to-scale economy). Let the equilibrium effective returns-to-scale vector in the baseline model be $\hat{\eta}$. The *fixed returns-to-scale economy* is an otherwise identical economy in which the returns to scale of all firms are exogenously set to $\eta_{il} = \hat{\eta}_i$ for all i and l and do not respond to changes in the environment. All other quantities are chosen optimally.

Endogenous returns to scale do two main things in our model: they create dispersion in η_{il} within a sector, and they allow η_{il} to respond to changes in parameters. The fixed returns-to-scale economy, by construction, shuts down both of these channels. By comparing our baseline model to this counterfactual, we can therefore isolate the full impact of endogenous scalability on economic outcomes.¹⁶ In what follows, we use $\tilde{\cdot}$ to denote quantities in the fixed returns-to-scale economy.

5.1 Endogenous returns to scale and the level of GDP

We first examine the impact of endogenous returns to scale on GDP by comparing its level in the baseline and the fixed returns-to-scale economies. Since both economies share the same Domar weights, any difference in GDP must arise from differences in sectoral productivity. Comparing that quantity in the baseline model (Z) with its counterpart in the fixed returns-to-scale economy ($\tilde{Z}$), we find that¹⁷

$$\log Z_i(\hat{\eta}_i) - \log \tilde{Z}_i(\hat{\eta}_i) := \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right) > 0. \quad (25)$$

¹⁶In Section 7, we disentangle the separate impact of these two channels in the calibrated economy.

¹⁷See the proof of Proposition 5 for the derivation.

Thus, sectoral productivity is always larger in the model with endogenous returns to scale. Intuitively, when returns to scale are fixed, high- ε_{il} firms are no longer able to adjust their scalability to take advantage of their high productivity. This limits how much they produce, and sectoral productivity falls as a result. The superstar effect is completely shut down in that case. Equation (25) shows that the difference between the two economies is particularly pronounced when the effective dispersion φ_i is large and the effective returns to scale $\hat{\eta}_i$ is low. In those circumstances, highly productive firms in the baseline model can deviate strongly from $\hat{\eta}_i$ and thus contribute more to sectoral productivity.

The next result characterizes the aggregate impact of endogenous returns to scale.

Proposition 5. *The difference in log GDP between the baseline model and the fixed returns-to-scale economy is given by*

$$y - \tilde{y} = \sum_{i=1}^N \omega_i \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right) > 0. \quad (26)$$

The gain in GDP from endogenous returns to scale is simply the Domar-weighted gain in sectoral productivity. We see from (26) that $y - \tilde{y}$ is particularly large if the sectors in which high-productivity firms can raise their returns to scale more easily (high φ_i and low $\hat{\eta}_i$) are also important suppliers (high ω_i). In the calibrated economy of Section 7, we will see that the impact of endogenous returns to scale on the level of GDP can be sizable.

5.2 GDP response to changes in the environment

In addition to its impact on the level of GDP, endogenous returns to scale also affect how GDP responds to changes in the environment.

Proposition 6. *Let partial derivatives indicate that returns to scale $\{\eta_{il}\}$ are taken as fixed. In equilibrium, the following holds.*

1. *An increase in average productivity μ_j raises GDP:*

$$\frac{dy}{d\mu_j} = \frac{\partial y}{\partial \mu_j} = \omega_j > 0. \quad (27)$$

2. *An increase in productivity dispersion σ_j^2 raises GDP:*

$$\frac{dy}{d\sigma_j^2} = \frac{\partial y}{\partial \sigma_j^2} = \omega_j \left(\frac{1}{2} \frac{1}{1 - \hat{\eta}_j} + \frac{1 - \hat{\eta}_j}{4\gamma_j} \frac{1}{1 - \varphi_j} \right) > 0. \quad (28)$$

3. *An increase in entry cost κ_j lowers GDP:*

$$\frac{dy}{d \log \kappa_j} = \frac{\partial y}{\partial \log \kappa_j} = -\omega_j (1 - \hat{\eta}_j) < 0.$$

4. An increase in the returns-to-scale productivity cost γ_j lowers GDP:

$$\frac{dy}{d\gamma_j} = \frac{\partial y}{\partial \gamma_j} = -\omega_j \left(\frac{1}{1 - \hat{\eta}_j} + \frac{1 - \hat{\eta}_j}{2\gamma_j} \frac{\varphi_j}{1 - \varphi_j} \right) < 0. \quad (29)$$

Proof. The result follows directly from the envelope theorem. $\square$

Since the equilibrium $\hat{\eta}$ maximizes GDP, any *marginal* adjustment in returns to scale must have no impact on GDP. This implies that GDP responds to marginal changes in the environment *as if* returns to scale were fixed. Consequently, Hulten's (1978) theorem applies: the first-order impact of a productivity shock $d\mu_j$ is simply the Domar weight ω_j of the affected sector. Note, however, that Domar weights themselves are endogenous in our model and depend on the incentives shaping returns to scale. Similarly, the impact of a change in entry costs $d\log \kappa_j$ is determined by its direct effect on sector j 's productivity z_j , captured by $1 - \hat{\eta}_j$, weighted by that sector's importance, ω_j .

The impacts of σ_j^2 and γ_j operate in a similar way, but here endogenous returns to scale feature more prominently. Consider first an increase in σ_j^2 . This has two positive effects on sector j 's productivity z_j . First, the higher variance in ε_{il} implies a larger mass of very productive firms, with positive consequences for GDP. This dispersion effect is captured by the term $\frac{1}{2} \frac{\omega_j}{1 - \hat{\eta}_j}$ in (28) and would be at work even without within-sector dispersion in η_{il} (i.e., in the fixed returns-to-scale economy). Second, increasing σ_j^2 interacts with the superstar effect. Since high- ε_{il} firms already have high returns to scale, the increase in dispersion has a disproportionate impact on them. This effect is captured by the remaining term in (28). Overall, increasing σ_j^2 always has a positive effect on GDP, and endogenous returns to scale makes that effect larger, even to a first order, as it creates dispersion in within sector firm-level returns to scale.

Conversely, an increase in the cost of adjusting returns to scale γ_j has two adverse effects on GDP. The first effect is mechanical: a higher γ_j directly increases the average productivity cost $-a_j(\hat{\eta}_j)$ in sector j , which lowers GDP. This effect is captured by the first term in (29). Second, γ_j interacts with the superstar effect. Since the biggest producers have the highest returns to scale, they suffer particularly strongly from an increase in γ_j . Indeed, recall that $a_j(\eta_{jl}) = -\gamma_j / (1 - \eta_{jl})$ such that for $\eta_{jl} \approx 1$, a marginal increase in γ_j has a particularly severe impact on productivity. This effect is captured by the second term in (29). Proposition 6 shows that increasing γ_j always hurts y , and all the more so when the superstar effect is stronger.

5.3 Second-order impact of productivity shocks

In our baseline model, the equilibrium is efficient, which implies that GDP responds to infinitesimal productivity shocks as if returns to scale were held fixed. For larger shocks, however, the Domar weights themselves respond, leading to a further impact on GDP.

Proposition 7. *The response of log GDP y to a shock $\Delta\mu_i$ is given by*

$$\Delta y = \omega_i \Delta\mu_i + \frac{1}{2} \frac{d\omega_i}{d\mu_i} (\Delta\mu_i)^2 + o\left((\Delta\mu_i)^2\right). \quad (30)$$

Furthermore, the second-order term is non-negative, $d\omega_i/d\mu_i \geq 0$ and given by (24).

Equation (30) provides a second-order approximation of the GDP response to a shock $\Delta\mu_i$. While the first-order term is the standard Domar-weight effect from Hulten's theorem, the second-order term captures a novel channel driven by the endogenous adjustment in returns to scale. That channel operates through the response of Domar weights to the shocks.

The intuition is straightforward. Recall that a positive productivity shock in sector i propagates downstream, lowering input costs for i 's customer sectors. These sectors, in turn, are incentivized to increase their own returns to scale to capitalize on the cheaper inputs. This shift towards greater scalability makes the production network more reliant on sector i , which increases its Domar weight. As a result, the derivative $d\omega_i/d\mu_i$ is positive. This implies that the second-order term is always positive, adding *convexity* to the GDP response. In other words, endogenous returns to scale amplify the impact of positive productivity shocks and dampen the adverse impact of negative shocks.

To identify the sources of this amplification and dampening, we can use Lemma 6 to express the second-order coefficient as

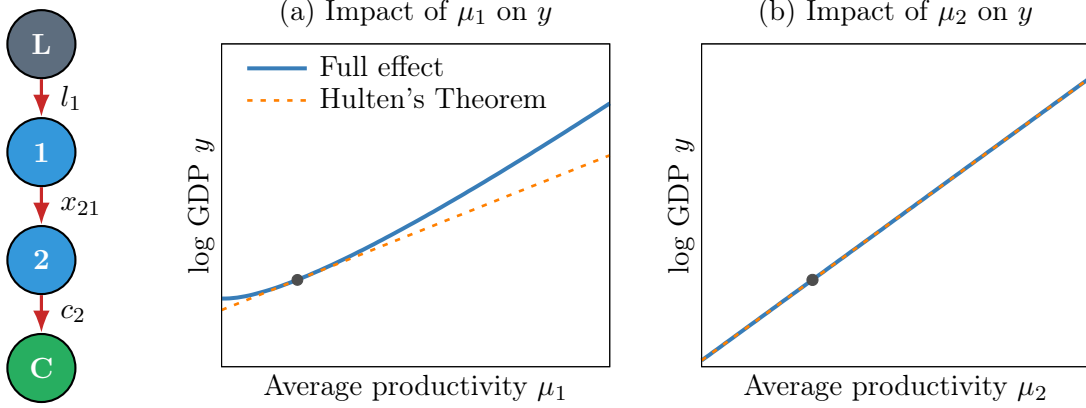
$$\frac{d\omega_i}{d\mu_i} = - \sum_{k=1}^N \omega_k \Psi_k^{-1} \mathcal{K}_{ki}^2 \geq 0. \quad (31)$$

The second-order effect is therefore stronger when the shock hits a sector i that is a key supplier (large $|\mathcal{K}_{ki}|$) to sectors that are large (high ω_k) and have elastic returns to scale (low $|\Psi_k|$, meaning a low cost of adjusting scalability). Those sectors more strongly increase their cost share of good i after the increase in μ_i , contributing to a larger increase in ω_i . Crucially, it is the square of $\mathcal{K}_{ki}$ that shows up in (31), implying that the convex response of GDP is disproportionately stronger for shocks to the most important suppliers.

Proposition 7 implies that whether shocks hit sectors that are upstream or downstream in the supply chain matters for their impact on GDP. The following example illustrates the mechanism.

Example. Consider the vertical economy depicted in Figure 5. An upstream sector (Sector 1) sells its entire output to a downstream sector (Sector 2), which in turn sells to the final consumer. Panel (a) shows the impact of a productivity shock μ_1 to the *upstream* sector. The solid line, representing the response of GDP, is clearly convex and lies above the linear, first-order approximation from Hulten's theorem (dashed line). This illustrates the amplification effect: as μ_1 increases, the price of good 1 falls, inducing firms in the downstream Sector 2 to increase their returns to scale to capitalize on cheaper inputs. This change in production processes makes Sector 1 a more important supplier, increasing its Domar weight and thus magnifying the aggregate benefit of its higher productivity. Panel (b) shows a starkly different result for a productivity shock μ_2 to the *downstream* sector. In this case, the full GDP response is linear and coincides with Hulten's theorem approximation. Because Sector 2 is at the bottom of the supply chain, a fall in its price provides no benefit to any other sectors. There are no downstream customers to re-optimize their scalability choices, and thus no structural amplification. This example shows a key implication of our model: the macroeconomic impact of a productivity shock depends crucially on a sector's position in the production network.

Figure 5: The impact of sectoral productivity on GDP in a vertical economy



5.4 Extension: The role of wedges

So far, our analysis has focused on an efficient equilibrium where the envelope theorem holds, meaning that adjustments in $\hat{\eta}$ have only second-order effects on GDP. We now show that in the presence of frictions, markups or other distortions, this is no longer the case, and changes in returns to scale can have first-order effects on GDP.

We consider a setup with general wedges in Appendix D.4, but to illustrate the forces at work transparently, we focus here on distortionary sales wedges. Specifically, we assume that firm l in sector i retains only a fraction $1 - \tau_i^S$ of its revenue, where $\tau_i^S \in [0, 1)$ is the wedge. From the firm's perspective, these wedges are equivalent to a reduction in productivity, leading to an inefficient adjustment in scalability. We assume that the proceeds from the wedges are fully rebated lump-sum to the household, ensuring that they act as pure distortions with no direct loss in resources.

Sales wedges directly affect scalability choices, as the following lemma shows.

Lemma 8. *An increase in τ_j^S decreases the returns to scale in all downstream sectors:*

$$\frac{d\hat{\eta}_i}{d\tau_j^S} = -\frac{1}{1 - \tau_j^S} \Psi_i^{-1} \mathcal{K}_{ij} \leq 0. \quad (32)$$

Intuitively, the sales wedge acts like a markup of $(1 - \tau_j^S)^{-1}$, creating a gap between marginal costs and market prices. As τ_j^S rises, output prices increase as well, making intermediate inputs more expensive for downstream firms. Facing higher variable input costs, these firms optimally substitute away from input-intensive technologies by reducing their returns to scale. Since firms do not internalize that the revenue from wedges is rebated to the household, this adjustment leads to an equilibrium with inefficiently low returns to scale.

Wedges also have important implications for GDP. Propositions 6 and 7 show that without wedges, changes in returns to scale only have second-order effects on GDP. This is no longer the case when wedges are present.

Proposition 8. *In the presence of sales wedges, the impact of a parameter $\chi \in \{\mu_j, \sigma_j, \kappa_j, \gamma_j\}$ on GDP is given by*

$$\frac{dy}{d\chi} = \underbrace{\frac{\partial y}{\partial \chi}}_{\text{Direct effect}} + \underbrace{\sum_{i=1}^N \frac{\partial y}{\partial \hat{\eta}_i} \frac{d\hat{\eta}_i}{d\chi}}_{\text{Structural change effect}}, \quad (33)$$

where $\partial y / \partial \chi$ is given by Proposition 6, $d\hat{\eta}_i / d\chi$ is given by Lemmas 6, 10 and 11, and $\partial y / \partial \hat{\eta}_i \geq 0$.

This proposition shows that the total effect of a shock is now the sum of a direct effect (the standard Hulten-like term) and a new scalability effect. Since sales wedges push $\hat{\eta}$ to be inefficiently low, any shock that incentivizes firms to increase their returns to scale ($d\hat{\eta}/d\chi > 0$) now generates an additional, first-order welfare gain.

Consider, for example, a positive productivity shock $d\mu_j > 0$. As in the efficient case, this shock directly raises GDP through the Hulten term. However, the lower input costs that it generates also induce firms to increase their $\hat{\eta}$. Because the economy started from an inefficiently low level of returns to scale, this structural adjustment is no longer a second-order refinement but a first-order improvement. This implies that $dy/d\mu_j \geq \omega_j$. Productivity shocks have a larger impact on GDP in this distorted economy because they not only improve productivity but also partially correct the pre-existing distorted scalability structure.

The nature of this new first-order effect depends on the type of distortion affecting the economy. While sales wedges lead to inefficiently low returns to scale, other distortions can have the opposite effect. For example, as we show in Appendix D.4, a corporate profit tax effectively raises the cost of entry, which incentivizes incumbent firms to choose inefficiently high returns to scale. In such an economy, a productivity shock that further raises $\hat{\eta}$ could actually be welfare-reducing at the margin.

5.5 Implications for growth

Endogenous scalability also accelerates long-run economic growth. To illustrate this mechanism transparently, we consider a continuous-time setup where the equilibrium of a single-sector version of our baseline model is repeated at each instant. Growth is driven by a constant rate of productivity improvement, $d\mu/dt = g_\mu > 0$.

We first characterize the growth rate of returns to scale in this environment.

Corollary 1. *The growth of effective returns to scale $\hat{\eta}$ is given by*

$$\frac{d\hat{\eta}}{dt} = \Psi^{-1} \mathcal{K} g_\mu > 0. \quad (34)$$

Furthermore, as $t \rightarrow \infty$, effective returns to scale $\hat{\eta}$ converges to 1.

The constant improvements in sectoral productivity μ result in cheaper intermediate inputs, which push firms to increase their returns to scale, so that $d\hat{\eta}/dt > 0$.¹⁸ The strength of that

¹⁸Several empirical studies document rising returns to scale over time. De Loecker et al. (2020) estimate that

mechanism relies on how fast productivity improves (g_μ), but also on the importance of intermediate inputs in production, as captured by the term $-\alpha/(1 - \hat{\eta}\alpha) = \mathcal{K}$. When α is large, intermediates are more important and $\hat{\eta}$ grows faster, all else equal. In addition, $d\hat{\eta}/dt$ depends on the elasticity of $\hat{\eta}$ in response to a change in input prices. As in the comparative statics exercise of Section 4, this effect is captured by $1/\Psi$. An economy that is more flexible and faces stronger incentives will more rapidly reconfigure itself toward a more scalable structure.

The evolution of the economy's returns to scale matters for the growth rate of GDP.

Proposition 9. *The growth rate of GDP is given by*

$$\frac{dy}{dt} = \frac{g_\mu}{1 - \alpha} \times \left(1 - \frac{1}{\sqrt{\frac{1}{\gamma} \frac{1 - \alpha}{\alpha} \frac{g_\mu}{1 - \varphi} t + K}} \right) > 0, \quad (35)$$

where $K > 0$ is a time-invariant term given in the proof of the proposition.

This proposition provides a closed-form solution for the growth rate of GDP along its entire transition path. The growth rate dy/dt can be understood as the product of two terms: a *potential long-run growth rate*, $g_\mu/(1 - \alpha)$, and a convergence factor (the term in parentheses) that starts below one and asymptotically approaches it as $t \rightarrow \infty$. This means that the economy's growth rate is always increasing, accelerating towards its long-run potential, which it reaches asymptotically.

The speed of this convergence is governed by the terms inside the square root. The economy adapts its structure and accelerates more quickly when the forces driving reorganization are stronger. Specifically, the transition is faster when: 1) the pace of innovation (g_μ) is high, providing a strong and persistent incentive to adapt, and 2) the economy is structurally flexible, meaning the net cost of adjusting scale is low (low γ and high φ).

Figure 6 provides an example of the dynamics of the growth rate of GDP. The left panel shows that from its initial condition $\hat{\eta}(0) = \hat{\eta}_0$, growth increases before converging to its long-run level $g_\mu/(1 - \alpha)$. The right panel depicts the same economy but with a higher γ . In this case, increasing returns to scale is more costly, and the progression to the long-run growth rate is slower.¹⁹

Finally, we can compare the growth rate of GDP in the model with endogenous returns to scale to its counterpart in the fixed returns-to-scale economy.

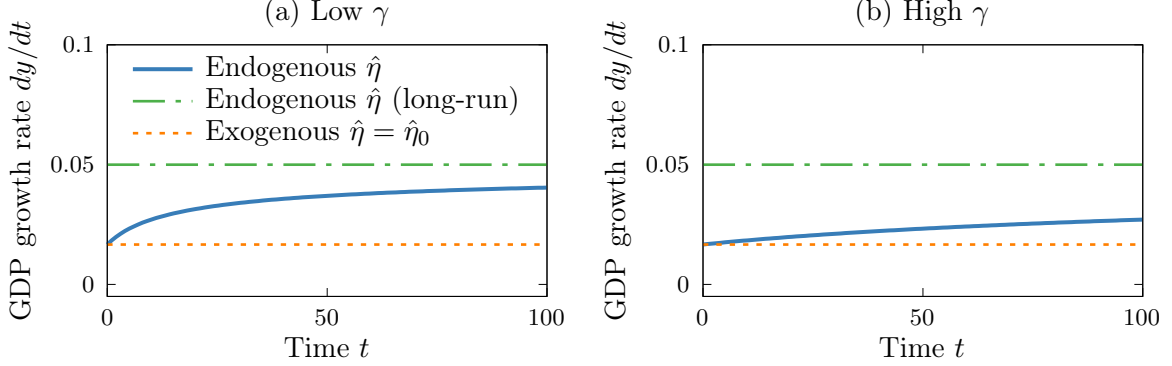
Corollary 2. *For any $t > 0$, GDP grows faster in the economy with endogenous returns to scale. In the limit as $t \rightarrow \infty$, the long-run growth rates satisfy*

$$\lim_{t \rightarrow \infty} \frac{dy}{dt} = \frac{1}{1 - \alpha} g_\mu > \frac{1}{1 - \hat{\eta}_0 \alpha} g_\mu = \lim_{t \rightarrow \infty} \frac{d\tilde{y}}{dt},$$

firm-level returns to scale have increased over the last few decades in the United States. Chiavari and Goraya (2025) show that this finding holds even after accounting for intangible capital. Lashkari et al. (2024) provide evidence that the decline in IT prices led to an increase in returns to scale in France.

¹⁹While growth is always accelerating ($d^2y/dt^2 > 0$), the rate of acceleration is in general modest. One can show that the second derivative of log GDP scales with g_μ^2 . Thus, for realistic calibrations of annual productivity growth (e.g., $g_\mu \approx 1\%$), the acceleration is a gradual, slow-moving process that might be hard to notice over short horizons.

Figure 6: Endogenous returns to scale accelerate growth



where $\tilde{y}$ is log GDP in the fixed returns-to-scale economy, and where $\hat{\eta}_0$ is effective returns to scale in the baseline economy at $t = 0$.

This result captures the key growth implication of our model: an economy that can adapt its returns to scale grows faster. In the fixed-scale economy, the benefits of technological progress are constrained by a static production structure. In our model, in contrast, as sectoral productivity increases, the economy adopts more scalable technologies and becomes more interconnected.²⁰ This increases Domar weights and magnifies the benefit of the higher productivity. This effect grows larger over time, with an increasing gap between the growth rates of the two economies. Our model therefore suggests that the macroeconomic consequences of scaling decisions, like Ford's introduction of the moving assembly line, are not a one-off level effect, but a persistent force that reshapes the economy's long-run growth trajectory.

The quantitative implications of endogenous returns to scale for growth can be substantial. For instance, using parameters calibrated to the Spanish economy (where $\alpha \approx 0.67$ and $\hat{\eta}_0 \approx 0.83$, see the next sections), a 1% annual rate of underlying technological progress (g_μ) would translate into a 2.2% growth rate in the fixed- η economy. With endogenous returns to scale, however, the long-run growth rate converges to 3.0%.

6 Empirical evidence

Our theoretical results describe how returns to scale respond to the economic environment. In this section, we use detailed firm-level data from Spain to provide empirical evidence for these predictions at the firm, sector, and aggregate levels. Consistent with the model, we show a robust

²⁰Because a sector's intermediate-input share of gross output increases with $\hat{\eta}$ (see Section 3.1), Corollary 1 predicts that this share rises gradually over time. At first glance this may seem at odds with the data: the aggregate U.S. intermediate-input share of gross output has been roughly stable since 1947. That stability, however, reflects two offsetting movements: a rising *within-sector* share and a *between-sector* reallocation toward less intermediate-intensive (service) sectors. The within-sector share has in fact risen by about 2.5 percentage points since 1947, of the same order as the increase implied by our calibration (Appendix B.4.3 provides the comparison). Our model is designed to capture this within-sector margin; it abstracts, for tractability, from the between-sector reallocation that is well known to accompany economic growth.

positive correlation between productivity and returns to scale in the cross-section of firms. We further use panel data and within-firm variation to demonstrate that returns to scale respond to incentives: firms increase scalability as they grow and reduce it when facing higher input costs. At the sector level, the theory predicts that industries with stronger endogenous scalability should exhibit fatter firm-size tails. We find strong support for this mechanism in the data. Finally, our theory implies that endogenous scalability is beneficial for GDP. To test this prediction, we extend our analysis to 24 countries, and show that the strength of the mechanism is indeed a predictor of long-run economic development.

6.1 Data

Our primary source of firm-level data is Moody’s Orbis Historical database, which covers the near-universe of Spanish firms between 1995 and 2019. This dataset provides detailed information on sales, labor costs, capital stocks, and material costs. After cleaning, our sample comprises 9,754,405 firm-year observations.²¹ We use these data to estimate firm-level returns to scale using a production function approach. We briefly describe our estimation procedure below and provide additional details in Appendix A.1.

We complement our analysis with firm-level data from 23 additional countries. We use data from Orbis for 21 European countries with good coverage of the key variables needed for production function estimation. For developing economies, we rely on China’s National Bureau of Statistics (NBS) firm-level database and India’s Annual Survey of Industries (ASI). Both datasets are censuses of manufacturing firms above specific size thresholds. The details of data cleaning and variable construction are provided in Appendix A.9.

6.2 Cross-sectional evidence

Our theory predicts that within a sector, firms of similar size should have similar returns to scale. We take advantage of this prediction to estimate returns to scale across the firm-size distribution. Specifically, for each sector i and year t , we group firms into 10 deciles based on their 7-year moving average of firm-level log sales (years $t - 3$ to $t + 3$). This construction smooths out short-run fluctuations and measurement error and thus yields a more reliable measure of a firm’s position in the size distribution.

In our baseline approach, we assume that all firms in sector i , year t and decile d_t share the same Cobb-Douglas production technology $Q_{ilt} = \tilde{A}_{ilt} K_{ilt}^{\beta_{i,d_t(l),t}^K} L_{ilt}^{\beta_{i,d_t(l),t}^L} M_{ilt}^{\beta_{i,d_t(l),t}^M}$, where K_{ilt} , L_{ilt} and M_{ilt} denote capital, labor and materials, respectively. We then estimate the output elasticities for each cell (i, t, d_t) using the Blundell and Bond (2000) IV-GMM estimator on a 7-year rolling-window sample. The estimated returns to scale η_{ilt} for a firm l in sector i and year t is therefore given by the sum of these elasticities: $\eta_{ilt} = \hat{\beta}_{i,d_t(l),t}^K + \hat{\beta}_{i,d_t(l),t}^L + \hat{\beta}_{i,d_t(l),t}^M$. We use the Blundell–Bond

²¹We deflate all nominal variables using the Spanish GDP deflators and drop any firm-year observation whose average revenue product for any input (fixed assets, wage bills, or material costs) lies above the 99th percentile or below the 1st percentile of that year’s distribution.

estimator as our baseline because it can estimate the gross output production function by leveraging moment conditions that exploit input persistence, without requiring assumptions on input timing. We then use the estimated returns to scale for the years 1997-2019 in our empirical analysis and in the calibration of Section 7.²²

To compare productivity in the cross-section despite heterogeneous production technologies, we follow best practices in the development accounting literature (Caves et al., 1982a; Feenstra et al., 2015) and construct a comparable measure, $\hat{z}_{ilt}$, using a Törnqvist productivity index. Specifically, we define a hypothetical “average firm” for each sector-year with mean (log) sales, (log) inputs, and output elasticities. The measure $\hat{z}_{ilt}$ then compares productivity between firm l and the average firm by looking at how much more output one produces relative to the other, adjusting for differences in technology and input use. We formally define this productivity index and provide more detail in Appendix A.3.²³

A key prediction of our theory is that larger, more productive firms operate technologies with higher returns to scale. Figure 7 supports this pattern in the data. Panel (a) plots returns to scale against firms’ sales percentiles within a sector-year. It shows that the largest firms (100th percentile) operate with returns to scale of around 0.91, compared to 0.79 for the smallest firms (1st percentile).²⁴ Furthermore, panel (b) shows that higher productivity is associated with higher returns to scale, which is a distinctive feature of our endogenous returns-to-scale mechanism. In addition, the empirical relationship between returns to scale and productivity is concave. This supports our assumption of a rising marginal cost of scalability (concave a_i) and qualitatively matches the theoretical prediction illustrated in Figure 2.

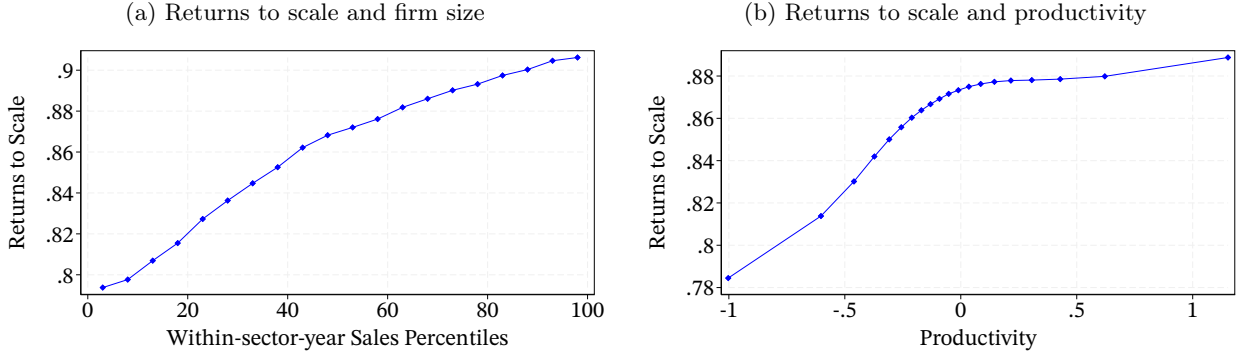
In Appendix A.4, we confirm the robustness of our results using a wide range of alternative strategies. These include: (i) alternative production function estimators, including Olley and Pakes (1996) and Levinsohn and Petrin (2003); (ii) controls for market power to alleviate the measurement errors in output using the Akerberg et al. (2015) estimator; and (iii) grouping firms by rolling sales percentiles or by contemporaneous rather than moving-average sales. Our main empirical patterns are robust across all of these alternative estimation strategies. One additional worry is that in reality firms might deviate from our model’s price-taking assumption, in which case our estimates measure revenue rather than quantity returns to scale. To address this, in Appendix A.5 we assume that firms face CES or Kimball demand and find that the returns-to-scale–size gradient is stronger than under our baseline specification, suggesting that the baseline provides a conservative estimate of this relationship. Finally, a Monte-Carlo exercise in Appendix E applies the Blundell–Bond estimator to panels simulated from a dynamic extension of the baseline model. It accurately recovers the true

²²We do not estimate the elasticities for 1995 and 1996 as there are few observations in each sector-year-decile cell in those years. Our 1997-2019 sample covers 9,424,952 firm-year observations.

²³In our calibrated economy of Section 7, the within-sector correlation between this Törnqvist index and ε_{il} is above 0.99. The same number for $\varepsilon_{il} + a_i(\eta_{il}) + \log \zeta_{il}$ is about 0.92. The Törnqvist index is therefore a good proxy for productivity for reduced-form estimates.

²⁴Hubmer et al. (2026) document similar patterns in Canadian firms and US manufacturing firms. Gao and Kehrig (2025) and McAdam et al. (2024) report that industries with larger average firm size also have higher returns to scale in the United States and in European countries, respectively. Using production data to infer firm- and industry-level returns to scale has a long tradition since Hall (1990) and Basu and Fernald (1997).

Figure 7: Returns to scale, productivity, and firm size in the cross-section



Notes: Panel (a) presents a binned scatter plot of firm-level returns to scale against within-sector-year sales percentiles. Panel (b) presents a binned scatter plot of firm-level returns to scale against productivity, controlling for sector-year fixed effects; the unconditional mean of returns to scale is added back for interpretability. Sectors are defined as the 62 sectors in the Input–Output table of the Annual Spanish National Accounts. Both panels are constructed using a sample of Spanish firms from Orbis. See Appendix A.1 for details on variable construction and sample selection.

size profile and sectoral effective returns to scale.

6.3 Within-sector evidence: firm growth

This cross-sectional evidence shows that, on *average*, larger and more productive firms are more scalable. But the theory also predicts that the returns to scale of *individual firms* should respond to changes in the economic environment. Indeed, Lemma 3 establishes that beneficial shocks, whether higher productivity or lower input prices, induce firms to adopt more scalable technologies. Since these same shocks also drive firm expansion, the model implies that returns to scale and sales should co-move positively within a firm over time. We test this prediction in this subsection.

Testing this prediction requires a different empirical strategy. The baseline procedure of Section 6.2 estimates a single returns-to-scale parameter for all firms in a sector-year-decile cell, and is therefore not designed to detect changes in technology within firms. Instead, we extend the Blundell–Bond estimator to allow output elasticities to vary with a firm’s growth history, interacting each input with indicators of how much the firm has grown since entering the panel. Apart from these interactions, the moment conditions closely follow the design of Section 6.2. Monte-Carlo simulations presented in Appendix E.6 validate our approach.

We measure a firm’s growth, G_{it} , as the difference between the firm’s recent log sales, averaged over years $t - 5$ to $t - 3$, and its average log sales during its first two years in the panel. Because its most recent component is dated $t - 3$ and its entry benchmark is fixed once and for all, G_{it} is predetermined with respect to the shocks that affect the firm’s current sales and inputs. Within each sector-year, we sort firms into terciles of G_{it} , with the middle tercile as the reference group. Appendix A.6 provides the formal definition and the implementation details.

We then estimate, separately for each sector i , the following equation:

$$q_{ilt} = \sum_{X \in \{K, L, M\}} \left(\beta_i^X + \varphi_i^X t + \sum_{b \neq 2} \theta_{bi}^X T_{ilt}^b \right) x_{ilt}^X + \sum_{b \neq 2} \delta_{bi} T_{ilt}^b + \alpha_i + \gamma_{it} + \varepsilon_{ilt}, \quad (36)$$

where q_{ilt} is log sales, x_{ilt}^X is log input X , and T_{ilt}^b indicates that firm l belongs to growth tercile b in sector-year it . Because the regression is estimated sector by sector, all coefficients, including the year fixed effects γ_{it} , are sector-specific. The trend interaction $\varphi_i^X t$ allows each input's elasticity to drift linearly over the 24-year panel, absorbing slow technological change. Estimation follows the baseline procedure, with the interactions and tercile indicators instrumented in the same way as the inputs. For sector i , the estimated difference in returns to scale between tercile b and the middle tercile is $\text{RTS}_i(T_b) - \text{RTS}_i(T_2) = \sum_{X \in \{K, L, M\}} \theta_{bi}^X$, and we average these sector-level differences into a single economy-wide estimate.²⁵

Panel (a) of Figure 8 reports the estimated returns-to-scale differences across growth terciles. Firms in the bottom growth tercile operate returns to scale 0.012 below the middle tercile of their own sector, while firms in the top tercile operate 0.010 above it — a spread of 0.022 between the most- and least-grown firms within the same sector. Because the growth terciles are formed from past sales and the comparison is within sector, the rising gradient is not driven by contemporaneous shocks to sales or by cross-sector composition. It thus confirms a key model implication: as firms grow, their returns to scale rise.²⁶

6.4 Within-firm evidence: input costs and the network

We next examine how returns to scale respond to changes in input costs. Lemma 3 predicts that an increase in the price H_i of the variable-input bundle induces firms to choose technologies with lower returns to scale. In the model, this price is also the channel through which the production network shapes technology choices. Because the bundle includes intermediate inputs, its price depends on the prices set by a firm's suppliers and, through them, on costs further upstream. A cost shock anywhere in the network should therefore propagate downstream and lower the returns to scale of the firms it reaches.

To test these predictions, we need variation in input costs that firms do not control. We use changes in the world price of crude oil, propagated to each sector through the Spanish input–output network. The variation we use therefore comes from movements in a world energy price that individual Spanish producers cannot influence.

²⁵We use the mean-group estimator of Pesaran and Smith (1995), that is, the unweighted average of the sector-level estimates, and compute standard errors from the cross-sector dispersion of the estimates. See Ibragimov and Müller (2010) for the formal argument of inference based on group-level estimates.

²⁶In Appendix A.6, we report our implementation details and show that the results are robust to alternative definitions of firm growth and to restricting the sample to firms above the median size in their sector. One additional worry is that firms with permanently high growth rates might also have high returns to scale. To address this concern, we control in the same appendix for the firm's permanent growth tercile (interacted with the inputs), so that identification comes only from within-firm changes in growth.

We first measure each sector's exposure to oil prices. Let α_{ij} denote the share of sector i 's variable costs spent on inputs from sector j , computed from the 2010 Spanish input–output table used in the calibration, and let $\mathcal{E}$ denote the set of energy-producing sectors: coke and refined petroleum, and electricity, gas, and steam. We define the network energy exposure ℓ_i of sector i as

$$\ell_i = \sum_{j \in \mathcal{E}} \alpha_{ij} + \sum_{j \notin \mathcal{E}} \alpha_{ij} \ell_j.$$

The first term captures sector i 's direct purchases from energy producers, while the second captures the energy embodied in the inputs it buys from non-energy sectors. This recursion is the standard Leontief price equation; solving it forward sums the energy content of every supply chain that reaches sector i , so that ℓ_i measures the total energy share, direct and indirect, of sector i 's variable costs.²⁷ Across the non-energy sectors of the baseline design, ℓ_i ranges from 0.03 to 0.27 with an average of 0.11.

We then define the sector-level cost shifter $\log P_{it}^E := \ell_i \Delta \log p_t^{\text{oil}}$, where $\Delta \log p_t^{\text{oil}}$ is the annual change in the log euro price of Brent crude oil. The cost shifter measures the energy-driven change in the log price of sector i 's entire variable-input bundle. It is therefore the empirical counterpart of a change in $\log H_i$, the quantity that governs the returns-to-scale choice in the model. The same oil-price movement raises this price more in sectors that use more energy, either directly or through their suppliers.

The estimating equation replaces the growth-tercile interactions in (36) with interactions between each input and this continuous cost shifter,

$$q_{ilt} = \sum_{X \in \{K, L, M\}} (\beta_i^X + \varphi_i^X t + \theta_i^X \log P_{it}^E) x_{ilt}^X + \alpha_t + \gamma_{it} + \varepsilon_{ilt}, \quad (37)$$

which we again estimate separately for each sector, following the baseline procedure (see Appendix A.7). Because the shifter is common to all firms in a sector-year, its main effect is absorbed by the year effects. The implied sector-specific response of returns to scale is $\partial \eta_{it} / \partial \log P_{it}^E = \sum_{X \in \{K, L, M\}} \theta_i^X$.

Averaging the sector-level responses as before yields a coefficient of -0.066 ($t = -3.8$), which implies that a 10 percent energy-driven increase in the price of the variable-input bundle lowers returns to scale by about 0.007.²⁸ The negative and precisely estimated response supports the model's supply-chain mechanism which predicts that higher variable-input costs induce firms to choose technologies with lower returns to scale.

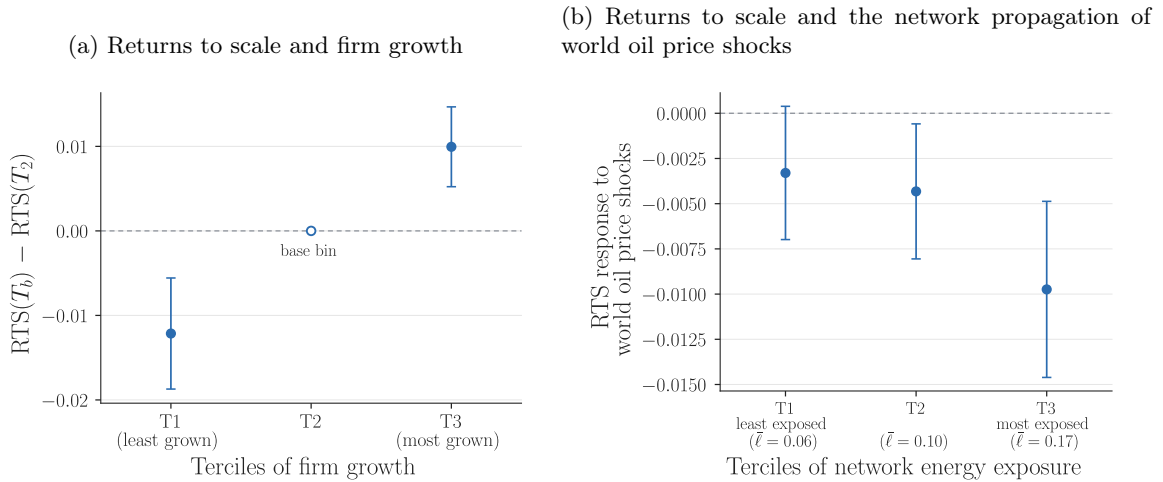
A negative average response, however, could in principle reflect any aggregate force that comoves with the world oil price. The network mechanism makes a sharper prediction: the same oil-price movement should lower returns to scale more in sectors with greater network exposure ℓ_i , since

²⁷This interpretation of ℓ_i assumes full cost pass-through and one-for-one movements between energy-producer prices and the world oil price.

²⁸We exclude the two energy-producing sectors from this average because the oil price is an output price for them rather than an input cost.

their input bundles absorb more of the shock. Panel (b) of Figure 8 tests this prediction. We sort the non-energy sectors into terciles of ℓ_i and average, within each tercile, the sector-level response of returns to scale to the oil price itself. These responses are expressed per unit log change in the oil price rather than in the bundle price, so they are mechanically smaller in magnitude than the estimate above. As the figure shows, the response strengthens monotonically with exposure and reaches -0.010 in the top tercile, three times the magnitude of the bottom-tercile estimate. This suggests that energy-cost shocks propagate, in proportion to sectors' direct and indirect exposure, to the technology choices of downstream firms, as the model predicts.²⁹

Figure 8: Within-firm evidence: growth of returns to scale and the input-cost response



Notes: Panel (a) reports $RTS(T_b) - RTS(T_2)$ by tercile of predetermined firm growth since its first observation in the panel. The middle tercile is omitted and therefore equals zero by construction. The production-function interactions are estimated separately for each of the 55 sector cells using the system-GMM and minimum-distance procedure. Panel (b) reports the response of returns to scale to a unit log increase in the euro price of Brent crude oil, by tercile of fixed network energy exposure ℓ_i . The two energy-producing cells are excluded because the oil price is an output price rather than an input cost, leaving 53 non-energy sector cells. In both panels, points are unweighted mean-group estimates across the relevant sector cells (Pesaran and Smith, 1995), and whiskers are 90 percent confidence intervals based on the cross-cell dispersion of the estimates (Fama and MacBeth, 1973; Ibragimov and Müller, 2010). See the Appendix for variable construction, implementation details, and additional results.

Taken together, the within-firm results provide direct evidence that returns to scale adjust endogenously. They rise as firms grow and fall when upstream cost shocks raise input prices. The same technology margin responds in both settings, supporting the model's central premise that firms adjust their returns to scale in response to changes in economic conditions.

6.5 Sectoral evidence

Our model also has implications for cross-industry variation.³⁰ For instance, because high-productivity firms adopt more scalable technologies to leverage their productivity advantage, sectors with greater productivity dispersion should also exhibit greater dispersion in returns to scale

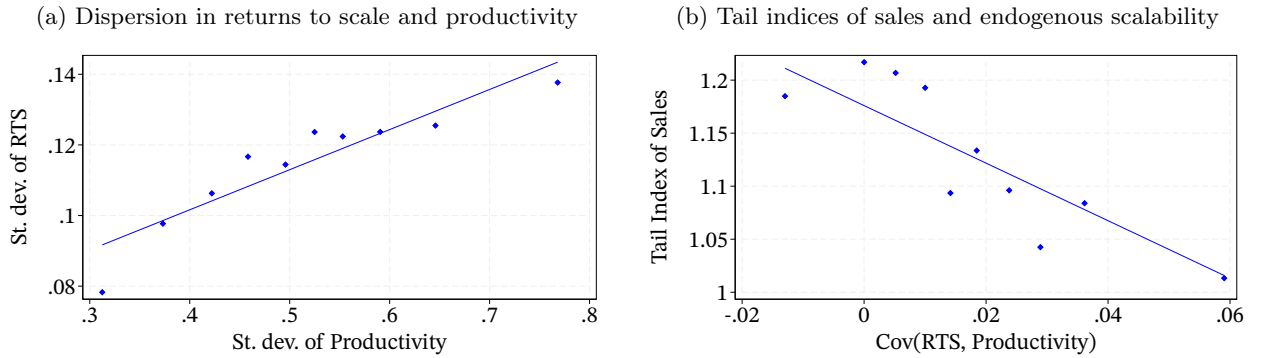
²⁹In Appendix A.7, we show that the results are robust to alternative measures of network energy exposure and to including the energy-producing sectors.

³⁰Here and in the exercises below, we rely on the cross-sectional returns-to-scale estimates of Section 6.2.

(Lemma 5). The first panel of Figure 9 confirms this prediction in the Spanish data. Sectors in the top decile of productivity dispersion have a standard deviation of returns to scale that is 0.06 higher than those in the bottom decile. This pattern supports the model’s “double blessing” mechanism, which gives rise to superstar firms through the adoption of high-scalability technologies.

This mechanism also implies that the shape of the firm-size distribution should vary systematically across sectors. Specifically, Proposition 1 shows that sectors where adopting high scalability is easier (high φ_i) should have a thicker right tail of the sales distribution. Since φ_i is not directly observable, we construct a proxy for the strength of the mechanism: the within-sector covariance between returns to scale and the productivity index, $\text{Cov}(\eta_{il}, \varepsilon_{il} + a_i(\eta_{il}))$. Theoretically, this covariance should be positive when returns to scale are endogenous ($\varphi_i > 0$), but zero if returns to scale are fixed.³¹ Using this measure, panel (b) of Figure 9 confirms the model prediction: sectors where firms can more easily adjust their returns to scale exhibit significantly thicker tails in their sales distribution.³²

Figure 9: Dispersion in returns to scale, productivity and market concentration



Notes: Panel (a) presents a binned scatter plot of the standard deviation of firm-level returns to scale against the standard deviation of firm-level productivity, for all sector-year observations. Panel (b) presents a binned scatter plot of the tail index of firm sales against the covariance between firm-level returns to scale and productivity, for all sector-year observations with at least 50 firms. Year fixed effects are controlled for in both panels and unconditional means are added back for interpretability. See Appendix A.1 for details on variable construction and sample selection.

6.6 Cross-country development and endogenous scalability

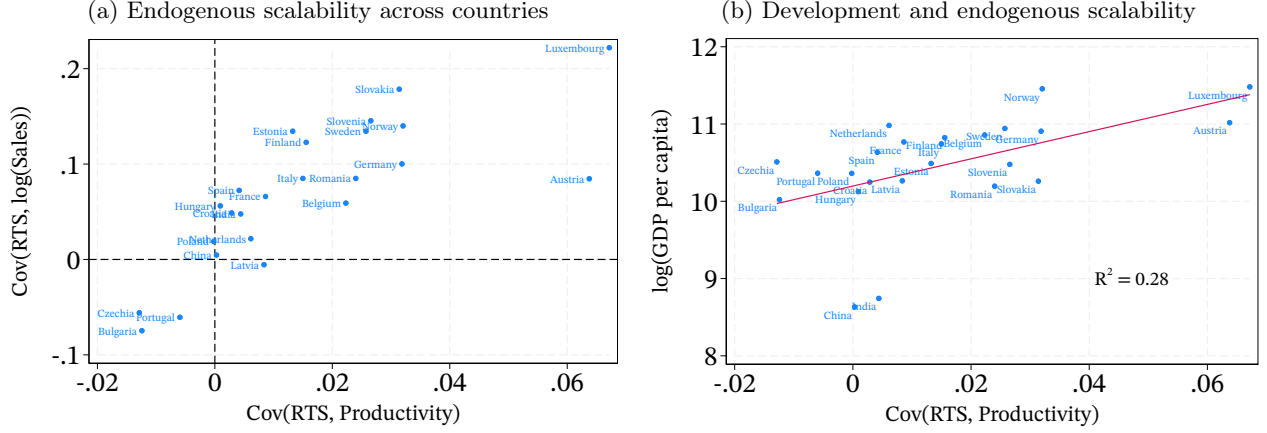
We conclude this section by providing cross-country evidence for the importance of endogenous returns to scale in our sample of 24 countries. To ensure comparability, our main cross-country analysis focuses on manufacturing firms only. For each country, we select the seven-year window with the largest number of firm-year observations between 2001 and 2019.³³ Within this window, we estimate size-decile-specific production functions using the same Blundell–Bond procedure described above to construct firm-level measures of returns to scale and productivity.

³¹One can show that in the model this quantity is increasing in φ_i .

³²We use the tail index estimator of Gabaix and Ibragimov (2011). See Appendix A.8 for details.

³³For several countries, the quality of the data is uneven over the sample period. We therefore restrict our analysis to the time window with the most observations to improve the quality of our estimates. We then apply identical data cleaning and estimation procedures to all country samples.

Figure 10: Cross-country evidence on productivity, sales, and returns to scale



Notes: Panel (a) plots, for each country, the covariance between firm-level returns to scale and log sales against the covariance between firm-level returns to scale and productivity; dashed lines denote zero covariances. Panel (b) plots the seven-year-average of log GDP per capita against the covariance between firm-level returns to scale and productivity; the solid line reports the fitted linear relationship and the figure reports the associated R^2 . Each marker corresponds to a country. See Appendix A.9 for details on variable construction and sample selection.

Figure 10 summarizes our results. Panel (a) reports the covariance between returns to scale and log sales, as well as between returns to scale and productivity. Mirroring the Spanish evidence in Section 6.2, we find that in most countries, larger and more productive firms systematically operate with higher returns to scale. This suggests that endogenous scalability is at work across a broad set of countries.

Panel (b) relates the strength of the endogenous returns-to-scale mechanism to economic development. As before, we measure the intensity of endogenous scalability using the firm-level covariance between productivity and returns to scale. Plotting this measure against log GDP per capita reveals a clear relationship: countries with stronger endogenous scalability are systematically richer.³⁴ This is consistent with the model prediction that endogenous scalability increases the level of GDP. Quantitatively, variation in this measure accounts for approximately one quarter of the cross-country variation in GDP per capita. This suggests that the capacity of productive firms to adopt more scalable technologies might be an important driver of long-run economic development. Consistent with this interpretation, Chen (2026) documents that richer economies exhibit thicker right tails of the firm size distribution, the pattern that Proposition 1 associates with stronger endogenous scalability.

7 Calibration to the Spanish economy

To evaluate the quantitative importance of endogenous returns to scale, we calibrate the model to the Spanish economy. We rely on the detailed firm-level data introduced in the previous section

³⁴India and China are two outliers but since those countries are at an earlier stage of development, their lower GDP per capita level might be explained by their distance to the technology frontier.

to discipline the model parameters. Our calibration strategy is summarized here, and more details are in Appendix B.

7.1 Calibration strategy

We calibrate the household preference vector β and the input-share matrix α directly to match the sectoral final consumption shares and intermediate expenditure shares observed in the 2010 Spanish Input-Output tables.³⁵ The firm-level returns to scale η_{il} are taken from our estimates of Section 6. We then compute each sector’s effective returns to scale $\hat{\eta}_i$ as the sales-weighted average of η_{il} . We find substantial heterogeneity in $\hat{\eta}_i$ across sectors, ranging from 0.54 to 0.98, with a mean and median of 0.83 and 0.82, respectively. Figure 19 in Appendix B.4 shows $\hat{\eta}_i$ for all sectors.

The productivity dispersion σ_i and the cost of scalability γ_i are jointly identified by targeting within-sector moments that are informative about scalability choices. As we show in Appendix B, the model implies a mapping from the pair (σ_i, γ_i) to the cross-sectional dispersion of firm-level returns to scale η_{il} and profits Π_{il} . Accordingly, we choose σ_i and γ_i for each sector to match the interquartile ranges of these two variables in the data.³⁶ Appendix B.4 reports the calibrated values (Figure 18) and shows that the model matches the targeted moments well (Figure 17). By construction, the calibrated model also matches the empirical effective returns to scale $\hat{\eta}$ perfectly.³⁷

To validate the calibration, we test the model’s ability to reproduce untargeted moments of the firm distribution. Recall that our mechanism implies that larger firms operate technologies with higher returns to scale. Panel (a) of Figure 11 compares the relationship between these quantities in the model and the data. Although this moment was not targeted, the model replicates the empirical pattern reasonably well. Two differences, however, are worth noting. First, the fit is less precise for very small firms. But since these producers account for a negligible share of aggregate output, they have little influence on the counterfactual exercises that follow. Second, the relationship between sales and returns to scale is steeper in the model than in the data. This is likely because firms in the model adjust their returns to scale instantly in response to productivity shocks, whereas real-world adjustments are subject to frictions and delays.

Panel (b) of Figure 11 shows that the model closely matches the empirical link between productivity (measured using the Törnqvist index) and sales. This is reassuring, as the link between these variables in the model is driven by the endogenous returns-to-scale parameter η_{il} .³⁸ Finally, panel (c) shows that the tail of the firm-size distribution is thicker in sectors where the endogenous scalability mechanism is stronger. Once again, the model matches the data reasonably well.

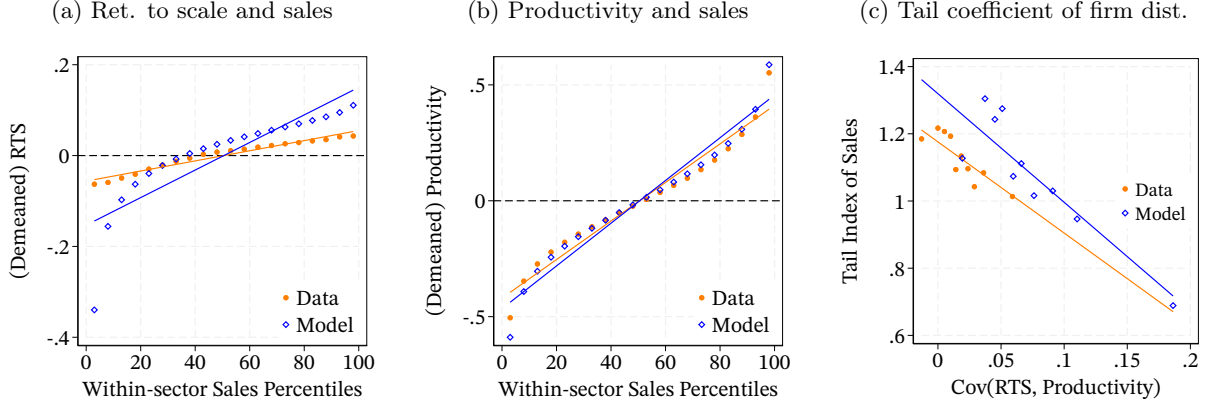
³⁵We calibrate the model to the 2010 Spanish economy because it is close to the midpoint of our sample period and provides detailed input-output tables that can be cleanly matched to the Orbis firm-level data at the NACE 2-digit level. Over time, changes in the national-accounts base complicate consistent harmonization of detailed input-output tables across years.

³⁶To be consistent with the model, we compute the profits of firm l in sector i as $(1 - \eta_{il})P_iQ_{il}$. We target interquartile ranges instead of variances to reduce the impact of outliers.

³⁷As we study *changes* to the environment, the results presented below are independent of the average productivity μ and the entry cost κ , and so there is no need to specify their values. Given all the other parameters, we can always find a combination of μ and κ to match $\hat{\eta}$ perfectly.

³⁸Productivity and returns to scale are also positively correlated in the calibrated model, as they are in the data.

Figure 11: RTS, sales, productivity, and tail coefficients in the model and in the data



Notes: *Data* correspond to Spanish firms in Orbis; *Model* corresponds to simulated firm-level outcomes from the calibrated model. Simulated firm observations are reweighted at the sector level so that the sector composition matches the data. Panels (a) and (b) report binscatter plots of returns to scale and productivity against sales percentiles. The y-axis variables are residualized by sector–year fixed effects in the *Data* series and sector fixed effects in the *Model* series. Panel (c) plots the tail index of sales against the within-sector-year covariance of returns to scale and productivity: the *Data* series uses sector–year statistics (computed only for sector–years with at least 50 firms) and includes year fixed effects, whereas the *Model* series uses sector-level statistics computed from simulated data and includes no fixed effects.

7.2 Contribution of endogenous returns to scale to GDP

Using our calibrated model, we first evaluate the importance of endogenous returns to scale for the level of GDP. To do so, we compare our calibrated baseline economy to the “fixed returns-to-scale” counterfactual where each firm’s returns to scale is exogenously fixed at its sector’s average, $\eta_{il} = \hat{\eta}_i$ (Definition 2). As shown in Proposition 5, the difference in log GDPs between those two economies is given by (26). Using our calibrated parameters, we find that this gain is 11.7%, so that allowing high-productivity firms to choose more scalable technologies has a substantial effect on GDP. Figure 20 in Appendix B.4 decomposes this aggregate gain, showing the contribution of each sector.

Next, we evaluate the importance of our mechanism for long-run macroeconomic growth. We conduct an experiment in which we increase the mean productivity μ_i of all sectors by one percentage point every year. We then compare the response of GDP in our full model to two counterfactual benchmarks. The purpose of this exercise is to decompose the full effect of endogenous scalability into two components: the gain from the initial *static dispersion* in returns to scale, and the additional gain from allowing these returns to scale to *dynamically adjust* over time. Our first benchmark is the fixed returns-to-scale economy where $\eta_{il}(t) = \hat{\eta}_i(0)$ for all firms, shutting down both channels. Our second benchmark is a *dispersed returns-to-scale* economy, where η_{il} is fixed at each firm’s initial level $\eta_{il}(t) = \eta_{il}(0)$, thus featuring the initial dispersion but not the dynamic adjustment.³⁹

Panel (a) of Figure 12 shows the evolution of GDP over time in each economy, relative to the fixed returns-to-scale benchmark. The initial gap at $t = 0$ reflects the level effect from (26). The solid blue line captures the full effect of endogenous returns to scale on GDP. We see that after one

³⁹We formally analyze this economy in Appendix D.6.

hundred years, GDP has grown an extra 5.5% in the full model. Panel (c) illustrates the underlying mechanisms. As μ increases, the price of intermediate inputs falls, which incentivizes firms to adopt more scalable technologies. This increase in returns to scale leads to higher Domar weights, which makes the increase in productivity more impactful.⁴⁰

Figure 12 allows us to decompose the 5.5% gain in GDP. It shows that the dispersed returns-to-scale economy grew by an extra 2.8% over a century compared to the fixed benchmark. This implies that the static reallocation channel and the dynamic adjustment channel each contribute roughly half of the total effect. The static channel is powerful because, even when individual η_{il} cannot change, the economy-wide productivity boom disproportionately benefits the firms that can scale more easily—that is, those that started with higher η_{il} (Lemma 4). As these firms expand their output disproportionately, their sales shares increase. This compositional shift has two beneficial effects. First, it endogenously raises effective returns to scale $\hat{\eta}$ and the Domar weights ω , which amplifies growth even though no firm changes its technology. Second, since these high- η firms are also the most productive, this reallocation of market share directly raises aggregate productivity.

Figure 12 also shows that the evolution of log GDP over time in the full model is convex, with growth constantly accelerating (panel b). This indicates that the effects of endogenous returns to scale become increasingly important over time, with the gap between the three curves in panel (a) increasing indefinitely. In the long-run, as $t \rightarrow \infty$, the baseline economy converges to a growth rate of 3.1% under our parametrization. In contrast, in the fixed returns-to-scale economy, where the endogenous returns-to-scale mechanism is shut down, long-run growth is only 2.3%. The large gap between these two numbers suggests that endogenous scalability might play a significant role in shaping long-run economic outcomes.⁴¹

7.3 Distorted economy

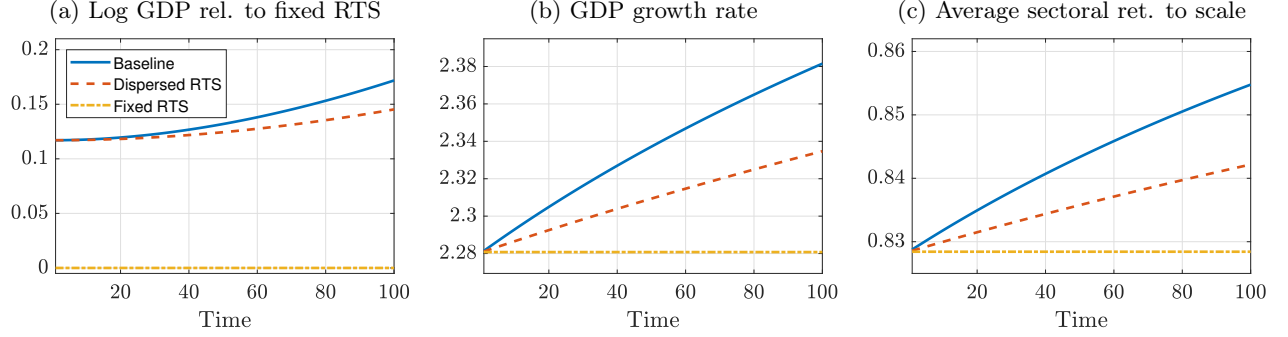
One implication of our model is that the ability of high-productivity firms to adopt more scalable technologies is important for the level and growth rate of GDP. Yet, several studies have documented that larger firms often suffer from higher wedges (e.g., Restuccia and Rogerson, 2008). With endogenous returns to scale, those wedges would disproportionately distort the scalability decisions of those superstar firms, with potentially large consequences for welfare. In this section, we provide a simple exercise to quantify this new adverse effect of wedges.⁴²

⁴⁰Over one hundred years, effective returns to scale in the calibrated model increase by about 0.025, which is within the range found by empirical studies. For instance, De Loecker et al. (2020) find that firm-level returns to scale increased between 0.01 and 0.1 over periods of 40 to 60 years, depending on the estimation method. Furthermore, the calibrated model implies that a sector’s intermediate-input share rises gradually over time (about 1.7 percentage points per century). Appendix B.4.3 shows this is consistent with U.S. data (1947–2023) once the offsetting compositional shift toward services is separated out (see Figure 21). Spain’s shorter industry data, 1995–2019, point the same way.

⁴¹As $t \rightarrow \infty$, the *growth rates* (but not the levels) of GDP in the baseline model and the dispersed returns-to-scale economy converge. In that limit, output in both economies is dominated by a vanishingly small fraction of extremely productive firms operating with near-constant returns to scale. In that case, the gains from the *dynamic channel* of endogenous returns to scale are exhausted, and both economies grow at the same pace of 3.1%.

⁴²Hubmer et al. (2026) also look at the impact of wedges in an economy with heterogeneous but exogenous returns to scale.

Figure 12: Endogenous returns to scale and productivity: Implications for GDP



Notes: Panel (a) shows $y(t) - \bar{y}(t)$ (“Baseline”) and $y^d(t) - \bar{y}(t)$ (“Dispersed RTS”) as the productivity μ of all sectors grows at 1% per year, where y , $\bar{y}$, and y^d are log GDPs in the economies with fully flexible returns to scale, fixed returns to scale, and dispersed returns to scale that do not respond to changes in μ , respectively. Panel (b) shows the growth rates of log GDP and panel (c) shows average returns to scale $\sum_{i=1}^N \hat{\eta}_i(t)/N$.

Following Hsieh and Klenow (2009), we compute the sales wedge $\hat{\tau}_{il}^S$ for firm l in sector i as the ratio of the marginal revenue product of labor to the wage:

$$\frac{1}{1 - \hat{\tau}_{il}^S} = \frac{\eta_{il} \left(1 - \sum_{j=1}^N \alpha_{ij}\right) P_i Y_{il}}{W L_{il}}.$$

Consistent with the literature, we find these wedges to be large. The average sales-weighted wedge in a sector is 0.40. Furthermore, wedges are positively correlated with firm size: the average within-sector correlation between $\hat{\tau}_{il}^S$ and firm sales is 0.27.

To study the impact of these wedges in our calibrated economy, we assume that each firm is subject to a sales wedge τ_{il}^S given by

$$\log(1 - \tau_{il}^S) = \log(1 - \tau_i^S) - b_i(\varepsilon_{il} - \mu_i), \quad (38)$$

where $\log(1 - \tau_i^S)$ is an average sector-wide term and $-b_i(\varepsilon_{il} - \mu_i)$ captures size-dependent distortions, with $b_i > 0$ implying that more productive firms face stronger distortions. As in the case where the wedge corresponds to a tax or a markup, we assume that net revenues collected from τ_{il}^S are rebated to the household lump-sum. We calibrate the intercept τ_i^S and the slope b_i to match the average sectoral wedge and the covariance between firm profits and the estimated wedges in the data. We estimate $b_i > 0$ in nearly all sectors.⁴³ After fixing τ_i^S and b_i , we recalibrate the remaining model parameters so that the economy continues to match our empirical targets. Further details on this procedure are provided in Appendix B.3.

To evaluate the importance of these distortions, we conduct an experiment in which we remove all wedges.⁴⁴ The first two columns of Table 1 show that, in the baseline model, this leads to a large

⁴³Our estimates of size-dependent distortions (b_i) are consistent with the cross-country evidence in Ayerst et al. (2024) and with recent estimates for India in Hsieh et al. (2025).

⁴⁴For some sectors, removing wedges would push φ_i above one. For these sectors, we set $\varphi_i = 0.99$. We conduct sensitivity analysis to the value of this threshold in Appendix B.5.

increase in both returns to scale and GDP. To understand the mechanisms behind this result, the table also reports the outcome of the same experiment in the dispersed and the fixed returns-to-scale economies. In the dispersed returns-to-scale economy, where firms choose their returns to scale in the presence of wedges but cannot adjust them once wedges are removed, the impact of eliminating wedges is substantially smaller. In this case, firms at the top of the distribution cannot increase their scalability to fully benefit from the removal of distortions, which limits welfare gains.⁴⁵ In the fixed returns-to-scale economy, where the endogenous scalability mechanism is shut down entirely, the effect of wedges is even more muted. Because all firms operate with the same returns to scale, the firm-size distribution is more compressed and GDP is produced more evenly across firms. Distorting high-productivity producers is thus less damaging in that economy.

Table 1: Returns to scale and GDP when wedges are removed

	Size-dependent wedges		Flat wedges	
	Δ RTS	Δ log GDP	Δ RTS	Δ log GDP
Baseline economy	0.067	167%	0.020	62%
Dispersed ret. to scale	0.046	138%	0.010	60%
Fixed ret. to scale	0	70%	0	58%

Notes: Increases in average effective returns to scale, $\Delta \left[\sum_{i=1}^N \hat{\eta}_i / N \right]$, and in log GDP, Δy , due to removal of sales wedges in the baseline economy, and in the economies with dispersed and fixed returns to scale. The “Size-dependent wedges” columns report the results when sales taxes are correlated with firm productivity. The “Flat wedges” columns report the results when sales wedges are identical for all firms within a sector.

To further show that wedges that disproportionately affect the top firms are particularly harmful when those firms endogenously choose the scalability of their operation, the last two columns of Table 1 repeat this analysis for an economy with flat wedges ($b_i = 0$). In this setting, productive firms face no additional penalty relative to smaller firms, so removing distortions yields much smaller GDP gains. Moreover, the results are nearly identical across the three model specifications. This confirms that the interaction between endogenous scalability and size-dependent distortions is the primary driver of our results.

In summary, our quantitative analysis suggests that endogenous returns to scale might play a substantial role in shaping both the level and the growth rate of GDP. When highly productive firms are able to adopt more scalable technologies as the technological frontier advances, they can expand disproportionately, with substantial gains for welfare. Taxes or distortions that fall on those firms, however, can disrupt this process. As our results show, size-dependent distortions discourage the adoption of high-scale technologies, stifle the growth of superstar firms, and generate efficiency losses that can exceed those in standard models. Policy interventions that disproportionately burden high-productivity firms may therefore be particularly harmful.

⁴⁵Effective sectoral returns to scale still increase in this case since high- η_{il} firms become larger.

8 Conclusion

We develop a theory in which returns to scale are endogenous equilibrium objects driven by incentives. At the micro level, this mechanism gives rise to superstar firms and fat-tailed firm-size distributions. At the macro level, it endows the economy with greater resilience by dampening the impact of adverse shocks while amplifying favorable ones. It also provides an engine for long-run growth, as the economy’s organizational structure co-evolves with its technological frontier. Input-output connections between sectors play a crucial role in these mechanisms.

Several extensions would be worth pursuing. First, extending the model to a dynamic setting could help connect it more closely to the data. We have worked out a simple dynamic version of the model in Appendix E, but other mechanisms may also be at work. For example, introducing capital would change the incentives to increase returns to scale. Since capital can be accumulated, its presence might affect the growth properties of the model. Second, the superstar firms that emerge in our model might, in reality, gain market power, creating a feedback effect that would further increase their incentives to scale. Third, modeling individual margins through which firms achieve scale would allow for a tighter connection to the data and a deeper understanding of the importance of scalability for macroeconomic outcomes.

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Online Appendix for “Endogenous Returns to Scale”

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Not for publication

This document is an online supplement for the paper “Endogenous Returns to Scale”. Part [A](#) provides the details of the data exercises of Section [6](#) along with several robustness checks. Part [B](#) provides additional information about the calibration exercise of Section [7](#). Part [C](#) provides the proofs of the formal results. Part [D](#) contains several extensions of the model along with robustness exercises. Part [E](#) presents a dynamic extension of the model and uses it to validate our returns-to-scale estimation procedure on simulated data.

A Appendix for Section [6](#)

This appendix contains details about the reduced-form results of Section [6](#).

A.1 Details of the Spanish Orbis data

Our Spanish firm-level data are drawn from the Orbis Historical Disk. Orbis is commonly regarded as the most comprehensive cross-country firm database, covering both public and private firms’ financial statements and measures of real activity (Kalemli-Özcan et al., [2024](#)). We focus on Spain because firm coverage is close to universal—capturing over 95% of total industry gross output after 2010—making it well-suited for economy-wide analysis. Our sample spans 1995–2019.^{[46](#)}

Sample cleaning Our sample construction closely mirrors the cleaning steps used in our earlier work (Kopytov et al., [2024](#)). We begin by merging each firm’s descriptive information with its financial statements using the unique BVD firm identifier (BVDID). We then restrict our analysis to Spanish firms, defined as firms that satisfy two criteria: 1) their latest address is in Spain and 2) their BVDID starts with the ISO-2 code **ES**. In the resulting Orbis Spain sample, we implement the following standard cleaning procedure:

⁴⁶Orbis offers good coverage of the Spanish economy starting from 1995. Moreover, the most recent observations in the version of Orbis Historical Disk Product that we use are from 2021. We therefore use 2019 as the last year of the sample since there is usually a two-year reporting lag for some variables (see Kalemli-Özcan et al. ([2024](#)) for details).

1. We harmonize the calendar year of each firm-year observation using the variable `closing_date`: if the closing date is on or after July 1, the current year is assigned as the calendar year. Otherwise, the previous year is assigned.⁴⁷
2. In a given year, a firm in the Orbis database might have multiple financial statements from different sources (local registry, annual report, or others), for consolidated or unconsolidated accounts. When several source-consolidation combinations exist for a firm, we deduplicate by selecting the account that, in order of priority, 1) shows the most consistent reporting frequency (closest to regular annual reporting), 2) offers the longest non-missing time series for key financial variables (fixed assets and/or sales), and 3) is consolidated, if the first two criteria are tied.
3. We only keep firm-year observations with non-missing and positive sales (`operating_revenue_turnover`), fixed assets (`fixed_assets`), wage bills (`costs_of_employees`), and material costs (`material_costs`). We also harmonize the units of all monetary values to be in current euros.
4. To prevent outliers from affecting the production function estimation, we exclude any firm-year observation whose average revenue product of any input (fixed assets, wage bills, or material costs) lies above the 99th percentile or below the 1st percentile of that year’s distribution.⁴⁸

A.2 Details of the production function and RTS estimation

This appendix describes in detail how we implement the production-function estimation procedure that delivers the results used in the main text.⁴⁹

We use the Blundell and Bond (2000) IV-GMM estimator to estimate the production functions as our benchmark. This estimator is designed for dynamic panel settings with persistent firm-level variables and, under standard moment conditions, delivers consistent estimates of output elasticities. Our model imposes a competitive output market in a sector. In this setting, the identifying assumptions are most plausible when there is sufficient persistent variation in predetermined inputs and in the cost of flexible inputs, so that observed input choices are not collinear with unobserved productivity. Recent work by De Ridder et al. (2022) further shows, through Monte Carlo simulations, that this approach performs well when such identifying variation is strong.

Our empirical strategy builds on the model’s implication that, within a sector, firms that are similar in size should operate under similar production technologies and therefore exhibit similar returns to scale. We

⁴⁷This adjustment matters little for the Spanish sample, as 99% of firms close their books on December 31.

⁴⁸One might worry that trimming on average revenue products of inputs could mechanically remove observations corresponding to extreme returns to scale, since average revenue products of variable inputs are inversely related to the chosen η . In practice, however, the extreme tails in the data are unlikely to reflect meaningful limits of η ; they typically coincide with implausible or mismeasured inputs and would require implausibly extreme η to rationalize.

⁴⁹Many recent papers have used production-function estimation to recover heterogeneity in returns to scale at the firm or industry level, including De Loecker et al. (2020), Ruzic and Ho (2023), Chiavari (2024), McAdam et al. (2024), Savagar and Kariel (2024), Demirel (2025), Hubmer et al. (2026) and Gao and Kehrig (2025).

use this prediction to estimate returns to scale across the firm-size distribution. For each sector i and year t , we rank firms by a smoothed measure of size: the 7-year moving average of firm-level log sales computed over the window from $t - 3$ to $t + 3$.⁵⁰ We then assign firms to 10 deciles, $d_t = \{1, \dots, 10\}$, based on this sector-year ranking.⁵¹ Using a moving average reduces the influence of short-run fluctuations and measurement error in annual sales, and thus provides a more stable proxy for the scale of a firm's production. For each sector-decile-year cell (i, d_t, t) (using the 7-year rolling sample around t), we assume firms share a common Cobb-Douglas technology:

$$\begin{aligned} q_{ilt} &= \beta_{i,d_t(l),t}^L l_{ilt} + \beta_{i,d_t(l),t}^K k_{ilt} + \beta_{i,d_t(l),t}^M m_{ilt} + \gamma_t^{i,d_t(l)} + \kappa_{il}^{i,d_t(l)} + \varsigma_{ilt}^{i,d_t(l)}, \\ \varsigma_{ilt}^{i,d_t(l)} &= \rho^{i,d_t(l)} \varsigma_{il,t-1}^{i,d_t(l)} + e_{ilt}^{i,d_t(l)}, \quad |\rho^{i,d_t(l)}| < 1, \\ e_{ilt}^{i,d_t(l)} &\sim MA(0), \end{aligned}$$

where q_{ilt} , l_{ilt} , k_{ilt} , m_{ilt} are the logs of output, labor, capital, material inputs for firm l . These are measured as deflated values of, respectively, sales, wage bills, fixed assets and material costs using the GDP deflator in the Annual Spanish National Accounts. We assume a firm's productivity contains three components: a year-specific component $\gamma_t^{i,d_t(l)}$, a firm-specific effect $\kappa_{il}^{i,d_t(l)}$ and an autoregressive component $\varsigma_{ilt}^{i,d_t(l)}$ with i.i.d. innovation $e_{ilt}^{i,d_t(l)}$. The model admits the following dynamic representation:

$$\begin{aligned} q_{ilt} &= \rho^{i,d_t(l)} q_{il,t-1} + \beta_{i,d_t(l),t}^L l_{ilt} - \rho^{i,d_t(l)} \beta_{i,d_t(l),t}^L l_{il,t-1} + \beta_{i,d_t(l),t}^K k_{ilt} - \rho^{i,d_t(l)} \beta_{i,d_t(l),t}^K k_{il,t-1} \\ &\quad + \beta_{i,d_t(l),t}^M m_{ilt} - \rho^{i,d_t(l)} \beta_{i,d_t(l),t}^M m_{il,t-1} + \gamma_t^{*i,d_t(l)} + \kappa_{il}^{*i,d_t(l)} + e_{ilt}^{i,d_t(l)}, \end{aligned} \quad (39)$$

where $\gamma_t^{*i,d_t(l)} := \gamma_t^{i,d_t(l)} - \rho^{i,d_t(l)} \gamma_{t-1}^{i,d_t(l)}$ and $\kappa_{il}^{*i,d_t(l)} := (1 - \rho^{i,d_t(l)}) \kappa_{il}^{i,d_t(l)}$. Therefore, we can estimate the following dynamic specification with current and lagged variables:

$$\begin{aligned} q_{ilt} &= \alpha^{i,d_t(l)} q_{il,t-1} + \beta_{i,d_t(l),t}^{L0} l_{ilt} + \beta_{i,d_t(l),t}^{L1} l_{il,t-1} + \beta_{i,d_t(l),t}^{K0} k_{ilt} + \beta_{i,d_t(l),t}^{K1} k_{il,t-1} \\ &\quad + \beta_{i,d_t(l),t}^{M0} m_{ilt} + \beta_{i,d_t(l),t}^{M1} m_{il,t-1} + \gamma_t^{*i,d_t(l)} + \kappa_{il}^{*i,d_t(l)} + e_{ilt}^{i,d_t(l)}, \end{aligned} \quad (40)$$

where, under our assumption, the AR(1) productivity structure implies restrictions across coefficients (e.g., $\alpha^{i,d_t(l)} = \rho^{i,d_t(l)}$ and $\beta_{i,d_t(l),t}^{x1} = -\rho^{i,d_t(l)} \beta_{i,d_t(l),t}^{x0}$ for $x \in \{L, K, M\}$).

⁵⁰We use the centered window whenever feasible. Near the sample boundaries and when firm-year observations are missing, we use the longest available window to preserve coverage.

⁵¹For the production function estimation we pool seven pairs of closely related sectors in which one sector has few firms: (i) coke and refined petroleum products (input-output sector 10) with chemicals and chemical products (11); (ii) air transport (33) with warehousing and transportation support services (34); (iii) financial services (41) with insurance and pension funding (42); (iv) architectural and engineering services (46) with scientific research and development (47); (v) employment services (51) with security, investigation, and building services (53); (vi) public administration and defence (54) with education (55); and (vii) membership organisations (60) with other personal services (62). Every other estimation sector coincides with a sector of the 2010 input-output table.

Blundell-Bond system-GMM moments We now describe the system-GMM moment conditions we exploited to estimate the model in (40). Our choice of moment conditions follows the exact implementation in Table III, column 5 of Blundell and Bond (2000), where we treat $\{q, l, k, m\}$ as potentially endogenous and use two sets of moments:

(i) Difference equation (levels dated $t - 2$ and earlier):

$$\mathbb{E}\left[x_{il,t-s} \Delta e_{ilt}^{i,d_t(l)}\right] = 0 \quad \text{for } x \in \{q, l, k, m\} \text{ and } s \geq 2. \quad (41)$$

(ii) Levels equation (first differences dated $t - 1$ only):

$$\mathbb{E}\left[\Delta x_{il,t-1} \left(\kappa_{il}^{*i,d_t(l)} + e_{ilt}^{i,d_t(l)}\right)\right] = 0 \quad \text{for } x \in \{q, l, k, m\}. \quad (42)$$

Moreover, year dummies are included as controls and treated as exogenous instruments in the levels equation. We implement this estimation using the `xtabond2` command in Stata.

Obtaining the Firm-level Returns-to-Scale Estimates After estimating (40), we then use the minimum distance estimator by Söderbom (2009) to impose the AR(1)-implied restrictions and get the restricted parameter estimates $\left(\hat{\rho}^{i,d_t(l)}, \hat{\beta}_{i,d_t(l),t}^K, \hat{\beta}_{i,d_t(l),t}^L, \hat{\beta}_{i,d_t(l),t}^M\right)$. The estimated returns to scale η_{ilt} for a firm l in sector i and year t is therefore given by the sum of these elasticities:

$$\eta_{ilt} = \hat{\beta}_{i,d_t(l),t}^K + \hat{\beta}_{i,d_t(l),t}^L + \hat{\beta}_{i,d_t(l),t}^M.$$

Because 1995 and 1996 contain relatively few firm-year observations for production-function estimation, we report results using only the 1997–2019 estimates matched to the firm-level data for our analysis.

A.3 Constructing the Törnqvist productivity index

To compare productivity across firms, we rely on the Törnqvist productivity index. We provide here theoretical results about that index to link our model with our estimation procedure.

Lemma 3 shows that returns to scale are increasing in ε_{il} , but we do not observe ε_{il} directly in the data. In addition, comparing measured productivity $e^{\varepsilon_{il}} A_i(\eta_{il}) \zeta(\eta_{il})$ across firms with different technologies faces well-known issues about the choice of units. When going to the data, we rely instead on a Törnqvist productivity index, which is commonly used to compare productivities across firms or countries with different production functions (Caves et al., 1982a; Caves et al., 1982b) and recently in Penn World Table by Feenstra et al. (2015). Specifically, we use the multilateral Törnqvist productivity index by Caves et al. (1982a) that has been extensively used in the firm dynamics context (Aw et al., 2001).

Definition 3 (Multilateral Törnqvist productivity index). Consider a sector i in year t . Let N_{it} be the number of firms observed in (i, t) . Define the sector-year reference firm's moments as $\overline{\log Q_{it}} = \frac{1}{N_{it}} \sum_l \log Q_{ilt}$,

$\overline{\log O_{it}} = \frac{1}{N_{it}} \sum_l \log O_{ilt}$, $\overline{\beta_{O,it}} = \frac{1}{N_{it}} \sum_l \beta_{O,ilt}$ where $O \in \{K, L, M\}$ and $\beta_{O,ilt}$ are firm-level output elasticity of input O . The multilateral Törnqvist productivity index of firm l is defined as

$$z_{ilt} := (\log Q_{ilt} - \overline{\log Q_{it}}) - \sum_{O \in \{K, L, M\}} \frac{1}{2} (\beta_{O,ilt} + \overline{\beta_{O,it}}) (\log O_{ilt} - \overline{\log O_{it}}).$$

For any two firms k and l in sector i and year t , we say firm k is more productive than firm l if $z_{ikt} > z_{ilt}$.

Intuitively, the measure z_{ilt} compares productivity between firm l and the reference firm by looking at how much more output one produces relative to the other, adjusting for differences in technology and input use. It is “multilateral” because z_{ilt} is defined relative to a common sector-year reference firm constructed from *all firms* in (i, t) , so productivity comparisons $z_{ikt} - z_{ilt}$ are base-firm invariant and can be consistently ranked across all firm pairs. In our benchmark case, we set all $\beta_{O,ilt} = \hat{\beta}_{i,d_t(l),t}^O$ to obtain the estimated productivity index $\hat{z}_{ilt}$ and use it as our measured productivity in all cross-sectional exercises that involve comparisons between firms within a sector-year. We find the Törnqvist index to be a good proxy for productivity in model-simulated data. In our calibrated economy of Section 7, the within-sector correlation between the Törnqvist index and ε_{il} is above 0.99. The same number for $\varepsilon_{il} + a_i(\eta_{il}) + \log \zeta_{il}$ is about 0.92.

A.4 Robustness of the production function and RTS estimation

This appendix shows that the documented positive RTS-size and RTS-productivity relationships, across firms in the cross-section (Section 6.3), are not driven by a particular estimator, choice of IV-GMM instruments, or grouping design. We first vary the Blundell and Bond (2000) system-GMM specification by changing the treatment of year dummies and the internal instrument set, following the implementation in De Ridder et al. (2022) (Appendix A.4.1). We then re-estimate production functions using standard control-function approaches—Olley and Pakes (1996) and Levinsohn and Petrin (2003)—to verify that our results are not specific to IV-GMM (Appendix A.4.2). Moreover, we account for potential market power by adding markup controls (proxied by sales shares) within an Akerberg et al. (2015) estimator (Appendix A.4.2). Finally, we show that our conclusions are robust to alternative ways of forming size groups (Appendix A.4.3).

We report coefficients from simple regressions to summarize the robustness of our empirical findings across firms with these alternative estimates of returns to scale and productivity. To show robustness for Figure 7, which documents the cross-sectional pattern that larger and more productive firms have higher returns to scale within a sector-year, we estimate two simple regressions of returns to scale on log sales and productivity:

$$\eta_{ilt} = \beta_0 \log(\text{Sales}_{ilt}) + \delta_{it} + \epsilon_{ilt}, \quad \eta_{ilt} = \beta_1 \hat{z}_{ilt} + \delta_{it} + \epsilon_{ilt}, \quad (43)$$

where δ_{it} denotes sector-year fixed effects. The estimated coefficients of β_0 and β_1 are displayed in Table 2 across all specifications.

Results using our benchmark estimator are reported in column (1) of Table 2. We now describe the

alternative estimators and grouping designs used in the robustness checks.

Table 2: Across-firm variation in returns to scale, productivity, and firm size with alternative production function estimators

	Dependent variable: Firm-level RTS						
	(1) BB baseline	(2) BB alternative	(3) OP	(4) LP	(5) ACF market power	(6) Av.-size percentiles	(7) Cur.-size deciles
log (Sales _{it})	0.023*** (0.001)	0.028*** (0.001)	0.044*** (0.001)	0.037*** (0.001)	0.050*** (0.002)	0.019*** (0.001)	0.008*** (0.001)
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952
R ²	0.688	0.688	0.644	0.806	0.572	0.728	0.655
$\hat{z}_{it}$	0.050*** (0.002)	0.058*** (0.003)	0.083*** (0.002)	0.081*** (0.003)	0.091*** (0.005)	0.034*** (0.002)	0.021*** (0.002)
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952	9,424,952
R ²	0.655	0.648	0.569	0.760	0.515	0.684	0.652

Notes: This table reports coefficients from cross-sectional regressions of firm-level returns to scale (RTS) on (i) log sales and (ii) firm productivity, $\hat{z}_{it}$, each including sector-year fixed effects. Column (1) uses our benchmark Blundell–Bond (BB) estimates; column (2) uses an alternative BB specification as implemented in De Ridder et al. (2022). Columns (3) and (4) use the Olley–Pakes (OP) and Levinsohn–Petrin (LP) control-function estimators, respectively. Column (5) reports results from the Akerberg–Caves–Frazer (ACF) estimator with market power controls (proxied by firms’ sales shares). Columns (6) and (7) use alternative grouping methods for estimating elasticities: rolling average-size percentiles and contemporaneous size deciles. The regressions use a sample of Spanish firms from Orbis. See Appendix A.1 for details on variable construction and sample selection. Standard errors (in parentheses) are two-way clustered at the firm and sector-year level. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.4.1 With alternative Blundell-Bond specifications

Our baseline specification follows Blundell and Bond (2000) and includes year dummies. Including year effects is recommended in dynamic-panel GMM applications because it absorbs economy-wide shocks and thereby reduces cross-firm correlation in the regression residuals. At the same time, once common year shocks are removed, identification of flexible-input elasticities relies on variation that is not common across firms in a group. In practice, this shifts weight toward persistent within-year differences in flexible input costs or wedges across firms. If such variation is interpreted literally as firm-specific input prices, it can raise concerns about measurement—because input quantities constructed from expenditures may mechanically inherit noise from unobserved firm-level prices.⁵²

As a robustness check, we therefore also implement the Blundell–Bond estimator specification used by De Ridder et al. (2022), which omits year dummies and uses a more conservative internal-instrument set.

⁵²However, if the relevant heterogeneity operates through non-monetary wedges—e.g., distortions that affect effective input costs without changing the recorded unit prices paid by the firm, in the spirit of Hsieh and Klenow (2009)—then this concern is mitigated because observed input quantities are not mechanically distorted by unobserved prices.

Concretely, our baseline estimates a dynamic sales equation with current and one-lag terms for labor, capital, and materials, includes year fixed effects, and instruments the endogenous variables with lags starting at $t - 2$ (and deeper) in the transformed equation, while treating the year dummies as standard instruments in the levels equation. In contrast, the De Ridder et al. (2022) specification removes year dummies and restricts the GMM-style instruments to a single deeper lag (the third lag) for output and inputs. It also uses an eleven-year rolling window $[t - 5, t + 5]$ rather than the seven-year window used in our baseline, and forms the size deciles from the eleven-year average of log sales. Relative to our baseline, this alternative places less weight on within-year cross-sectional price/wedge variation as the driver of instrument relevance and also reduces instrument proliferation by construction. The results using the alternative Blundell-Bond estimates are reported in column (2) of Table 2.

A.4.2 With different production function estimators

We also use other commonly used production function estimators as a robustness check. In particular, we consider the control-function approach and implement the Olley and Pakes (1996) and Levinsohn and Petrin (2003) estimators.

The Olley–Pakes estimator We first implement the Olley and Pakes (1996) (OP) estimator to estimate the production function for each sector–decile–year cell. The Olley and Pakes (1996) estimator is a semiparametric control-function method that addresses simultaneity bias, since unobserved productivity affects firms’ input choices. It assumes that investment is a function of capital and productivity and, under a monotonicity condition, can be inverted to express unobserved productivity in terms of observed investment and capital. Substituting this inverted control function into the production function, the method first estimates the elasticities of freely adjustable inputs (labor and materials in our case) while controlling for productivity, and then uses a Markov assumption on productivity to recover the coefficient on the quasi-fixed input, capital. To implement this approach, we measure real investment as the change in the capital stock net of depreciation, and we recognize that this can generate zero or negative investment values, which reduces the usable sample for production function estimation.⁵³ The results using the OP estimator are reported in column (3) of Table 2.

The Levinsohn–Petrin estimator Because investment can be lumpy in practice and the Olley and Pakes (1996) procedure may force us to drop observations with zero or negative investment, we also apply the Levinsohn and Petrin (2003) (LP) estimator as an additional robustness check. Instead of using investment, this method uses intermediate inputs (materials in our case) as the control variable. It assumes that materials are flexibly chosen after observing productivity, while capital is still treated as quasi-fixed. Under the assumption that materials demand is a function of capital and productivity and is monotone in productivity

⁵³This reduces the production function estimation sample within each sector–decile–year cell, but we apply the resulting coefficients to the common baseline panel, so the main regression sample is unchanged.

(conditional on capital), the materials demand function can be inverted to recover unobserved productivity. This control function allows consistent estimation of the labor elasticity, and additional moment conditions then recover the elasticities of capital and materials. The results using the LP estimator are reported in column (4) of Table 2.

ACF estimator with market power controls Our model abstracts from market power and markups, but these forces could hinder the identification of output elasticities and obscure the positive returns-to-scale-size relationship we identify in the cross section. When firms have market power, elasticities estimated from deflated sales are revenue elasticities rather than physical output elasticities. We discuss the distinction between revenue and quantity returns to scale, and why it strengthens rather than weakens our conclusions, in Appendix A.5 below. Here we verify robustness to explicit market-power controls.

Following Baqaee and Farhi (2019) and De Loecker et al. (2020), we control for markups using firms’ sales shares (measured at the NACE 3-digit and 4-digit levels) and estimate production functions using the Akerberg et al. (2015) (ACF) estimator, treating price variation as an additional measurement component to be partialled out. The results using the ACF estimator are reported in column (5) of Table 2.

A.4.3 With different size-based grouping methods

Grouping firms by 7-year average sales percentiles Our benchmark approach groups firms in sector i and year t into deciles based on their 7-year average log sales. While straightforward, this discretization can generate non-smooth variation across firm sizes. As a robustness check, we therefore implement a rolling-percentile approach based on firms’ 7-year average sales. For each sector-year, we rank firms into 100 percentiles using their 7-year average (log) sales. In each sector i , for each percentile p_t , we construct a local sample consisting of firms whose percentile rank lies between $p_t - 15$ and $p_t + 15$ in year t . We then estimate output elasticities for each cell (i, t, p_t) using the Blundell–Bond estimator on the corresponding 7-year rolling-window sample. The results using the rolling-percentile grouping approach are reported in column (6) of Table 2.

Grouping firms by contemporaneous sales deciles Alternatively, we group firms into deciles based on contemporaneous firm-level (log) sales in year t , rather than the 7-year average. We then estimate output elasticities for each cell (i, t, d_t) using the same Blundell–Bond estimator on a 7-year rolling-window sample. The results using the contemporaneous sales-decile grouping approach are reported in column (7) of Table 2.

A.4.4 Summary

Overall, the main empirical patterns are robust. Across all alternative production-function estimators (alternative Blundell–Bond specifications, Olley–Pakes, Levinsohn–Petrin, and ACF with market-power controls) and alternative grouping methods (rolling percentiles and contemporaneous deciles), we continue to

find a positive relationship between firm-level returns to scale and firm size, as well as between returns to scale and productivity, in the cross section (within sector-years). While magnitudes vary across specifications, the sign and statistical significance of these relationships are stable (see Table 2).

A.5 Revenue versus quantity returns to scale

In our model, firms within a sector face the same price, and they take it as given. Consequently, revenue and quantity returns to scale coincide. If firms have market power, they internalize that as they expand the quantity sold, the price falls, such that their revenue rises by less than their quantity. In this case, the estimated returns to scale is a revenue elasticity, η^R , which understates its quantity counterpart η^Q (Klette and Griliches, 1996; Bond et al., 2021). This subsection shows that our cross-sectional empirical finding of returns to scale rising with firm size survives the two leading forms of imperfect competition: (1) Under CES demand, markups are constant and revenue returns to scale differs from quantity returns to scale by a multiplicative constant, which is the same for all firms. (2) When the demand elasticity falls with size, so that larger firms charge higher markups, the correction itself grows with size. In both cases, the quantity returns-to-scale-size gradient is steeper than its revenue counterpart.

A.5.1 CES demand

Suppose that, instead of taking prices as given, firms are monopolistic competitors facing a constant elasticity of substitution ϑ , with markup $\mu = \vartheta/(\vartheta - 1)$. Specifically, let sector output aggregate firm quantities with a CES demand, $Y_{it} = \left(\sum_{l=1}^{N_i} Q_{ilt}^{\frac{\vartheta-1}{\vartheta}} \right)^{\frac{\vartheta}{\vartheta-1}}$, so that each firm faces the demand curve $P_{ilt} = P_{it} (Y_{it}/Q_{ilt})^{1/\vartheta}$. With a quantity production function $Q_{ilt} = \tilde{A}_{ilt} K_{ilt}^{\tilde{\beta}_{il}^K} L_{ilt}^{\tilde{\beta}_{il}^L} M_{ilt}^{\tilde{\beta}_{il}^M}$, the equilibrium revenue function is

$$P_{ilt}Q_{ilt} = P_{it}Y_{it}^{1/\vartheta} \tilde{A}_{ilt}^{1-1/\vartheta} K_{ilt}^{(1-1/\vartheta)\tilde{\beta}_{il}^K} L_{ilt}^{(1-1/\vartheta)\tilde{\beta}_{il}^L} M_{ilt}^{(1-1/\vartheta)\tilde{\beta}_{il}^M}.$$

The term $P_{it}Y_{it}^{1/\vartheta}$ is common to all firms in a sector-year and thus is absorbed by the year effects in any within-sector regression. Then input elasticities can be estimated using the same steps as in Appendix A.2, with each estimated coefficient being $\beta_{il}^X = (1-1/\vartheta)\tilde{\beta}_{il}^X$ for $X \in \{K, L, M\}$. The sum of the input coefficients is the revenue returns to scale,

$$\beta_{il}^L + \beta_{il}^M + \beta_{il}^K = \eta_{il}^R = \eta_{il}^Q \left(1 - \frac{1}{\vartheta} \right), \quad (44)$$

where $\eta_{il}^Q = \tilde{\beta}_{il}^K + \tilde{\beta}_{il}^L + \tilde{\beta}_{il}^M$ is the quantity returns to scale.

Two consequences follow. First, even though the financial statements can only identify η^R alone, with a calibrated elasticity of substitution ϑ , we can still recover the quantity returns to scale via $\eta^Q = \eta^R \frac{\vartheta}{\vartheta-1}$. Second, as the elasticity is constant within a sector, the correction is a common multiplication of $\frac{\vartheta}{\vartheta-1}$, which leaves the shape of the estimated profile exactly intact: the ordering of firms, the sector ranking under a common ϑ , and every ratio are unchanged, while levels and level differences, including the size gradient, scale by the common factor. The finding that returns to scale rise with firm size is thus invariant to ϑ : it

holds in quantities if and only if it holds in revenues.

Figure 13 plots the estimated quantity returns to scale for $\vartheta \in \{6, 8, 10\}$. The gradient scales by the same factor, from 0.124 at $\vartheta = 10$ to 0.134 at $\vartheta = 6$, against 0.112 in revenue returns to scale. Therefore, against different calibrated CES alternatives, every cross-firm comparison survives, and only the level of the implied technology depends on the assumed ϑ .

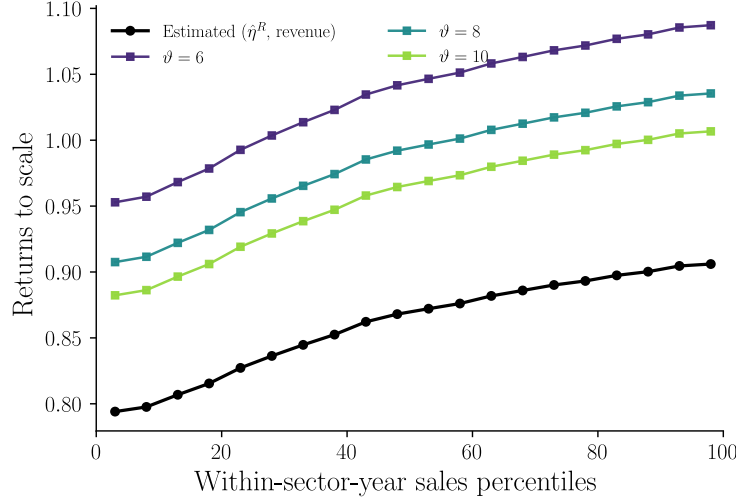


Figure 13: Implied quantity returns to scale under a constant elasticity of substitution

Notes: Black: binned means of the firm-level revenue estimates against within-sector-year sales percentiles (twenty bins), constructed exactly as in Figure 7(a). Colored: the same profile multiplied by $\mu = \vartheta/(\vartheta - 1)$, the exact correction implied by (44). The correction is multiplicative, so all curves share the same shape.

A.5.2 Non-constant demand elasticities and endogenous markups

Another empirically relevant alternative is that demand elasticity falls with firm size, so that larger firms charge higher markups. Under Kimball demand (Kimball, 1995), sector output Y_{it} is defined implicitly by $\sum_{l=1}^{N_i} \Upsilon(Q_{ilt}/Y_{it}) = 1$ with Υ increasing and concave, and each firm faces the inverse demand $P_{ilt} = D_{it}\Upsilon'(x_{ilt})$, where $x_{ilt} = Q_{ilt}/Y_{it}$ is the firm's relative size and D_{it} a sector-year demand index common to all firms. A commonly used parametrization of Kimball demand is the Klenow-Willis family (Klenow and Willis, 2016),

$$\Upsilon'(x) = \frac{\bar{\vartheta} - 1}{\bar{\vartheta}} \exp\left(\frac{1 - x^{\epsilon/\bar{\vartheta}}}{\epsilon}\right),$$

under which the demand elasticity falls with relative size, $\vartheta(x) = \bar{\vartheta}x^{-\epsilon/\bar{\vartheta}}$, and the markup $\mu(x) = \vartheta(x)/(\vartheta(x) - 1)$ rises with it. The local revenue elasticity is therefore $\eta^R(x) = \eta^Q/\mu(x)$: the correction from revenue to quantity units now grows with firm size.

Both the relative size and the markup can be imputed from a firm's own sales. Revenue relative to the sector-year average, $\tilde{r}_{ilt} := \log R_{ilt} - \overline{\log R_{it}}$, satisfies

$$\tilde{r}_{ilt} = \log x_{ilt} + \log \Upsilon'(x_{ilt}) + c_i, \quad \frac{d\tilde{r}_{ilt}}{d \log x_{ilt}} = \frac{1}{\mu(x_{ilt})} > 0,$$

so relative revenue is strictly increasing in relative size and the mapping can be inverted firm-year by firm-year. We calibrate the demand system to the main specification of Edmond et al. (2023) ($\bar{\vartheta} = 10.86$, superelasticity $\epsilon = 1.74$), and calibrate the sector constant c_i so that the sector's cost-weighted average markup equals the Edmond et al. (2023) aggregate markup of 1.15. We can then obtain the relative quantity $\hat{x}_{ilt}$ and markup $\hat{\mu}(\hat{x}_{ilt})$ for every firm-year. We then re-estimate quantity returns to scale cell by cell, via the estimating equation (40) of Appendix A.2, using the inverted log quantity $\log \hat{x}_{ilt}$ as the outcome variable in place of log deflated sales, with the same inputs, instruments, and minimum-distance step.

Figure 14 reports the re-estimated quantity estimates. The revenue profile rises from 0.794 in the bottom five percentiles to 0.906 in the top five. Re-estimating with the imputed quantity at the Edmond et al. (2023) calibration raises it to 0.837 and 1.007, a gradient of 0.171 against 0.112 in revenue, steeper because the correction grows with size.

The correction is insensitive to the assumed curvature within the range the empirical literature supports: Beck and Lein (2020) estimate a superelasticity of 1.39 from European scanner data, against 1.74 in Edmond et al. (2023), and the two deliver the same corrected gradient of 0.171.

Summing up, when markups rise with firm size, the revenue returns-to-scale-size profile is a compressed image of its quantity counterpart. Every departure from price taking that we consider moves the estimates in the same direction: measured in revenue, the rising returns-to-scale-size profile is not an artifact of market power. Instead, it is always an understatement.

A.6 Details of the within-firm growth design

This appendix provides additional details on the growth design of Section 6.3: the construction of the predetermined growth measure G_{lt} , the estimating equation and its system-GMM moment conditions, the pooling procedure, and robustness to alternative measures and samples.

A.6.1 Construction of the predetermined growth measure

Let t_l^0 denote the first year in which firm l appears in the panel and t_l^1 the second such year, and let q_{lt} denote its log sales. The predetermined growth measure used in Section 6.3 is

$$G_{lt} = \frac{1}{3} \sum_{j=3}^5 q_{l,t-j} - \frac{1}{2} (q_{l,t_l^0} + q_{l,t_l^1}), \quad (45)$$

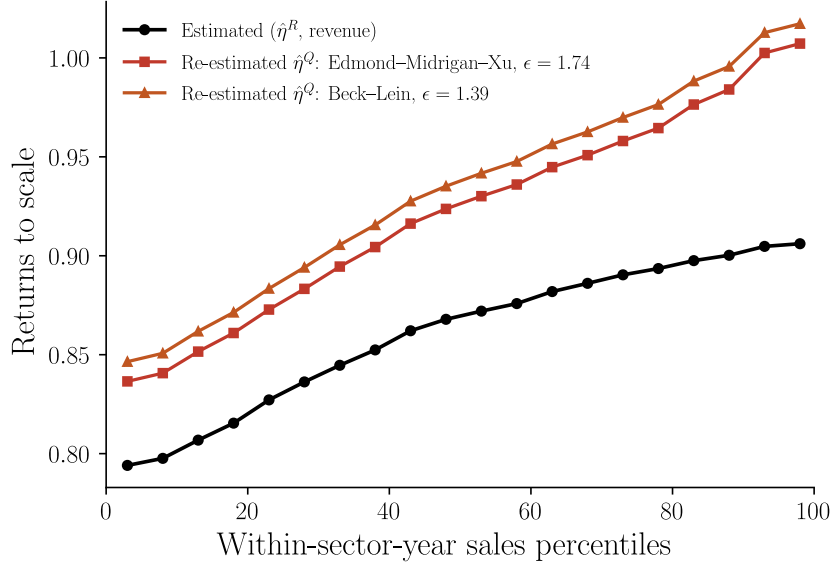


Figure 14: Estimated revenue and implied quantity returns to scale under Kimball demand

Notes: Black: binned means of the firm-level revenue estimates against within-sector-year sales percentiles (twenty bins), constructed exactly as in Figure 7(a). Colored: the quantity returns to scale re-estimated cell by cell with the imputed log quantity as the outcome, at $\epsilon = 1.74$ (Edmond et al., 2023) and $\epsilon = 1.39$ (Beck and Lein, 2020), attached to firm-years through the panel's cell assignment.

the difference between the firm's recent average log sales and its average log sales during its first two years in the panel.

The trailing component averages log sales over calendar years $t - 3$, $t - 4$, and $t - 5$: lags are matched on calendar years, so reporting gaps do not shift the window, and the average is taken over the lags that exist, with G_{lt} missing when none does. The entry component averages the firm's first two observed years, which need not be consecutive; for firms already active when the panel begins, G_{lt} therefore measures growth since first observation rather than since birth. The trailing component is dated $t - 3$ and earlier, and the entry benchmark is a firm-level constant formed once from those two years, so all time variation in G_{lt} is predetermined with respect to the innovations in the estimating equation. Firms are assigned to terciles of G_{lt} within each sector-year by rank, with the middle tercile as the reference group.

A.6.2 Estimating equation and moment conditions

We restate the estimating equation (36) of the main text:

$$q_{ilt} = \sum_{X \in \{K, L, M\}} \left(\beta_i^X + \varphi_i^X t + \sum_{b \neq 2} \theta_{bi}^X T_{ilt}^b \right) x_{ilt}^X + \sum_{b \neq 2} \delta_{bi} T_{ilt}^b + \alpha_l + \gamma_{it} + \varepsilon_{ilt}. \quad (46)$$

Estimation follows the baseline Blundell–Bond procedure exactly. As in the baseline, log productivity contains a year effect, a firm effect, and an AR(1) component with autocorrelation ρ_i and innovation e_{ilt} ; quasi-differencing therefore turns (46) into a dynamic equation in which every regressor enters together with its one-year lag, with the coefficient on each lag restricted to $-\rho_i$ times the coefficient on the current value. We estimate the dynamic equation without this restriction by system GMM and then impose it by minimum distance, exactly as in the baseline.

Our moment conditions follow the implementation in Appendix A.2, extended to the interacted regressor set: we treat sales, the inputs, their tercile and trend interactions, and the tercile indicators as potentially endogenous — every variable

$$a_{ilt} \in \{q_{ilt}\} \cup \{x_{ilt}^X, t x_{ilt}^X, T_{ilt}^b x_{ilt}^X, T_{ilt}^b : X \in \{K, L, M\}, b \neq 2\},$$

— and use two sets of moments. (i) Difference equation (levels dated $t - 2$ and earlier):

$$\mathbb{E}[a_{il,t-s} \Delta e_{ilt}] = 0 \quad \text{for } s \geq 2, \quad (47)$$

where a ranges over the variables in this list. (ii) Levels equation (first differences dated $t - 1$ only):

$$\mathbb{E}[\Delta a_{il,t-1} (\kappa_{il}^* + e_{ilt})] = 0. \quad (48)$$

Year dummies are included as controls and treated as exogenous instruments in the levels equation. Because the interacted regressor set is large, we collapse the instrument matrix: for each instrumenting variable and each lag distance s , the instruments are summed across time periods, so the set contains one instrument per variable and lag distance rather than one per variable, period, and lag. The tercile indicators that enter the instrument set — $T_{il,t-2}^b$ and deeper in the difference equation, and the first difference $T_{il,t-1}^b - T_{il,t-2}^b$ in the levels equation — carry no information dated after their own period: a firm-year at s with a defined G_{ls} must already have been observed at $s - 3$, so its first two observed years are dated s or earlier. They therefore satisfy the same timing restrictions as the lagged input levels.

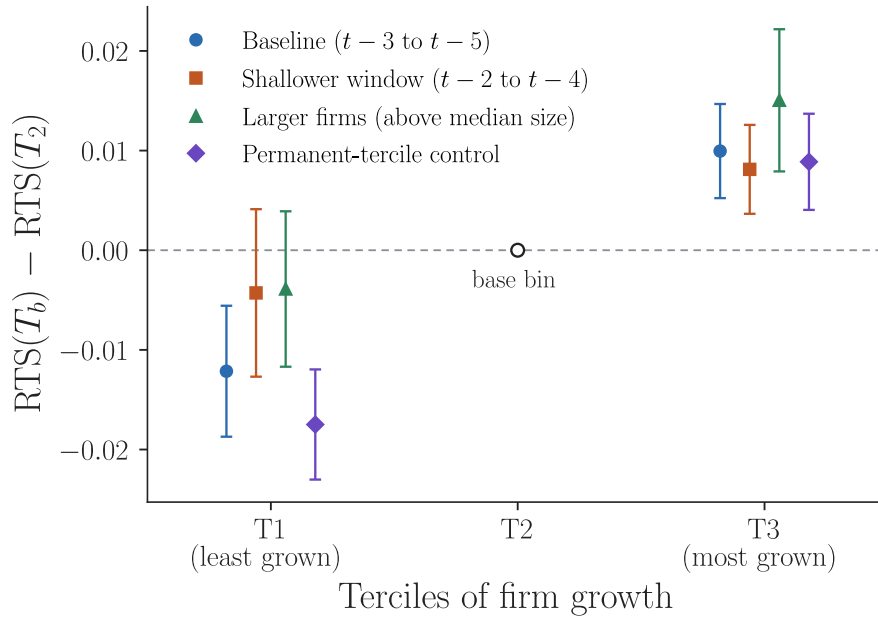
A.6.3 Pooling across sectors

We estimate (46) separately for each of the 55 sector cells of the baseline procedure, which covers the 62 input–output sectors by merging seven pairs of small sectors; cells with fewer than 120 firms are not estimated. The sector-level tercile differences $\text{RTS}_i(T_b) - \text{RTS}_i(T_2) = \sum_X \theta_{bi}^X$ are pooled with the mean-group estimator (Pesaran and Smith, 1995) — the unweighted average of the cell estimates — and no cells are trimmed. Standard errors of the pooled differences are computed from the cross-cell dispersion of the estimates, following Fama and MacBeth (1973); Ibragimov and Müller (2010) provide the justification for inference based on group-level estimates.

A.6.4 Robustness

Figure 15 plots the tercile profile under alternative measures and samples, re-estimating the full system for each variant. Shortening the trailing window of G_{it} to $t - 2$ through $t - 4$ — moving the growth measure closer to the estimation innovations — attenuates the profile: -0.004 ($t = -0.8$) below and $+0.008$ ($t = +3.0$) above the middle tercile. Keeping only firms whose average sales exceed the median in their sector leaves the profile monotone, with the top-tercile gap at $+0.015$ ($t = +3.5$) and the spread at $+0.019$ ($t = +3.6$). Lastly, one concern is that firms with permanently higher returns to scale might also be fast-growing on average. To ensure that the profile reflects within-firm adjustment rather than a mechanical association between fixed firm types and growth, we add the firm’s permanent growth tercile, its median tercile over the years it is observed, interacted with the inputs. The estimates of interest are then identified from firms whose tercile position moves, and the profile survives intact: -0.017 ($t = -5.2$) below and $+0.009$ ($t = +3.0$) above the middle tercile, with the spread at $+0.026$ ($t = +6.2$). In every case the profile rises from the least- to the most-grown tercile, as the model predicts. Appendix E.6 validates these designs on simulated panels.

Figure 15: Growth design: tercile profile under alternative measures and samples



Notes: Pooled mean-group differences in returns to scale relative to the middle tercile of predetermined growth, with 90 percent confidence bands from the cross-cell dispersion of the estimates. The baseline dates the trailing average at $t - 3$ through $t - 5$ and the shallower window at $t - 2$ through $t - 4$; the third variant keeps only firms whose average sales exceed the median in their sector. The fourth variant adds the firm’s permanent growth tercile interacted with the inputs.

A.7 Details of the network energy-price design

This appendix provides additional details on the network energy-price design of Section 6.4: the construction of the network exposure ℓ_i , the estimating equation and its system-GMM moment conditions, and robustness to the exposure convention and the sample of cells.

A.7.1 Construction of the network exposure

Collecting the variable-cost shares α_{ij} in a matrix Ω and partitioning the sectors into the energy block $\mathcal{E}$ — coke and refined petroleum, and electricity, gas, and steam — and its complement $\mathcal{E}^c$, the fixed point $\ell_i = \sum_{j \in \mathcal{E}} \alpha_{ij} + \sum_{j \notin \mathcal{E}} \alpha_{ij} \ell_j$ of Section 6.4 has the closed form

$$\ell_{\mathcal{E}^c} = (I - \Omega_{\mathcal{E}^c \mathcal{E}^c})^{-1} \Omega_{\mathcal{E}^c \mathcal{E}} \mathbf{1}. \quad (49)$$

The shares α_{ij} come from the 2010 Spanish input-output table. Each row sums to the sector's intermediate share of variable cost, defined as materials plus the wage bill. The oil price is the annual Brent spot price in euros, entering in log changes over 1997–2019. Each firm's exposure $\ell_i \Delta \log p_t^{\text{oil}}$ is assigned using the firm's own input-output sector. We exclude the two energy-producing estimation cells, mining and quarrying, and electricity, gas, and steam, from the baseline pooling: for these sectors the oil price moves the output price, not just the input cost.

A.7.2 Estimating equation and moment conditions

We restate the estimating equation (37) of the main text:

$$q_{ilt} = \sum_{X \in \{K, L, M\}} (\beta_i^X + \varphi_i^X t + \theta_i^X \log P_{it}^E) x_{ilt}^X + \alpha_l + \gamma_{it} + \varepsilon_{ilt}. \quad (50)$$

The main effect of $\log P_{it}^E$ is common to all firms of a sector-year and is absorbed by the year effects. Estimation is identical to the growth design: every regressor enters together with its one-year lag under the common-factor restriction, and we use the same two sets of moment conditions, treating as potentially endogenous every variable

$$a_{ilt} \in \{q_{ilt}\} \cup \{x_{ilt}^X, t x_{ilt}^X, \log P_{it}^E x_{ilt}^X : X \in \{K, L, M\}\}.$$

(i) Difference equation (levels dated $t - 2$ and earlier):

$$\mathbb{E}[a_{il,t-s} \Delta e_{ilt}] = 0 \quad \text{for } s \geq 2, \quad (51)$$

where a ranges over the variables in this list. (ii) Levels equation (first differences dated $t - 1$ only):

$$\mathbb{E}[\Delta a_{il,t-1} (\kappa_{il}^* + e_{ilt})] = 0. \quad (52)$$

Year dummies are exogenous instruments in the levels equation, the instrument matrix is collapsed, and the common-factor restriction is imposed by minimum distance, exactly as in the growth design. The interactions inherit their timing validity from the shifter: ℓ_i is fixed at its 2010 value and the oil price is a world price that individual Spanish producers cannot influence. The sector-level responses $\sum_X \theta_i^X$ are pooled across the 53 non-energy cells with the same mean-group procedure and cross-cell dispersion standard errors as the growth design (Section A.6).

A.7.3 Robustness of the energy design

Table 3 reports the pooled response of returns to scale per unit log of the energy-driven variable-cost index under alternative exposure conventions and samples. Column (1) is the baseline of the main text: pooled over the 53 non-energy estimation cells, the response is -0.066 (standard error 0.017). Column (2) adds the two energy-producing cells. The estimate remains basically unchanged. Column (3) uses the raw Leontief inverse as the measure of energy exposure. This measure takes into account indirect exposure of energy sectors on themselves, thus leading to a roughly doubled measured exposure. Consistently, the regression coefficient halves to -0.032 .

Figure 16 repeats panel (b) of Figure 8 under each of the three measures of sectoral energy exposure. All three panels show the same pattern: the deepest response sits in the most-exposed tercile.

Table 3: Energy design: pooled returns-to-scale response under alternative exposure conventions

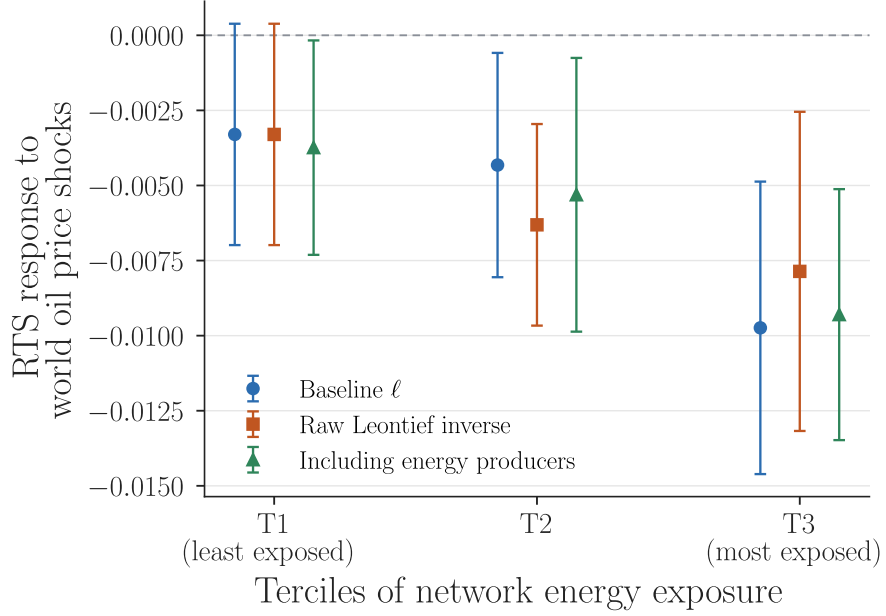
	(1)	(2)	(3)
	Baseline	All 55 cells	Raw Leontief
Pooled response	-0.066^{***} (0.017)	-0.066^{***} (0.017)	-0.032^{***} (0.008)

Notes: Each column reports the unweighted mean-group estimate of the response of returns to scale to a unit log increase in the energy-driven variable-cost index, with cross-cell dispersion standard errors in parentheses. (1): the pinned exposure ℓ_i of the main text on the 53 non-energy cells. (2): adds the two energy-producing cells, for which the oil price is an output price. (3): raw Leontief inverse, which roughly doubles the dose. *, **, *** indicate significance at the 10%, 5%, and 1% level.

A.8 Estimation of the tail index

This appendix describes how we estimate the tail index of the firm-size distribution in each sector-year using the log-rank estimator of Gabaix and Ibragimov (2011). For each sector i and year t , let S_{ilt} denote firm l 's sales, and let N_{it} be the number of firms observed in (i, t) . We assign ranks $r = 1, \dots, N_{it}$ according

Figure 16: Energy design: exposure-tercile dose–response under alternative exposure conventions



Notes: Pooled response of returns to scale per unit log of the world oil price by exposure tercile, with 90 percent confidence bands from the cross-cell dispersion of the estimates. Terciles rank the 53 non-energy cells (55 where indicated) under each exposure convention of Table 3; the plotted response does not depend on the ranking convention (up to the seven cells that merge two sectors), so the series differ only in how sectors are assigned to terciles.

to their sales, where $r = 1$ corresponds to the firm with the largest sales. Let $S_{i(1)t} \geq S_{i(2)t} \geq \dots \geq S_{i(N_{it})t}$ denote sales sorted in descending order within sector-year (i, t) .

We focus on the right tail of the sales distribution and select the tail sample as follows: If $N_{it} > 5000$, we use firms in the top 1% of the sales distribution in (i, t) . If $N_{it} \leq 5000$, we use the 50 firms with the largest sales in (i, t) .⁵⁴ For each sector-year (i, t) , we estimate the Pareto tail index ζ_{it} within the tail sample using the Gabaix and Ibragimov (2011) bias-corrected log-rank regression:

$$\log \left(r - \frac{1}{2} \right) = a_{it} - \zeta_{it} \log S_{i(r)t} + u_{irt}. \quad (53)$$

This regression relates the log bias-corrected rank $\log \left(r - \frac{1}{2} \right)$ to log sales. We recover $\hat{\zeta}_{it}$ as the negative of the OLS slope coefficient on $\log S_{i(r)t}$ and use it as the tail index of sales in Figure 9.

⁵⁴If fewer than 50 firms are observed, we use all available firms.

A.9 Details of the cross-country firm-level data

This appendix describes the firm-level data sources and sample construction for our cross-country analysis. We augment the analysis with firm-level data from a total of 24 countries (including Spain). For 22 European countries, we use Orbis and restrict attention to countries with good coverage of the variables required for production-function estimation. For developing countries, we use China’s National Bureau of Statistics (NBS) manufacturing firm database and India’s Annual Survey of Industries (ASI). Both the NBS and ASI datasets are censuses of above-scale manufacturing firms.⁵⁵ To ensure comparability across countries, we restrict all datasets to manufacturing firms. For each country, we select a seven-year window that maximizes the number of firm-year observations. We briefly discuss the data cleaning below.

Orbis For Orbis, we start from the raw firm-year panel for each country and apply the same four-step cleaning procedure used in Appendix A.1 for Spain.⁵⁶ We then (i) restrict the sample to manufacturing firms (corresponding to USSIC codes 2000-3999) and (ii) deflate all nominal financial variables using the country-specific GDP deflator from the World Bank. We implement the seven-year window selection described above on the full country panel, before the restriction in (i) and the deflation in (ii), and keep the window with the largest number of firm-year observations for each country.

India ASI Our Indian data come from the Annual Survey of Industries (ASI) for 1998–2017. We harmonize industry codes to NIC-2004 and then map them to the USSIC division level, retaining only manufacturing divisions. We measure sales using the gross sale value of all products. We measure capital using the average of the opening and closing gross book value of total capital. We measure labor using total wage bills. All variables are deflated using India’s GDP deflator from the World Bank. We then select the seven-year window with the largest number of firm-year observations (2011–2017).

China NBS The China data are annual firm-level surveys collected by the National Bureau of Statistics (NBS). We use the 1998–2007 sample period. We measure sales using product sales revenue, capital using total fixed assets, and labor using total annual wages payable. Firms are classified by a four-digit Chinese Industry Classification (CIC) code, which we harmonize to the USSIC division level. We retain manufacturing divisions only. All nominal variables are deflated using China’s GDP deflator from the World Bank. We then select the seven-year window with the largest number of firm-year observations within the available sample period (2001–2007).

Production function and returns to scale estimation We estimate production functions using the Blundell-Bond approach, following our baseline estimation strategy. We treat manufacturing as a single

⁵⁵For China, the NBS data also include state-owned enterprises below these thresholds. For India, the ASI includes a representative sample of small firms below the census cutoff.

⁵⁶Because the international panels are smaller, we exclude observations whose average revenue product of any input lies above the 99.5th or below the 0.5th percentile of that year’s within-country distribution, rather than the 99th and 1st percentiles used in the Spanish sample.

sector within each country. For each country c , let $[t_m(c)-3, t_m(c)+3]$ denote the selected seven-year window and $t_m(c)$ is the median year of that window. We only estimate production functions for firms existing in the median year $t_m(c)$. We group firms into deciles for year $t_m(c)$ based on their seven-year average log sales. We then estimate a decile-specific Cobb-Douglas production function using the full seven-year panel.

Let $\hat{\beta}_{c,d(l),t_m(c)}^O$ denote the estimated output elasticity of input $O \in \{K, L, M\}$ for country c and sales decile d . The returns to scale assigned to firm l in country c at year $t_m(c)$ is computed as the sum of the estimated input elasticities:

$$\eta_{clt_m(c)} = \hat{\beta}_{c,d(l),t_m(c)}^K + \hat{\beta}_{c,d(l),t_m(c)}^L + \hat{\beta}_{c,d(l),t_m(c)}^M.$$

We then construct the Törnqvist productivity index $\hat{z}_{clt_m(c)}$ using these estimates and compute the covariance between returns to scale and log sales, as well as between returns to scale and productivity $\hat{z}_{clt_m(c)}$ used in Figure 10 panel (a). In panel (b), we plot the seven-year average ($t_m(c) - 3$ to $t_m(c) + 3$) of log GDP per capita obtained from Penn World Table version 11.0 against the covariance between returns to scale and productivity $\hat{z}_{clt_m(c)}$.

B Appendix for Section 7

This section contains details about the calibration of Section 7.

B.1 Calibration data

This section describes the datasets used in the calibration and how the associated sectoral moments are computed.

1. We calibrate the sectoral parameters using the 2010 input-output table from the Spanish National Accounts. This table partitions the Spanish economy into 62 sectors which are usually defined at the 2-digit NACE industry level.⁵⁷ Conforming to the accounting conventions in the data, we calibrate the input elasticities of good s' in the production of sector s as

$$\hat{\alpha}_{ss'} = \frac{\text{Input from } s' \text{ at basic prices}_s}{\text{Total input at basic prices}_s} \times \frac{\text{Intermediate consumption at purchaser's prices}_s}{\text{Intermediate consumption at purchaser's prices}_s + \text{total labor expenditure}_s}$$

and the labor elasticity as

$$1 - \sum_{s'} \hat{\alpha}_{ss'} = \frac{\text{total labor expenditure}_s}{\text{Intermediate consumption at purchaser's prices}_s + \text{total labor expenditure}_s},$$

⁵⁷Sector 63 (household-related production activities) and sector 64 (services by extraterritorial organizations and bodies) are also present in the 2010 input-output table, but their input-output data is missing.

which corresponds to the labor share of total cost in the data.^{58,59} We calibrate the consumption share β_s to be the share of final consumption expenditure of good s in the sum of consumption expenditure spent on the 62 sectors.

2. We compute cross-sectional moments from the Orbis sample. After steps 1-4 in Appendix A.1 and the production function estimation in Appendix A.2, we perform a few additional steps:
 - (a) We winsorize the estimated returns to scale η_{ilt} at the top or bottom 0.5% of the firm-year distribution. In addition, we cap values above 0.99 at 0.99. Using firm-level returns to scale η_{ilt} , we compute each sector's effective returns to scale $\hat{\eta}_{it}$ as the sales-weighted average of these firm-level estimates.
 - (b) We compute profits as $\Pi_{ilt} = (1 - \eta_{ilt}) P_{it} Q_{ilt}$ and winsorize it at the top or bottom 0.5% within each sector-year.
 - (c) We then compute the interquartile ranges of $\log \Pi_{ilt}$ and η_{ilt} at the sector-year level.
 - (d) Finally, we average these sector-year moments over time to obtain sector-level moments used in our static model.

B.2 Interquartile ranges for returns to scale and profits

From (12), we have

$$\eta_{il} = 1 - \frac{1}{\frac{1-\varphi_i}{1-\hat{\eta}_i} + \frac{\varepsilon_{il}-\mu_i}{2\gamma_i}},$$

which implies⁶⁰

$$\text{IQR}(\eta_{il}) = \frac{1}{\frac{1-\varphi_i}{1-\hat{\eta}_i} + \frac{\sigma_i}{2\gamma_i} \Phi^{-1}(0.25)} - \frac{1}{\frac{1-\varphi_i}{1-\hat{\eta}_i} + \frac{\sigma_i}{2\gamma_i} \Phi^{-1}(0.75)}, \quad (54)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal random variable.

Profit of firm l in sector i is given by (62). Plugging (10) and (12) in this expression, we get

$$\log \Pi_{il} = \frac{1}{4\gamma_i} \left(2\gamma_i \frac{1-\varphi_i}{1-\hat{\eta}_i} - \mu_i + \varepsilon_{il} \right)^2 + \log H_i, \quad (55)$$

⁵⁸Because sector-to-sector data at purchasers' prices (i.e., adjusted for taxes less subsidies on products) are unavailable, we calibrate the intermediate-input expenditure share of inputs from sector s' used by sector s with the share computed in basic prices, $\frac{\text{Input from } s' \text{ at basic prices}_s}{\text{Total input at basic prices}_s}$.

⁵⁹We deliberately omit capital in this calibration because the Spanish input-output table does not distinguish the user cost of capital from profits.

⁶⁰For all firms with $\eta_{il} \in (0, 1)$, η_{il} is strictly increasing in ε_{il} . In the calibrated economy, the fraction of firms with $\eta_{il} \notin (0, 1)$ is very small.

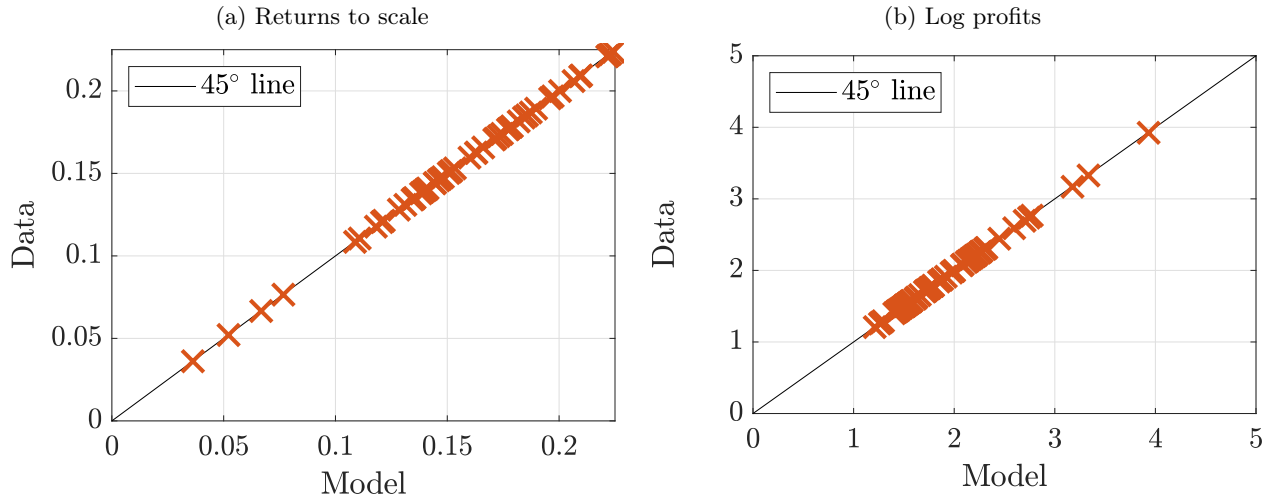
which implies⁶¹

$$\text{IQR}(\log \Pi_{il}) = \frac{\sigma_i^2}{4\gamma_i} \left(F_{\chi_1^2\left(\frac{2\gamma_i(1-\varphi_i)}{\sigma_i(1-\hat{\eta}_i)}\right)}^{-1}(0.75) - F_{\chi_1^2\left(\frac{2\gamma_i(1-\varphi_i)}{\sigma_i(1-\hat{\eta}_i)}\right)}^{-1}(0.25) \right), \quad (56)$$

where $F_{\chi_1^2(x)}(\cdot)$ is the cumulative distribution function of noncentral χ^2 distribution with one degree of freedom and the non-centrality parameter x , and $\varphi_i = \frac{\sigma_i^2}{2\gamma_i}$.

Equations (54) and (56) make clear that IQRs of returns to scale and log profits are functions of σ_i , γ_i , and $\hat{\eta}_i$. We can, therefore, use them to identify σ_i and γ_i . We choose σ_i and γ_i to minimize the distance between model-implied and empirical IQRs, with a constraint $\varphi_i \in [0, 1] \forall i$. Figure 17 shows that the calibrated model matches the targeted IQRs well.

Figure 17: Interquartile ranges in returns to scale and profits



Notes: Panels (a) and (b) report sectoral interquartile ranges in returns to scale and log profits in the calibrated model and in the data.

Figure 18 shows calibrated values of σ_i and γ_i for all sectors. The sector with most volatile productivity is “Petroleum”, with $\sigma_i = 3.09$. At the same time, this sector has a high cost of adjusting returns to scale, $\gamma_i = 7.24$, meaning that its effective productivity dispersion is not too large, $\varphi_i = 0.66$.

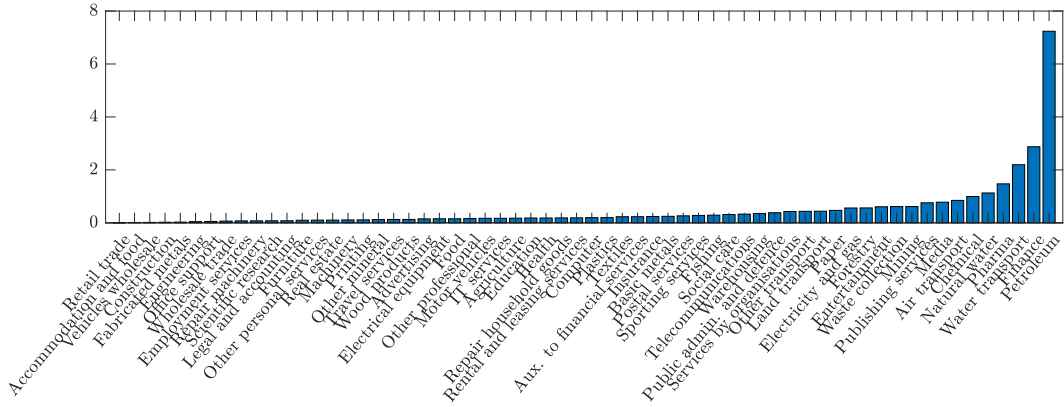
B.3 Calibration details for Section 7.3

We analyze the model with sales tax in Appendix D.5. In that appendix, we show that the model with sales taxes can be analyzed analogously to the main model if we properly redefine the mean and the variance

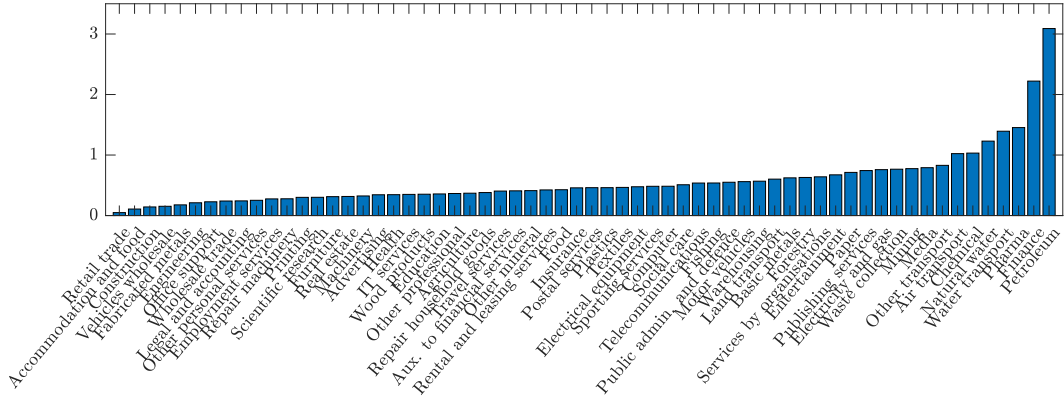
⁶¹From (10) and (12), $2\gamma_i \frac{1-\varphi_i}{1-\hat{\eta}_i} - \mu_i + \varepsilon_{il} > 0$ for all firms with $\eta_{il} \in (0, 1)$. For these firms, $\log \Pi_{il}$ is strictly increasing in ε_{il} . In the calibrated economy, the fraction of firms with $\eta_{il} \notin (0, 1)$ is very small.

Figure 18: Calibrated γ_i and σ_i

(a) Calibrated γ_i



(b) Calibrated σ_i



of sectoral shocks:

$$\tilde{\mu}_i = \mu_i + \log(1 - \tau_i^S) \quad \text{and} \quad \tilde{\sigma}_i = (1 - b_i) \sigma_i.$$

We can identify $\tilde{\sigma}_i$ and γ_i in the same way as described in Appendix B.2. The only difference is that we need to use after-tax profits in (56).

To pin down the parameters of the tax process (38), we proceed as follows. In the data, we compute the covariance of pre-tax profits with $\log(1 - \tau_{il}^S)$ for each sector. Using (55), we can compute the model

analogue of this quantity as

$$\begin{aligned} \text{Cov}(\log \Pi_{il}, \log(1 - \tau_{il}^S)) &= \text{Cov}\left(\frac{1}{4\gamma_i} \left(2\gamma_i \frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i} - \tilde{\mu}_i + \tilde{\varepsilon}_{il}\right)^2 - \log(1 - \tau_{il}^S), \log(1 - \tau_{il}^S)\right) = \\ &= -\frac{b_i}{1 - b_i} \tilde{\sigma}_i^2 \left(\frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i} + \frac{b_i}{1 - b_i}\right). \end{aligned}$$

We can identify b_i from this equation.

To compute τ_i^S , we rely on equation (97), derived in Appendix D.5, which we reproduce below:

$$\frac{1}{1 - \hat{\tau}_i^S} = \frac{1}{1 - \tau_i^S} \left(1 + \frac{\tilde{\varphi}_i}{\frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i}} \frac{b_i}{1 - b_i}\right) \exp\left(\frac{b_i}{1 - b_i} \frac{4\tilde{\varphi}_i\gamma_i \frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i} + \tilde{\sigma}_i^2 \frac{b_i}{1 - b_i}}{2} \frac{1}{1 - \tilde{\varphi}_i}\right). \quad (98)$$

In the data, we can observe $\hat{\tau}_i^S$ as the sales-weighted average tax rate in sector i . Then, (97) can be used to identify τ_i^S .

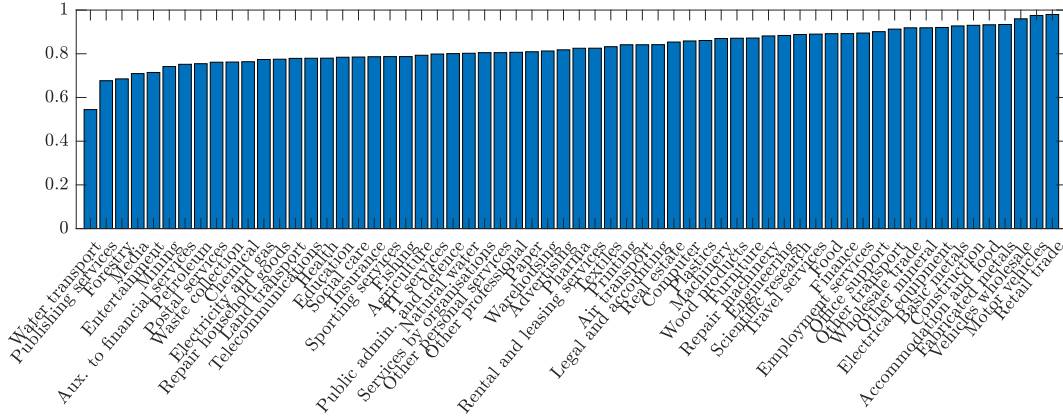
Finally, equation (96) makes clear that the proper measure of sectoral returns to scale $\hat{\eta}_i$ uses after-tax sales as weights.

B.4 Additional quantitative results

B.4.1 Sectoral effective returns to scale

Figure (19) shows effective sectoral returns to scale $\hat{\eta}_i$ for all sectors. In our data, the sector with lowest returns to scale is “Water transport” with $\hat{\eta}_i = 0.54$, and the sector with the highest returns to scale is “Retail trade” with $\hat{\eta}_i = 0.98$. The mean and median returns to scale are 0.83 and 0.82, respectively.

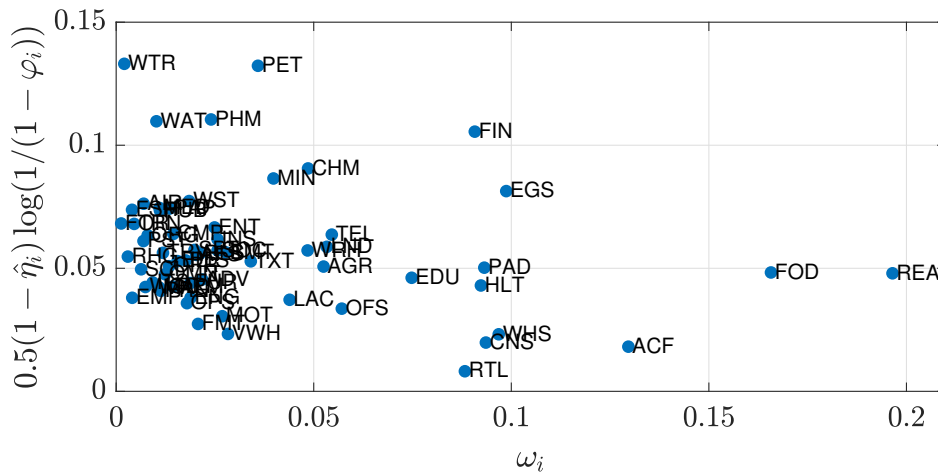
Figure 19: Effective returns to scale $\hat{\eta}_i$ across sectors



B.4.2 Sector-level contributions to GDP

Figure 20 decomposes the gap in GDP between our baseline model and the fixed returns-to-scale economy in its sectoral components. It reports the two terms in (26) that capture a sector’s importance: 1) its Domar weight ω_i and 2) the flexibility of its sectoral productivity $\frac{1}{2}(1 - \hat{\eta}_i) \log \frac{1}{1 - \varphi_i}$. We see that the “Water transport” sector is the most flexible one. However, since its Domar weight is only 0.0021, its importance for the economy is small. High-Domar-weights sectors like “Finance”, “Real estate”, and “Electricity and gas” that are also flexible are where the endogenous returns-to-scale mechanism has the most impact on GDP.

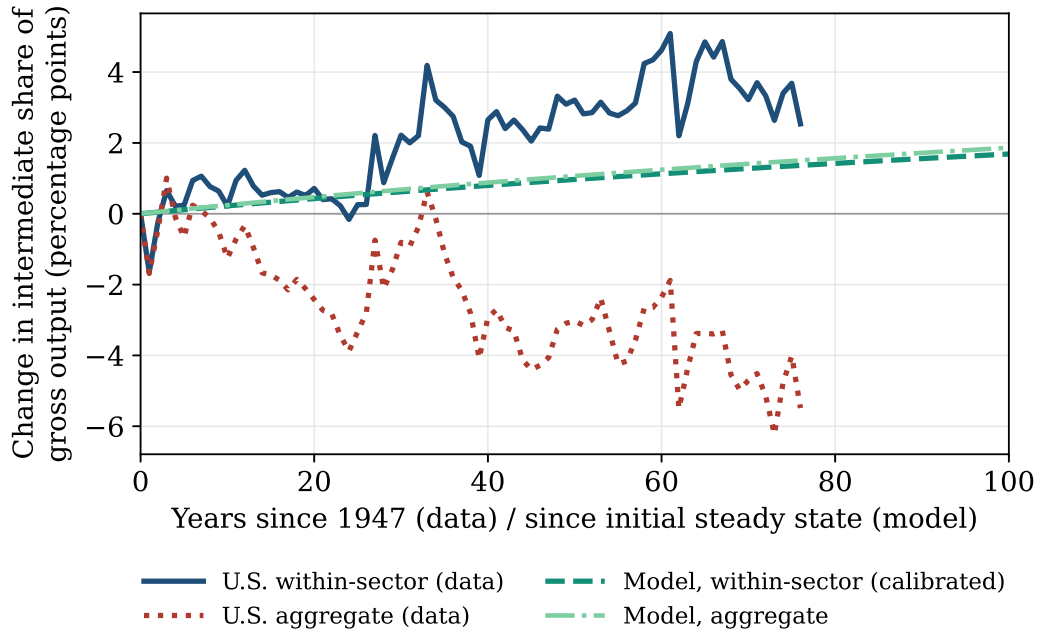
Figure 20: Domar weights, ω_i , and productivity gain due to endogenous returns to scale, $\frac{1}{2}(1 - \hat{\eta}_i) \log \frac{1}{1 - \varphi_i}$, across sectors



B.4.3 The intermediate-input share: model and data

Because scalable technologies are more intermediate-intensive, the model implies that a sector’s intermediate-input share of gross output rises with its effective returns to scale (Section 3.1), and hence over time as $\hat{\eta}$ increases (Corollary 1). One might worry that this conflicts with the “well-known” stability of the aggregate intermediate-input share. But the aggregate share combines a within-sector component, which is the model’s object, with a between-sector reallocation the model abstracts from. We separate the two in United States data, the only economy with consistent industry-level accounts reaching back to 1947. As Figure 21 shows, the aggregate US share actually *declines* over 1947–2023, but this masks two opposing movements. On the one hand, a compositional shift toward services, which rely less on intermediate inputs, lowers it by about 8 percentage points. On the other hand, within-sector intermediate share rises by about 2.5 points, close to the 1.7 p.p.-per-century speed implied by our calibration. Spain’s sectoral data begins only in 1995. Since then, its aggregate intermediate-input share has been roughly stable while its within-sector share has risen by about 4 percentage points.

Figure 21: Within- and between-sector decomposition of the intermediate share of gross output



Notes: The figure plots the cumulative change, in percentage points, of the nominal intermediate-input share of gross output, measured from the start of each series. The U.S. series are constructed from the BEA–BLS Integrated Industry-Level Production Account (44 industries, 1947–2023). The aggregate share is total intermediate inputs divided by total gross output across industries; the within-sector share instead holds each industry’s gross-output weight fixed at its 1947 value, isolating the change in industry-level intermediate shares from between-industry reallocation. The model series come from the calibrated growth transition of Section 7. The horizontal axis measures years since 1947 for the data and years since the initial steady state for the model.

B.5 Sensitivity analysis for Section 7.3

In Section 7.3, we experiment with removing wedges that are correlated with firm productivity. As we discuss there, removing these wedges leads to higher productivity dispersion. For some sectors, removing wedges would imply that $\varphi_i = \frac{\sigma_i^2}{2\gamma_i} > 1$, which is not allowed by our model. For these sectors, we set $\varphi_i = 0.99$. In this appendix, we explore how sensitive our results are to this threshold. Table 4 shows log GDP gains due to removal of sales wedges if we set $\varphi_{max} = 0.985, 0.99$ (main text), and 0.995. We see that the GDP gains become larger as φ_{max} increases. In the model, having sectors with $\varphi_i \rightarrow 1$ is particularly valuable because they feature a larger mass of firms with very high productivity draws operating at nearly constant returns to scale, which makes these sectors especially productive.

Table 4: Log GDP change after removal of sales wedges: Sensitivity analysis

	$\varphi_{max} = 0.985$	$\varphi_{max} = 0.99$	$\varphi_{max} = 0.995$
Baseline economy	160%	167%	177%
Dispersed RTS	134%	138%	142%
Fixed RTS	69%	70%	70%

Notes: Increases in log GDP due to removal of sales wedges in the baseline economy, and in the economies with fixed and dispersed returns to scale, for three values of maximum effective productivity dispersion φ .

C Proofs

C.1 Sectoral Domar weights

Multiplying the resource constraint for good i , given by (9), by P_i we get

$$P_i Q_i = P_i C_i + \sum_j P_i \int_0^{M_j} X_{ji,l} dl.$$

From the problem of the household we know that $P_i C_i = \beta_i \bar{P}Y$. It follows that

$$\frac{P_i Q_i}{\bar{P}Y} = \beta_i + \sum_j \frac{P_i}{\bar{P}Y} \int_0^{M_j} X_{ji,l} dl,$$

where we have divided by nominal GDP $\bar{P}Y$. Next, from the problem of firm l in sector j we know that

$$P_i X_{ji,l} = \alpha_{ji} \eta_{jl} P_j Q_{jl}.$$

Combining with the previous expression yields

$$\frac{P_i Q_i}{\bar{P}Y} = \beta_i + \sum_j \int_0^{M_j} \alpha_{ji} \eta_{jl} \frac{P_j Q_{jl}}{\bar{P}Y} dl,$$

or

$$\omega_i = \beta_i + \sum_j \alpha_{ji} \omega_j \hat{\eta}_j.$$

Solving this linear system leads to (18).

C.2 Proof of Lemma 1

Lemma 1. *The firm's marginal cost of production λ_{il} is given by*

$$\lambda_{il} = \frac{1}{e^{\varepsilon_{il}} A_i(\eta_{il})} H_i^{\eta_{il}} \Pi_{il}^{1-\eta_{il}}, \quad (3)$$

where $H_i := W^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}}$ is the price of the variable input bundle in sector i , and Π_{il} is profits,

$$\Pi_{il} = (1 - \eta_{il}) \lambda_{il} Q_{il}. \quad (4)$$

Proof. We tackle problem (2) through its cost minimization dual:

$$\min_{\eta_{il}, L_{il}, X_{ij,l}} W L_{il} + \sum_{j=1}^N P_j X_{ij,l}, \quad \text{subject to } F_i(L_{il}, X_{il}, \eta_{il}) \geq Q_{il}. \quad (57)$$

The Lagrangian is

$$\mathcal{L} = W L_{il} + \sum_{j=1}^N P_j X_{ij,l} - \lambda_{il} \left(e^{\varepsilon_{il}} A_i(\eta_{il}) \zeta(\eta_{il}) \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}} - Q_{il} \right),$$

and the first-order conditions with respect to L_{il} and $X_{ij,l}$ are

$$\eta_{il} \left(1 - \sum_{j=1}^N \alpha_{ij} \right) \lambda_{il} Q_{il} = W L_{il}, \quad (58)$$

$$\eta_{il} \alpha_{ij} \lambda_{il} Q_{il} = P_j X_{ij,l}. \quad (59)$$

Plugging back into the constraint, we find

$$\lambda_{il} = \frac{1}{(e^{\varepsilon_{il}} A_i(\eta_{il}))^{\frac{1}{\eta_{il}}}} H_i ((1 - \eta_{il}) Q_{il})^{\frac{1-\eta_{il}}{\eta_{il}}}. \quad (60)$$

Using the definition of Π_{il} from (4) yields the result.

Note also that the envelope theorem implies that λ_{il} is the marginal production cost of the firm. Notice that λ_{il} is increasing in Q_{il} for $\eta_{il} < 1$. As usual, we can then write the profit maximization problem of the firm as

$$\max_{Q_{il}} P_i Q_{il} - \int_0^{Q_{il}} \lambda_{il}(x) dx,$$

where the notation makes clear the dependence of $\lambda_{il}(Q_{il})$ on the size of the firm. This problem's first-order condition implies that $P_i = \lambda_{il}(Q_{il})$, so that the firm sets Q_{il} to equalize its marginal cost to the price of its good. $\square$

C.3 Proof of Lemma 2

Lemma 2. *At an interior solution, the firm chooses its returns to scale $\eta_{il} \in (0, 1)$ according to*

$$\frac{da_i(\eta_{il})}{d\eta_{il}} = \log H_i - \log \Pi_{il}, \quad (5)$$

where $a_i(\eta_{il}) := \log A_i(\eta_{il})$.

Proof. The first-order condition for η_{il} in the cost-minimization problem (57) is

$$\begin{aligned} \frac{dA_i(\eta_{il})}{d\eta_{il}} \zeta(\eta_{il}) \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}} + A_i(\eta_{il}) \frac{d\zeta(\eta_{il})}{d\eta_{il}} \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}} \\ + A_i(\eta_{il}) \zeta(\eta_{il}) \frac{d}{d\eta_{il}} \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}} = 0. \end{aligned} \quad (61)$$

Note that we do not include Lagrange multipliers for the constraints $0 \leq \eta_{il} \leq 1$ since we focus on interior solutions. Dividing by Q_{il} yields

$$\frac{d \log A_i(\eta_{il})}{d\eta_{il}} + \frac{d \log \zeta(\eta_{il})}{d\eta_{il}} + \frac{d}{d\eta_{il}} \log \left(L_{il}^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij,l}^{\alpha_{ij}} \right)^{\eta_{il}} = 0.$$

Combining this with (58) and (59) yields (5). □

C.4 Proof of Lemma 3

Lemma 3. *At an interior solution, the returns-to-scale parameter η_{il} satisfies⁶²*

$$\frac{d\eta_{il}}{d\varepsilon_{il}} = \frac{d\eta_{il}}{d \log P_i} = - \left[(1 - \eta_{il}) \frac{d^2 a_i}{d\eta_{il}^2} \right]^{-1} > 0, \quad \text{and} \quad \frac{d\eta_{il}}{d \log H_i} = \left[(1 - \eta_{il}) \frac{d^2 a_i}{d\eta_{il}^2} \right]^{-1} < 0.$$

Proof. We can combine (3) with the firm's optimality condition $\lambda_i = P_i$ to write

$$\log \Pi_{il} = \frac{1}{1 - \eta_{il}} (\log P_i + \varepsilon_{il} + a_i(\eta_{il}) - \eta_{il} \log H_i). \quad (62)$$

Together with (5), we can write the first-order condition with respect to η_{il} as

$$\underbrace{\log H_i - \log P_i - \varepsilon_{il}}_K = (1 - \eta_{il}) \frac{d \log A_i(\eta_{il})}{d\eta_{il}} + \log A_i(\eta_{il}), \quad (63)$$

⁶²When increasing P_i , we keep the price of the variable input bundle constant to distinguish the two channels that affect η_{il} .

where we use K as a temporary variable to denote the left-hand side of (63). Full differentiation yields

$$1 = -\frac{d\eta_{il}}{dK} \frac{d \log A_i(\eta_{il})}{d\eta_{il}} + (1 - \eta_{il}) \frac{d^2 \log A_i(\eta_{il})}{d\eta_{il}^2} \frac{d\eta_{il}}{dK} + \frac{d \log A_i(\eta_{il})}{d\eta_{il}} \frac{d\eta_{il}}{dK}.$$

Simplifying we find

$$\frac{d\eta_{il}}{dK} = \frac{1}{(1 - \eta_{il}) \frac{d^2 \log A_i(\eta_{il})}{d\eta_{il}^2}},$$

and the result follows. $\square$

C.5 Proof of Lemma 4

Lemma 4. *At an interior solution, the elasticity of output Q_{il} with respect to productivity ε_{il} is given by*

$$\frac{d \log Q_{il}}{d \varepsilon_{il}} = \underbrace{\frac{1}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d \varepsilon_{il}}}_{\text{Flexible } \eta \text{ effect}} > 0.$$

In addition, the elasticities of output Q_{il} with respect to prices are given by

$$\frac{d \log Q_{il}}{d \log P_i} = \underbrace{\frac{\eta_{il}}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d \log P_i}}_{\text{Flexible } \eta \text{ effect}} > 0, \quad \text{and} \quad \frac{d \log Q_{il}}{d \log H_i} = \underbrace{-\frac{\eta_{il}}{1 - \eta_{il}}}_{\text{Fixed } \eta \text{ effect}} + \underbrace{\frac{1}{1 - \eta_{il}} \frac{d\eta_{il}}{d \log H_i}}_{\text{Flexible } \eta \text{ effect}} < 0.$$

Proof. Profit maximization implies that the firm's marginal cost of production λ_i is equal to the price P_i , and so we can invert (3) and (4) to write

$$Q_{il} = \frac{1}{1 - \eta_{il}} (e^{\varepsilon_{il}} A_i(\eta_{il}))^{\frac{1}{1 - \eta_{il}}} \left(\frac{P_i}{H_i} \right)^{\frac{\eta_{il}}{1 - \eta_{il}}},$$

or, in log form, as

$$\log Q_{il} = -\log(1 - \eta_{il}) + \frac{1}{1 - \eta_{il}} \varepsilon_{il} + \frac{1}{1 - \eta_{il}} a_i(\eta_{il}) + \frac{\eta_{il}}{1 - \eta_{il}} (\log P_i - \log H_i). \quad (64)$$

Without endogenous returns to scale, it is immediate that

$$\frac{\partial \log Q_{il}}{\partial \varepsilon_{il}} = \frac{1}{1 - \eta_{il}} \quad \text{and} \quad \frac{\partial \log Q_{il}}{\partial \log P_i} = -\frac{\partial \log Q_{il}}{\partial \log H_i} = \frac{\eta_{il}}{1 - \eta_{il}}.$$

With endogenous returns to scale, we can combine (5) and (62) to find

$$-(1 - \eta_{il}) \frac{da_i(\eta_{il})}{d\eta_{il}} = \log P_i + \varepsilon_{il} + a_i(\eta_{il}) - \log H_i. \quad (65)$$

Combining (64) and (65), we get

$$\log Q_{il} = -\log(1 - \eta_{il}) - \frac{da_i(\eta_{il})}{d\eta_{il}} - (\log P_i - \log H_i). \quad (66)$$

Differentiating with respect to ε_{il} , we find

$$\frac{d \log Q_{il}}{d \varepsilon_{il}} = \frac{1}{1 - \eta_{il}} \frac{d \eta_{il}}{d \varepsilon_{il}} - \frac{d^2 a_i(\eta_{il})}{d \eta_{il}^2} \frac{d \eta_{il}}{d \varepsilon_{il}}.$$

Combining with Lemma 3 yields the result. The derivatives with respect to $\log P_i$ and $\log H_i$ can be computed in a similar way. The last part of the result follows from the signs of the derivatives in Lemma 3. $\square$

C.6 Proof of Proposition 1

Proposition 1. *Without endogenous returns to scale, the distribution of Q_{il} in sector i is log-normal. With endogenous returns to scale, the right tail of the distribution of Q_{il} behaves like a Pareto distribution with tail index $1/\varphi_i$, in the sense that*

$$\log(\mathbb{P}(Q_{il} > q)) \sim -\frac{1}{\varphi_i} \log q, \text{ as } q \rightarrow \infty.$$

Proof. Without endogenous returns to scale, the log of Q_{il} is given by (64). The only random term is ε_{il} and so Q_{il} is log-normal. We now turn to the case with endogenous returns to scale. Under Assumption 1, we can write (65) as

$$\frac{1}{1 - \eta_{il}} = \frac{\varepsilon_{il} + B_i}{2\gamma_i},$$

where we define $B_i := \log P_i - \log H_i$ as a temporary variable to simplify the notation. Combining with (66), we can write

$$\log Q_{il} = \log\left(\frac{\varepsilon_{il} + B_i}{2\gamma_i}\right) + \gamma_i \left(\frac{\varepsilon_{il} + B_i}{2\gamma_i}\right)^2 - B_i.$$

We want to characterize the right tail of Q_{il} . Because of the logarithm, we need to be careful about eventual bounds on ε_{il} . We impose here that $\varepsilon_{il} \sim \mathcal{N}(\mu_i, \sigma_i^2)$ is normally distributed with a truncation such that $\varepsilon_{il} > -B_i + 2\gamma_i$. We provide a full treatment of the model with truncated normal distribution in Appendix D.1. To simplify the notation, we drop the subscripts i and l from now on.

Step 1. We want to characterize the Complementary CDF (CCDF) $S_Q(q) = \mathbb{P}(Q > q)$ as $q \rightarrow \infty$. Let us define $g : (-B, \infty) \rightarrow \mathbb{R}$ as the function that maps ε to $\log Q$:

$$g(x) = \log\left(\frac{x + B}{2\gamma}\right) + \gamma \left(\frac{x + B}{2\gamma}\right)^2 - B.$$

One can show that g is a strictly increasing function. It is therefore invertible, and we can write

$$S_Q(q) = \mathbb{P}(\log Q > \log q) = \mathbb{P}(g(\varepsilon) > \log q) = \mathbb{P}(\varepsilon > g^{-1}(\log q)).$$

Given the properties of g , the right tail of Q corresponds to the right tail of ε .

Step 2. Let $y = g(x)$. We need to characterize the asymptotic behavior of $x = g^{-1}(y)$ as $y \rightarrow \infty$. Letting $X = \frac{x+B}{2\gamma}$, the equation $y = g(x)$ can be rewritten as

$$y + B = \log X + \gamma X^2.$$

As $y \rightarrow \infty$, it must be that $X \rightarrow \infty$. In this limit, the quadratic term γX^2 dominates $\log X$ and we can write⁶³

$$y + B \sim \gamma X^2, \text{ as } y \rightarrow \infty.$$

This implies that $X \sim \sqrt{y/\gamma}$ for large y .

Now, we relate this to $x = g^{-1}(y)$. Since $x = 2\gamma X - B$, we have

$$g^{-1}(y) = x = 2\gamma X - B \sim 2\sqrt{\gamma y}.$$

since the constant B is negligible as $y \rightarrow \infty$. We will come back to this expression momentarily.

Step 3. The CCDF of the truncated normal ε is given by

$$S_\varepsilon(x) = \frac{1}{K_1} S_{\bar{\varepsilon}}(x)$$

where $\bar{\varepsilon}$ is the untruncated normal with the same mean and variance, and where K_1 is a constant. It is well-known that approximating the Mills ratio implies that

$$S_{\bar{\varepsilon}}(x) \sim \frac{\sigma}{(x - \mu) \sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \text{ as } x \rightarrow \infty.$$

We can therefore write

$$\log S_\varepsilon(x) \sim \log\left(\frac{1}{K_1}\right) - \frac{(x - \mu)^2}{2\sigma^2} + \log\left(\frac{\sigma}{(x - \mu) \sqrt{2\pi}}\right).$$

As $x \rightarrow \infty$, the quadratic term dominates the others and thus

$$\log S_\varepsilon(x) \sim -\frac{x^2}{2\sigma^2}, \text{ as } x \rightarrow \infty.$$

Step 4. We now combine the results. Let $x_q = g^{-1}(\log q)$. From Step 1, $S_Q(q) = S_\varepsilon(x_q)$. From Step

⁶³As usual, we write “ $f(x) \sim g(x)$ as $x \rightarrow \infty$ ” if and only if $\lim_{x \rightarrow \infty} f(x)/g(x) = 1$.

3, for large q , and consequently large x_q ,

$$\log S_Q(q) \sim -\frac{x_q^2}{2\sigma^2}.$$

We can now substitute the asymptotic form for x_q from Step 2. Let $y = \log q$. As $q \rightarrow \infty$, $y \rightarrow \infty$ and

$$x_q = g^{-1}(\log q) \sim 2\sqrt{\gamma \log q}.$$

It is well-known that if $f \sim g$ then $f^r \sim g^r$ for r real. Therefore,

$$x_q^2 \sim 4\gamma \log q.$$

Since $\sim$ is transitive, we can substitute in the expression for $\log S_Q$ to find

$$\log S_Q(q) \sim -\frac{2\gamma}{\sigma^2} \log q, \text{ as } q \rightarrow \infty,$$

which is the result. □

C.7 Proof of Lemma 5

Lemma 5. *The returns to scale η_{il} of firm l in sector i are given by*

$$\frac{1}{1 - \eta_{il}} = \frac{1 - \varphi_i}{1 - \hat{\eta}_i} + \frac{\varepsilon_{il} - \mu_i}{2\gamma_i}. \quad (12)$$

Furthermore, the moments of the firm-level returns-to-scale distribution are given by

$$\mathbb{E}_i \left[\frac{1}{1 - \eta_{il}} \right] = \frac{1 - \varphi_i}{1 - \hat{\eta}_i}, \quad \mathbb{V}_i \left[\frac{1}{1 - \eta_{il}} \right] = \frac{\varphi_i}{2\gamma_i}, \quad \text{and} \quad \text{Cov}_i \left[\frac{1}{1 - \eta_{il}}, \varepsilon_{il} \right] = \varphi_i > 0. \quad (13)$$

Proof. Given Assumption 1, we can write the returns to scale first-order condition (63) as

$$\log P_i - \log H_i + \varepsilon_{il} = \frac{2\gamma_i}{1 - \eta_{il}},$$

Combining that expression with itself when $\varepsilon_{il} = \mu_i$ yields

$$\frac{1}{1 - \eta_{il}} = \frac{1}{1 - \eta_i(\mu_i)} + \frac{\varepsilon_{il} - \mu_i}{2\gamma_i},$$

and the result follows from combining with (72), derived below, and taking the moments. □

C.8 Proof of Proposition 2

Proposition 2. *The marginal cost of sector i is given by*

$$\lambda_i = \frac{1}{Z_i(\hat{\eta}_i)} W^{1-\hat{\eta}_i} \sum_{j=1}^N \alpha_{ij} \prod_{j=1}^N P_j^{\hat{\eta}_i \alpha_{ij}}, \quad (15)$$

where sectoral total factor productivity $Z_i(\hat{\eta}_i)$ is defined as

$$\log Z_i(\hat{\eta}_i) := \underbrace{\mu_i + a_i(\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1-\hat{\eta}_i}}_{\text{Exogenous returns to scale}} + \underbrace{\frac{1}{2} (1-\hat{\eta}_i) \log \left(\frac{1}{1-\varphi_i} \right)}_{\text{Superstar effect}} - \underbrace{(1-\hat{\eta}_i) \log \kappa_i}_{\text{Entry cost}}. \quad (16)$$

Furthermore, the effective returns to scale $\hat{\eta}_i$ is given by

$$\frac{1}{1-\hat{\eta}_i} = \frac{1}{2\gamma_i(1-\varphi_i)} (\mu_i + \log P_i - \log H_i). \quad (17)$$

Proof. Since firms in a sector all face the same sales price, they have the same marginal cost through profit maximization. We therefore define the marginal cost λ_i of a sector i as the marginal cost of any firm in that sector, such that $\lambda_i := \lambda_{il}$ for any l .

Together with (62), the free-entry condition (8) imposes that

$$\int_{-\infty}^{\infty} \underbrace{\left(\lambda_i \frac{e^{\varepsilon_{il}} A_i(\eta_{il})}{H_i^{\eta_{il}}} \right)^{\frac{1}{1-\eta_{il}}}}_{\Pi_{il}} f_i(\varepsilon_{il}) d\varepsilon_{il} = \kappa_i W, \quad (67)$$

where f_i is the probability density function of a normal distribution $\mathcal{N}(\mu_i, \sigma_i^2)$. Multiplying the term inside the parentheses by one, we find

$$\int_{-\infty}^{\infty} \left(\frac{\lambda_i}{H_i^{\eta_{il}}} \frac{H_i^{(1-\eta_{il}) \frac{\hat{\eta}_i}{1-\hat{\eta}_i}}}{H_i^{(1-\eta_{il}) \frac{\hat{\eta}_i}{1-\hat{\eta}_i}}} \frac{\lambda_i^{\frac{1-\eta_{il}}{1-\hat{\eta}_i}}}{\lambda_i^{\frac{1-\eta_{il}}{1-\hat{\eta}_i}}} e^{\varepsilon_{il}} A_i(\eta_{il}) \right)^{\frac{1}{1-\eta_{il}}} f_i(\varepsilon_{il}) d\varepsilon_{il} = \kappa_i W,$$

which can be reorganized as

$$\lambda_i = \frac{1}{\tilde{Z}_i} (\kappa_i W)^{1-\hat{\eta}_i} \left(W^{1-\sum_j \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}} \right)^{\hat{\eta}_i}, \quad (68)$$

where $\tilde{Z}_i$ is defined as

$$\tilde{Z}_i := \left[\int_{-\infty}^{\infty} \left(\left(\frac{\lambda_i}{H_i} \right)^{\frac{\eta_{il} - \hat{\eta}_i}{1 - \hat{\eta}_i}} e^{\varepsilon_{il}} A_i(\eta_{il}) \right)^{\frac{1}{1 - \eta_{il}}} f_i(\varepsilon_{il}) d\varepsilon_{il} \right]^{1 - \hat{\eta}_i}.$$

To simplify the notation, define $s_i := \log \lambda_i - \log H_i$. Using the definition of s_i and A_i , we can write

$$\tilde{Z}_i = \left[\int_{-\infty}^{\infty} \left(e^{s_i \frac{\eta_{il} - \hat{\eta}_i}{1 - \hat{\eta}_i} + \varepsilon_{il} - \frac{\gamma_i}{1 - \eta_{il}}} \right)^{\frac{1}{1 - \eta_{il}}} f_i(\varepsilon_{il}) d\varepsilon_{il} \right]^{1 - \hat{\eta}_i}. \quad (69)$$

For an arbitrary set of firm-level returns to scale $\{\eta_{il}\}$ this integral cannot be computed analytically, but we can do so here, given the relationship between ε_{il} and η_{il} implied by the model. Using Assumption 1, we can write the returns to scale first-order condition (63) as⁶⁴

$$\underbrace{\log \lambda_i - \log H_i}_{:=s_i} + \varepsilon_{il} = \frac{2\gamma_i}{1 - \eta_{il}} = -2a_i(\eta_{il}), \quad (70)$$

which implies that

$$\frac{1}{1 - \eta_{il}} = \frac{s_i + \varepsilon_{il}}{2\gamma_i}, \text{ and } \frac{\eta_{il}}{1 - \eta_{il}} = \frac{s_i + \varepsilon_{il} - 2\gamma_i}{2\gamma_i}.$$

Combining with $\tilde{Z}_i$, we find

$$\tilde{Z}_i = \left[\int_{-\infty}^{\infty} e^{\frac{(s_i + \varepsilon_{il})^2}{4\gamma_i} - \frac{s_i}{1 - \hat{\eta}_i}} f_i(\varepsilon_{il}) d\varepsilon_{il} \right]^{1 - \hat{\eta}_i}.$$

Given the structure of the normal distribution f_i , this integral can be computed when $2\gamma_i > \sigma_i^2$ and yields

$$\tilde{Z}_i = \left[\sqrt{\frac{2\gamma_i}{2\gamma_i - \sigma_i^2}} \exp \left(\frac{(s_i + \mu_i)^2}{2(2\gamma_i - \sigma_i^2)} - \frac{s_i}{1 - \hat{\eta}_i} \right) \right]^{1 - \hat{\eta}_i}.$$

We will rewrite this expression using $\hat{\eta}_i$. To do so, notice that we can write

$$\hat{\eta}_i = \frac{\int_l \eta_{il} P_i Q_{il} dl}{\int_l P_i Q_{il} dl} = 1 - \frac{\int_l (1 - \eta_{il}) P_i Q_{il} dl}{\int_l P_i Q_{il} dl} = 1 - \frac{\int_l \Pi_{il} dl}{\int_l \frac{1}{1 - \eta_{il}} \Pi_{il} dl}. \quad (71)$$

Using the profit expression (62), we can compute these integrals and find

$$1 - \hat{\eta}_i = 2\gamma_i \frac{1 - \varphi_i}{s_i + \mu_i} = (1 - \varphi_i) (1 - \eta_i(\mu_i)), \quad (72)$$

where $\eta_i(\mu_i)$ is the returns to scale chosen by the firm with $\varepsilon_{il} = \mu_i$ (computed from (70)). Notice that (72)

⁶⁴In equilibrium, the price charged by firms in sector i must be equal to their marginal costs, so that $\lambda_i = P_i$.

implies (17) because s_i is given by (70).

Combining (72) with our expression for $\tilde{Z}_i$, we find

$$\tilde{Z}_i = \left[\sqrt{\frac{1}{1-\varphi_i}} \exp \left(\frac{1-\varphi_i}{1-\hat{\eta}_i} a_i(\hat{\eta}_i) + \frac{\mu_i}{1-\hat{\eta}_i} \right) \right]^{1-\hat{\eta}_i}.$$

Taking the log yields

$$\log \tilde{Z}_i(\hat{\eta}_i) := \mu_i + a_i(\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1-\hat{\eta}_i} - (1-\hat{\eta}_i) \log \left(\sqrt{1-\varphi_i} \right),$$

where we have used the definition of φ_i and Assumption 1. The quantity $\tilde{Z}_i$ corresponds to the total factor productivity of sector i if we treat the mass of firms in that sector as an independent factor. But it will be often convenient to lump that input together with labor. In that case, we can rewrite (68) as

$$\lambda_i = \frac{1}{Z_i(\hat{\eta}_i)} W^{1-\hat{\eta}_i \sum_{j=1}^N \alpha_{ij}} \left(\prod_{j=1}^N P_j^{\alpha_{ij}} \right)^{\hat{\eta}_i},$$

where $Z_i := \tilde{Z}_i(\hat{\eta}_i) / \kappa_i^{1-\hat{\eta}_i}$, which completes the proof. $\square$

C.9 Proof of Proposition 3

Proposition 3. *The equilibrium price vector $P = (P_1, \dots, P_N)$ satisfies*

$$\log \frac{P}{W} = -\mathcal{L}(\hat{\eta}) z(\hat{\eta}), \quad (19)$$

where $z(\hat{\eta}) = (\log Z_1(\hat{\eta}_1), \dots, \log Z_N(\hat{\eta}_N))$ is the vector of log sectoral productivities (16). Furthermore, equilibrium log GDP $y := \log Y$ is given by

$$y(\hat{\eta}) = \underbrace{[\omega(\hat{\eta})]^\top z(\hat{\eta})}_{\text{Aggregate productivity}} + \underbrace{\log \bar{L}}_{\text{Labor endowment}}. \quad (20)$$

Proof. Since in equilibrium prices must be equal to marginal costs, we can use (15) to write

$$\frac{P_i}{W} = \frac{1}{Z_i(\hat{\eta}_i)} \prod_{j=1}^N \left(\frac{P_j}{W} \right)^{\hat{\eta}_i \alpha_{ij}}.$$

Taking the log of this equation leads to

$$\log \frac{P_i}{W} = -\log Z_i(\hat{\eta}_i) + \hat{\eta}_i \sum_{j=1}^N \alpha_{ij} \log \frac{P_j}{W}.$$

In vector notation, this becomes $\log(P/W) = -z(\hat{\eta}) + \text{diag}(\hat{\eta}) \alpha \log(P/W)$. Solving it for $\log(P/W)$ yields (19).

We now turn to the GDP equation. The budget constraint of the household is $\bar{P}Y = W\bar{L}$. Together with the definition of the price index, $\bar{P} = \prod_{i=1}^N P_i^{\beta_i} = 1$, we can therefore write

$$y = - \sum_{i=1}^N \beta_i \log \frac{P_i}{\bar{W}} + \log \bar{L},$$

and the result follows from combining this expression with (18) and (19). $\square$

C.10 Proof of Proposition 4

Proposition 4. *There exists a unique equilibrium, and it is efficient. Furthermore, the equilibrium vector of effective returns to scale $\hat{\eta}$ maximizes GDP $y(\hat{\eta})$.*

Proof. This proof proceeds in two steps. First, we write down the maximization problem of the social planner and show that its first-order conditions coincide with the equilibrium conditions. Since there exists at least one maximizer to the planner's problem, there is at least one solution to the planner's first-order conditions and so at least one efficient equilibrium exists. Second, we show that the equilibrium conditions imply that there can be at most one equilibrium.

Step 1. The planner maximizes

$$\max_{C, X, L, M, \eta} \sum_{i=1}^N \beta_i \log(C_i)$$

subject to the goods resource constraint

$$C_i + \sum_{j=1}^N M_j \int X_{ji}(\varepsilon) f_j(\varepsilon) d\varepsilon \leq M_i \int Q_i(\varepsilon) f_i(\varepsilon) d\varepsilon \quad \forall i \in \{1, \dots, N\} \quad (\text{multiplier } \lambda_i),$$

and the labor resource constraint

$$\sum_{i=1}^N M_i \int L_i(\varepsilon) f_i(\varepsilon) d\varepsilon + \sum_{i=1}^N M_i \kappa_i \leq \bar{L} \quad (\text{multiplier } \mu).$$

The first-order necessary conditions are as follows:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial C_i} &: \lambda_i = \frac{\beta_i}{C_i}, \\
\frac{\partial \mathcal{L}}{\partial L_i(\varepsilon)} &: \lambda_i \frac{\partial Q_i(\varepsilon)}{\partial L_i(\varepsilon)} - \mu = 0, \\
\frac{\partial \mathcal{L}}{\partial X_{ij}(\varepsilon)} &: \lambda_i \frac{\partial Q_i(\varepsilon)}{\partial X_{ij}(\varepsilon)} - \lambda_j = 0, \\
\frac{\partial \mathcal{L}}{\partial \eta_i(\varepsilon)} &: \frac{\partial Q_i(\varepsilon)}{\partial \eta_i(\varepsilon)} = 0, \\
\frac{\partial \mathcal{L}}{\partial M_i} &: \int \left[\lambda_i Q_i(\varepsilon) - \sum_{j=1}^N \lambda_j X_{ij}(\varepsilon) - \mu L_i(\varepsilon) \right] f_i(\varepsilon) d\varepsilon = \mu \kappa_i.
\end{aligned}$$

Now, we demonstrate that the competitive equilibrium allocation satisfies the planner's optimality conditions. To do this, we identify the planner's shadow prices with the equilibrium market prices. Set $\mu = W/\bar{P}Y$. Consequently, the planner's shadow price for good i , λ_i , corresponds to the market price $P_i/\bar{P}Y$. The first condition corresponds to the household's optimality condition (Section 2.4). The second and third optimality conditions correspond to the standard firm equilibrium optimality conditions (58) and (59). The planner's fourth optimality condition coincides with the firm equilibrium condition (61). Finally, the last optimality condition of the planner coincide with the free entry condition (8). Since the resource constraints are the same in the planner's problem and the equilibrium definition, we have shown that the planner's first-order conditions coincides with the equilibrium conditions. Since the planner's constraint set is closed and bounded, and that the objective function is continuous, the Extreme Value Theorem implies that there exists a maximizer to the planner's problem. This maximizer must satisfy the first-order necessary conditions. Therefore, there exists an equilibrium and that equilibrium is efficient.

Step 2. We now show that there can be at most one equilibrium. The equilibrium of the model boils down to equations (17) and (19). Indeed, if we let $p := \log(P/W)$ we can write these equations as

$$p = -\mathcal{L}(\hat{\eta}) z(\hat{\eta}), \quad (73)$$

where

$$z_i(\hat{\eta}) := \mu_i + a_i(\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1 - \hat{\eta}_i} + (1 - \hat{\eta}_i) \log \left(\frac{1}{\sqrt{1 - \varphi_i}} \right) - (1 - \hat{\eta}_i) \log \kappa_i, \quad (74)$$

and

$$\frac{1}{1 - \hat{\eta}_i} = \frac{1}{2\gamma_i(1 - \varphi_i)} \left(\mu_i + p_i - \sum_{j=1}^N \alpha_{ij} p_j \right). \quad (75)$$

There is a unique equilibrium if there are unique vectors $\hat{\eta}$ and p that solve these equations. We can combine these equations into a single one. Let us introduce the variable $v := (I - \alpha)p$ and a constant

$C_i = 2\gamma_i(1 - \varphi_i) > 0$. We can then rewrite (75) as

$$1 - \hat{\eta}_i = \frac{C_i}{\mu_i + v_i} \Leftrightarrow \hat{\eta}_i = 1 - \frac{C_i}{\mu_i + v_i}.$$

We are interested in equilibrium of the form $0 < \hat{\eta}_i < 1$ for all i .⁶⁵ This implies that we can restrict the relevant domain of v to be

$$\mu_i + v_i > C_i.$$

Using that notation, we can simplify the equation (74) for z_i as

$$z_i = \frac{\mu_i - v_i}{2} + \frac{C_i}{\mu_i + v_i} \log(K_i/\kappa_i),$$

where $K_i := 1/\sqrt{1 - \varphi_i}$. Next, we can premultiply (73) by $\mathcal{L}(\hat{\eta})^{-1} = (I - \text{diag}(\hat{\eta})\alpha)$ to find

$$(I - \text{diag}(\hat{\eta})\alpha)(I - \alpha)^{-1}v = -z$$

or

$$(I + \text{diag}(1 - \hat{\eta})\alpha(I - \alpha)^{-1})v = -z.$$

Substituting the expression for z and $1 - \hat{\eta}$, we find

$$F_i(v) := \frac{1}{2}(\mu_i + v_i)^2 + C_i \left(\left(\alpha(I - \alpha)^{-1}v \right)_i + \log(K_i/\kappa_i) \right) = 0.$$

There is a unique equilibrium if there is a unique solution v to the equation $F(v) = 0$. Recall that $p = (I - \alpha)^{-1}v$. Then

$$\hat{F}_i(p) := F_i(v(p)) = \frac{1}{2}(\mu_i + ((I - \alpha)p)_i)^2 + C_i((\alpha p)_i + \log(K_i/\kappa_i)).$$

The Jacobian of $\hat{F}$ is

$$M_{ik}(p) = (\mu_i + ((I - \alpha)p)_i)(I - \alpha)_{ik} + C_i\alpha_{ik}.$$

In matrix form,

$$M(p) = \text{diag}(\mu + v(p))(I - \alpha) + \text{diag}(C)\alpha.$$

The diagonal elements of M are

$$M_{ii} = (\mu_i + v_i)(1 - \alpha_{ii}) + C_i\alpha_{ii} > 0,$$

⁶⁵It is straightforward to write sufficient conditions on the parameters so that the equilibrium is of that form. In particular, large μ lead to higher equilibrium $\hat{\eta}$.

which is positive given our domain restriction that $\mu_i + v_i > C_i$. For off-diagonal terms $i \neq k$,

$$M_{ik} = -(\mu_i + v_i) \alpha_{ik} + C_i \alpha_{ik} = \alpha_{ik} (C_i - (\mu_i + v_i)) \leq 0,$$

such that M is a Z -matrix. Further notice that

$$\sum_{k \neq i} |M_{ik}| = ((\mu_i + v_i) - C_i) \sum_{k \neq i} \alpha_{ik}.$$

For M to be strictly diagonally dominant, it must be that

$$(\mu_i + v_i) (1 - \alpha_{ii}) + C_i \alpha_{ii} > ((\mu_i + v_i) - C_i) \sum_{k \neq i} \alpha_{ik},$$

which we can reorganize as

$$\mu_i + v_i > -C_i \frac{\sum_k \alpha_{ik}}{1 - \sum_k \alpha_{ik}}.$$

This condition is true since $C_i > 0$ and $\mu_i + v_i > C_i$. Therefore, M is diagonally dominant. It follows that $M(p)$ is a non-singular M -matrix for every p . Since nonsingular M -matrices are a subset of P -matrices, $M(p)$ is also a P -matrix for every p . By the Gale and Nikaido (1965) theorem, $\hat{F}(p)$ is therefore injective and can have at most one solution $\hat{F}(p) = 0$. There is therefore a unique p that solves our original system of equations. From the vector p , it is straightforward to recover all other equilibrium quantities in a unique fashion. There is therefore a unique equilibrium and it is efficient. $\square$

C.11 Proof of Lemma 6

Lemma 6. *An increase in average productivity μ_j increases returns to scale in all downstream sectors, such that*

$$\frac{d\hat{\eta}_i}{d\mu_j} = \Psi_i^{-1} \mathcal{K}_{ij} \geq 0. \quad (22)$$

Furthermore, the impact of productivity dispersion σ_j^2 on $\hat{\eta}_i$ is given by

$$\frac{d\hat{\eta}_i}{d\sigma_j^2} = \Psi_i^{-1} \left(\mathcal{K}_{ij} \frac{\partial z_j}{\partial \sigma_j^2} - \mathbb{1}(i = j) \frac{\partial^2 z_i}{\partial \sigma_i^2 \partial \hat{\eta}_i} \right), \quad (23)$$

where

$$\frac{\partial z_j}{\partial \sigma_j^2} = \frac{1}{2(1 - \hat{\eta}_j)} + \frac{1 - \hat{\eta}_j}{4\gamma_j(1 - \varphi_j)} > 0, \text{ and } \frac{\partial}{\partial \sigma_i^2} \left(\frac{\partial z_i}{\partial \hat{\eta}_i} \right) = \frac{1}{2(1 - \hat{\eta}_i)^2} - \frac{1}{4\gamma_i(1 - \varphi_i)}.$$

In particular, $d\hat{\eta}_i/d\sigma_j^2 \geq 0$ for $i \neq j$.

Proof. This proof proceeds as follows. First we derive the first-order conditions of the social planner. Second, we write down the derivative of the first-order conditions with respect to $\hat{\eta}_i$. Third, we use this expression

together with the implicit function theorem to derive the impact of μ_j and σ_j^2 on $\hat{\eta}_i$.

First step. Let us first compute the first-order conditions of the planner's problem. Differentiating (20) with respect to $\hat{\eta}_i$ and setting that expression to zero implies that

$$\frac{dy}{d\hat{\eta}_i} = \beta^\top \frac{d\mathcal{L}}{d\hat{\eta}_i} z(\hat{\eta}) + [\omega(\hat{\eta})]^\top \frac{dz(\hat{\eta})}{d\hat{\eta}_i} = 0.$$

Computing the derivative of $z(\hat{\eta})$, we find

$$\left(\frac{dz(\hat{\eta})}{d\hat{\eta}_i} \right)_j = \begin{cases} \frac{da_i(\hat{\eta}_i)}{d\hat{\eta}_i} + \frac{\sigma_i^2}{2} \frac{1}{(1-\hat{\eta}_i)^2} + \frac{1}{2} \log(1-\varphi_i) + \log \kappa_i & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases}$$

Next, the derivative of the Leontief inverse yields

$$\begin{aligned} \frac{d\mathcal{L}}{d\hat{\eta}_i} &= \frac{d(1 - \text{diag}(\hat{\eta})\alpha)^{-1}}{d\hat{\eta}_i} = -(1 - \text{diag}(\hat{\eta})\alpha)^{-1} \left[\frac{d(1 - \text{diag}(\hat{\eta})\alpha)}{d\hat{\eta}_i} \right] (1 - \text{diag}(\hat{\eta})\alpha)^{-1}. \\ &= \mathcal{L} [1_i 1_i^\top \alpha] \mathcal{L} = \mathcal{L}_{\cdot i} \alpha_i^\top \mathcal{L}. \end{aligned}$$

Putting the pieces together, we have

$$\omega_i(\hat{\eta}) \alpha_i^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta}) + \omega_i(\hat{\eta}) \left[\frac{da_i(\hat{\eta}_i)}{d\hat{\eta}_i} + \frac{\sigma_i^2}{2} \frac{1}{(1-\hat{\eta}_i)^2} + \frac{1}{2} \log(1-\varphi_i) + \log \kappa_i \right] = 0.$$

Since Domar weights are positive, we can write that condition as

$$\mathcal{F}_i := \alpha_i^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta}) + \frac{da_i(\hat{\eta}_i)}{d\hat{\eta}_i} + \frac{\sigma_i^2}{2} \frac{1}{(1-\hat{\eta}_i)^2} + \frac{1}{2} \log(1-\varphi_i) + \log \kappa_i = 0, \quad (76)$$

where we have defined $\mathcal{F}_i$.

Second step. The implicit function theorem states that

$$\frac{d\hat{\eta}}{d\mu} = - \left[\frac{\partial \mathcal{F}}{\partial \hat{\eta}} \right]^{-1} \left[\frac{\partial \mathcal{F}}{\partial \mu} \right].$$

First, let us compute the Jacobian matrix $\partial \mathcal{F} / \partial \hat{\eta}$. Consider an off-diagonal element $k \neq i$

$$\begin{aligned} \frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_k} &= \frac{\partial}{\partial \hat{\eta}_k} (\alpha_i^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta})) = \alpha_i^\top \left(\frac{\partial \mathcal{L}}{\partial \hat{\eta}_k} \right) z + \alpha_i^\top \mathcal{L} \left(\frac{\partial z}{\partial \hat{\eta}_k} \right) \\ &= \alpha_i^\top \mathcal{L} \mathbf{1}_k \mathbf{1}_k^\top \alpha \mathcal{L} z + \alpha_i^\top \mathcal{L} \mathbf{1}_k \frac{\partial z_k}{\partial \hat{\eta}_k}. \end{aligned}$$

Factoring this expression gives

$$\frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_k} = (\alpha_i^\top \mathcal{L} \cdot k) \left[\alpha_k^\top \mathcal{L} z + \frac{\partial z_k}{\partial \hat{\eta}_k} \right] = 0,$$

where the last equality follows since the term in bracket is the first-order condition of the planner.

For a diagonal element,

$$\frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_i} = \frac{\partial}{\partial \hat{\eta}_i} \left(\alpha_i^\top \mathcal{L} z + \frac{\partial z_i}{\partial \hat{\eta}_i} \right).$$

Through the logic above, the first term is 0, so we need only focus on the second part

$$\frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_i} = \frac{\partial}{\partial \hat{\eta}_i} \frac{\partial z_i}{\partial \hat{\eta}_i} = \frac{d^2 a_i}{d \hat{\eta}_i^2} + \frac{\sigma_i^2}{(1 - \hat{\eta}_i)^3} = (1 - \varphi_i) \frac{d^2 a_i}{d \hat{\eta}_i^2}. \quad (77)$$

Third step. Next, we can compute the element (i, j) of the matrix $\frac{\partial \mathcal{F}}{\partial \mu}$, which is $\frac{\partial \mathcal{F}_i}{\partial \mu_j}$. The FOC for sector i is $\mathcal{F}_i = \alpha_i^\top \mathcal{L} z + \frac{\partial z_i}{\partial \hat{\eta}_i}$. The parameter μ_j only enters through the vector z , specifically through its j -th element. Therefore,

$$\frac{\partial z}{\partial \mu_j} = \mathbf{1}_j.$$

Thus,

$$\frac{\partial \mathcal{F}_i}{\partial \mu_j} = \frac{\partial}{\partial \mu_j} (\alpha_i^\top \mathcal{L} z) = \alpha_i^\top \mathcal{L} \left(\frac{\partial z}{\partial \mu_j} \right) = \alpha_i^\top \mathcal{L} \mathbf{1}_j,$$

which is simply the (i, j) -th element of the matrix $\alpha \mathcal{L}$. Putting the pieces together,

$$\frac{d \hat{\eta}}{d \mu} = - \left(\frac{\partial \mathcal{F}}{\partial \hat{\eta}} \right)^{-1} (\alpha \mathcal{L}) = - \frac{(\alpha \mathcal{L})_{ij}}{\frac{d^2 a_i}{d \hat{\eta}_i^2} + \frac{\sigma_i^2}{(1 - \hat{\eta}_i)^3}} = - \left((1 - \varphi_i) \frac{d^2 a_i}{d \hat{\eta}_i^2} \right)^{-1} (\alpha \mathcal{L})_{ij}.$$

Fourth step. We now turn to the impact of σ_j^2 . We use the implicit function theorem once more. Note that

$$\frac{\partial \mathcal{F}_i}{\partial \sigma_j^2} = \alpha_i^\top \mathcal{L} \frac{\partial z}{\partial \sigma_j^2} + \frac{\partial}{\partial \sigma_j^2} \left(\frac{\partial z_i}{\partial \hat{\eta}_i} \right).$$

The vector $\partial z / \partial \sigma_j^2$ is zero everywhere except for its j -th element

$$\frac{\partial z_j}{\partial \sigma_j^2} = \frac{\partial}{\partial \sigma_j^2} \left(\frac{\sigma_j^2}{2(1 - \hat{\eta}_j)} - \frac{1 - \hat{\eta}_j}{2} \log(1 - \varphi_j) \right) = \frac{1}{2(1 - \hat{\eta}_j)} + \frac{1 - \hat{\eta}_j}{4\gamma_j(1 - \varphi_j)} > 0.$$

Similarly, $\frac{\partial}{\partial \sigma_j^2} \left(\frac{\partial z_i}{\partial \hat{\eta}_i} \right)$ is zero whenever $i \neq j$. We can compute

$$\frac{\partial}{\partial \sigma_j^2} \left(\frac{\partial z_j}{\partial \hat{\eta}_j} \right) = \frac{\partial}{\partial \sigma_j^2} \left(\frac{\sigma_j^2}{2(1 - \hat{\eta}_j)^2} + \frac{1}{2} \log(1 - \varphi_j) \right) = \frac{1}{2(1 - \hat{\eta}_j)^2} - \frac{1}{4\gamma_j(1 - \varphi_j)}.$$

Putting the pieces together, we find the result. □

C.12 Proof of Proposition 5

Proposition 5. *The difference in log GDP between the baseline model and the fixed returns-to-scale economy is given by*

$$y - \tilde{y} = \sum_{i=1}^N \frac{1}{2} \omega_i (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right) > 0. \quad (26)$$

Proof. We first compute GDP in the fixed returns-to-scale economy (denoted by $\tilde{\cdot}$), in which all firms in sector i have the same returns to scale $\eta_{il} = \hat{\eta}_i$. The free-entry condition is $E[\tilde{\Pi}_{il}] = \kappa_i \tilde{W}$. Using the expression for profit (62), this condition becomes

$$\int_{-\infty}^{\infty} \exp \left(\frac{1}{1 - \hat{\eta}_i} \left(\log \tilde{P}_i + \varepsilon_{il} + a_i (\hat{\eta}_i) - \hat{\eta}_i \log \tilde{H}_i \right) \right) f(\varepsilon_{il}) d\varepsilon_{il} = \kappa_i \tilde{W}.$$

Solving the integral and following the same aggregation steps as in the baseline model (Proposition 2), but without the endogenous choice of returns to scale, this condition yields a sectoral productivity of

$$\tilde{z}_i = \mu_i + a_i (\hat{\eta}_i) + \frac{\sigma_i^2}{2(1 - \hat{\eta}_i)} - (1 - \hat{\eta}_i) \log \kappa_i.$$

Because $\hat{\eta}_{il}$ is fixed, the term related to the choice of scale and the resulting amplified selection (i.e., the fourth term on the right-hand side of (16)) is absent. Since the sectoral production function and cost shares are still governed by $\hat{\eta}_i$, the pricing equation is analogous to the baseline model: $\log(\tilde{P}/\tilde{W}) = -\mathcal{L}(\hat{\eta}) \tilde{z}$. Log GDP is therefore given by:

$$\tilde{y} = [\omega(\hat{\eta})]^\top \tilde{z}(\hat{\eta}) + \log \bar{L}.$$

Note that the Domar weights $\omega(\hat{\eta})$ are identical to the baseline model because the sectoral input shares are the same in both economies.

Recall from (16) and (20) that in the baseline model

$$y = [\omega(\hat{\eta})]^\top z(\hat{\eta}) + \log \bar{L},$$

where

$$z_i = \mu_i + a_i (\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1 - \hat{\eta}_i} + \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right) - (1 - \hat{\eta}_i) \log \kappa_i.$$

As a result,

$$y - \tilde{y} = [\omega(\hat{\eta})]^\top (z(\hat{\eta}) - \tilde{z}(\hat{\eta})). \quad (78)$$

The difference in the sectoral productivity vectors, $z - \tilde{z}$, is a vector where the i -th element is

$$z_i(\hat{\eta}_i) - \tilde{z}_i(\hat{\eta}_i) = \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right).$$

Substituting in (78) yields (26). The inequality $y - \tilde{y} > 0$ holds since $0 < \varphi_i < 1$ for all i . $\square$

C.13 Proof of Proposition 7

Proposition 7. *The response of log GDP y to a shock $\Delta\mu_i$ is given by*

$$\Delta y = \omega_i \Delta\mu_i + \frac{1}{2} \frac{d\omega_i}{d\mu_i} (\Delta\mu_i)^2 + o\left((\Delta\mu_i)^2\right). \quad (30)$$

Furthermore, the second-order term is non-negative,

$$\frac{d\omega_i}{d\mu_i} = \left(- \sum_{k=1}^N \mathcal{K}_{ki} \omega_k \frac{d\hat{\eta}_k}{d\mu_i} \right) \geq 0.$$

Proof. The second-order expansion of y with respect to productivity shocks is

$$\Delta y = \sum_{i=1}^N \frac{dy}{d\mu_i} \Delta\mu_i + \frac{1}{2} \sum_{i,j} \frac{d^2 y}{d\mu_i d\mu_j} \Delta\mu_i \Delta\mu_j + o\left(\Delta^2 \mu\right).$$

By Proposition 6, $dy/d\mu_i = \omega_i$ which yields (30) when $\Delta\mu_j = 0$ for all $j \neq i$. Next, Lemma 7 implies that

$$\frac{d\omega_i}{d\mu_i} = - \sum_{k=1}^N \mathcal{K}_{ki} \omega_k \frac{d\hat{\eta}_k}{d\mu_i} \geq 0,$$

where the inequality follows since $d\hat{\eta}_k/d\mu_i \geq 0$ from Lemma 6. $\square$

C.14 Proof of Lemma 8

Lemma 8. *An increase in τ_j^S decreases the returns to scale in all downstream sectors:*

$$\frac{d\hat{\eta}_i}{d\tau_j^S} = - \frac{1}{1 - \tau_j^S} \Psi_i^{-1} \mathcal{K}_{ij} \leq 0. \quad (32)$$

Proof. See proof of Proposition 12 in Appendix D.4. $\square$

C.15 Proof of Proposition 8

Proposition 8. *In the presence of sales wedges, the impact of a parameter $\chi \in \{\mu_j, \sigma_j^2, \kappa_j, \gamma_j\}$ on GDP is given by*

$$\frac{dy}{d\chi} = \frac{\partial y}{\partial \chi} + \sum_{i=1}^N \frac{\partial y}{\partial \hat{\eta}_i} \frac{d\hat{\eta}_i}{d\chi},$$

where $\partial y/\partial \chi$ is given by Proposition 6, $d\hat{\eta}_i/d\chi$ is given by Corollaries 6 to 11, and $\partial y/\partial \hat{\eta}_i \geq 0$.

Proof. See proof of Proposition 14 in Appendix D.4. $\square$

C.16 Proof of Corollary 1

Corollary 1. *The growth of effective returns to scale $\hat{\eta}$ is given by*

$$\frac{d\hat{\eta}}{dt} = \Psi^{-1} \mathcal{K} g_{\mu} > 0. \quad (34)$$

Furthermore, as $t \rightarrow \infty$, effective returns to scale $\hat{\eta}$ converges to 1.

Proof. The first equation follows directly from (22). Note that the right-hand side is strictly positive for $0 < \hat{\eta} < 1$ and converges to 0 as $\hat{\eta} \rightarrow 1$. The second result follows. $\square$

C.17 Proof of Proposition 9

Proposition 9. *The growth rate of GDP is given by*

$$\frac{dy}{dt} = \frac{g_{\mu}}{1 - \alpha} \times \left(1 - \frac{1}{\sqrt{\frac{1}{\gamma} \frac{1 - \alpha}{\alpha} \frac{g_{\mu}}{1 - \varphi} t + K}} \right) > 0, \quad (35)$$

where $K > 0$ is a time-invariant term given in the proof of the proposition.

Proof. The envelope theorem implies that

$$\frac{dy}{dt} = (1 - \hat{\eta}\alpha)^{-1} g_{\mu}. \quad (79)$$

Therefore, to characterize dy/dt , we need to solve for $\hat{\eta}(t)$. Equation (34) can be written as

$$\frac{d\hat{\eta}}{dt} = \frac{1}{2\gamma - \sigma^2} \frac{\alpha (1 - \hat{\eta})^3}{1 - \hat{\eta}\alpha} g_{\mu}$$

and reorganized as

$$\frac{\alpha g_{\mu}}{(\gamma - \frac{\sigma^2}{2}) 2} dt = \left(\frac{1 - \alpha}{(1 - \hat{\eta})^3} + \frac{\alpha}{(1 - \hat{\eta})^2} \right) d\hat{\eta}.$$

Integrating on both sides yields

$$\frac{\alpha g_{\mu}}{(\gamma - \frac{\sigma^2}{2}) 2} t + R = \frac{1 - \alpha}{2(1 - \hat{\eta})^2} + \frac{\alpha}{1 - \hat{\eta}}, \quad (80)$$

where R is a constant that can be pinned down using an initial condition. Suppose that at $t = 0$, the

equilibrium is such that $\hat{\eta} = \hat{\eta}_0$. Then,

$$R = \frac{1 - \alpha}{2(1 - \hat{\eta}_0)^2} + \frac{\alpha}{(1 - \hat{\eta}_0)} > 0.$$

Equation (80) provides the evolution of $\hat{\eta}$ over time. Since $\gamma > \sigma^2/2$ by assumption, it shows that $\hat{\eta} \rightarrow 1$ as $t \rightarrow \infty$. Combining (80) with (79) yields

$$\frac{dy}{dt} = \frac{g_\mu}{1 - \alpha} \times \left(1 - \frac{1}{\sqrt{1 + \frac{1}{\gamma} \frac{1 - \alpha}{\alpha} \left(\frac{g_\mu}{1 - \varphi} t - \frac{1 - \alpha}{\alpha} a'(\hat{\eta}_0) - 2a(\hat{\eta}_0) \right)}} \right) > 0.$$

This expression can be written as (35). □

C.18 Proof of Corollary 2

Corollary 2. *For any $t > 0$, GDP grows faster in the economy with endogenous returns to scale. In the limit as $t \rightarrow \infty$, the long-run growth rates satisfy*

$$\lim_{t \rightarrow \infty} \frac{dy}{dt} = \frac{1}{1 - \alpha} g_\mu > \frac{1}{1 - \hat{\eta}_0 \alpha} g_\mu = \lim_{t \rightarrow \infty} \frac{d\tilde{y}}{dt},$$

where $\tilde{y}$ is log GDP in the fixed returns-to-scale economy, and where $\hat{\eta}_0$ is effective returns to scale in the baseline economy at $t = 0$.

Proof. In the economy with exogenous returns to scale, (20) implies that $\frac{dy}{dt} = \frac{1}{1 - \alpha \eta_0} g_\mu$. In the economy with endogenous returns to scale, the envelope theorem implies, at any point in time we have $\frac{dy}{dt} = \frac{1}{1 - \alpha \eta(t)} g_\mu$. This implies that the two economies have the same growth rate at $t = 0$ since $\eta(t) = \eta_0$ by definition. But since $d\eta/dt > 0$ by Corollary 1, the growth rate of the economy with endogenous returns to scale is larger for any $t > 0$. The second part of the result follows from taking the limit $t \rightarrow \infty$ in (35). □

D Robustness, extensions, and additional analysis

In this appendix, we provide additional analysis of the benchmark model presented in the main text. We also show this model can be extended in different ways.

D.1 Truncated normal shocks

In the baseline model, we assume that productivity shocks ε_{it} follow a normal distribution. While this allows for a tractable analytical solution, it theoretically permits firms to draw arbitrarily low productivity shocks, which could imply returns to scale $\eta_{it} \notin (0, 1)$. In this appendix, we solve the model assuming that

productivity follows a *Truncated Normal* distribution. We show that the equilibrium conditions converge to those of the baseline model as the truncation point goes to negative infinity.

Specifically, productivity shocks of firms in industry i follow truncated normal distribution with support $[\underline{\varepsilon}_i, \infty)$. We assume that $\underline{\varepsilon}_i$ is sufficiently high such that

$$\underline{\varepsilon}_i > 2\gamma_i - s_i,$$

where s_i is given in (70). Under this restriction, all firms choose $\eta_{il} \in (0, 1)$, as is evident from (70). The analogue of the free-entry condition, given by (8) in the main text, is

$$\exp(-s_i) \int_{\underline{\varepsilon}_i}^{\infty} \exp\left\{\frac{\varepsilon_{il}^2 - s_i^2}{4\gamma_i} + \frac{s_i^2 + \varepsilon_{il}s_i}{2\gamma_i}\right\} \exp\left(-\frac{(\varepsilon_{il} - \mu_i)^2}{2\sigma_i^2}\right) \frac{1}{1 - \Phi\left(\frac{\underline{\varepsilon}_i - \mu_i}{\sigma_i}\right)} d\varepsilon_{il} = \kappa_i \frac{W}{\lambda_i}.$$

Simplifying this expression, we get

$$\frac{W}{\lambda_i} \kappa_i = \frac{W^{1-\sum_j \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}}}{\lambda_i} \sqrt{\frac{1}{1-\varphi_i}} \exp\left(\frac{1}{2} \frac{(s_i + \mu_i)^2}{2\gamma_i(1-\varphi_i)}\right) T_{1i}, \quad (81)$$

where, as above, $\varphi_i = \frac{\sigma_i^2}{2\gamma_i}$, and

$$T_{1i} = \frac{1 - \Phi\left(\frac{\sqrt{1-\varphi_i}}{\sigma_i} \left(\underline{\varepsilon}_i - \mu_i - \frac{\varphi_i}{1-\varphi_i} (\mu_i + s_i)\right)\right)}{1 - \Phi\left(\frac{1}{\sigma_i} (\underline{\varepsilon}_i - \mu_i)\right)}.$$

In the baseline model, $\underline{\varepsilon}_i = -\infty$, and $T_{1i} = 1$.

Next, following the same steps as in Appendix C.8, we can compute $\hat{\eta}_i$:

$$\hat{\eta}_i = \frac{\int_l \eta_{il} P_i Q_{il} dl}{\int_l P_i Q_{il} dl} = 1 - \frac{1 - \varphi_i}{\frac{s_i + \mu_i}{2\gamma_i} + (1 - \varphi_i) T_2} = 1 - \frac{(1 - \varphi_i)(1 - \eta_i(\mu_i))}{1 + (1 - \varphi_i)(1 - \eta_i(\mu_i)) T_{2i}},$$

where

$$T_{2i} = \frac{\sigma_i}{2\gamma_i} \frac{1}{\sqrt{1-\varphi_i}} \frac{\phi\left(\frac{\sqrt{1-\varphi_i}}{\sigma_i} \left(\underline{\varepsilon}_i - \mu_i - \frac{\varphi_i}{1-\varphi_i} (\mu_i + s_i)\right)\right)}{1 - \Phi\left(\frac{\sqrt{1-\varphi_i}}{\sigma_i} \left(\underline{\varepsilon}_i - \mu_i - \frac{\varphi_i}{1-\varphi_i} (\mu_i + s_i)\right)\right)}.$$

Clearly, if $\underline{\varepsilon}_i = -\infty$, then $T_{2i} = 0$, and we are back to the baseline model (see Equation (72)).

Finally, we can derive the analogue of (19). Following the same steps as in Appendix C.8, we get

$$\log \frac{P}{W} = - (I - [\text{diag}(\varphi) + (I - \text{diag}(\varphi)) \text{diag}(\eta(\mu))])^{-1} \times \left[\mu + a(\eta(\mu)) - (I - \text{diag}(\varphi)) (I - \text{diag}(\eta(\mu))) \left(\frac{1}{2} \log(1 - \varphi) + \log \kappa - \log T_1 \right) \right].$$

Again, if $\underline{\varepsilon}_i = -\infty$, we are back to the baseline model.

Clearly, if firms with $\varepsilon_{il} = \mu_i$ choose $\eta_{il} \in (0, 1)$, $\underline{\varepsilon}_i$ can be chosen such that $\underline{\varepsilon}_i < \mu_i$. Furthermore, if σ_i is sufficiently small, T_{1i} is arbitrarily close to one and T_{2i} is arbitrarily close to zero. In that case, the mass of firms choosing $\eta_{il} \notin (0, 1)$ in our baseline model is negligible, and the baseline economy is almost equivalent to the model with truncated normal shocks.

D.2 TFP shifter and sales distribution

In this appendix, we relax Assumption 1 and explore how the shape of the TFP shifter affects the sales distribution.

Assumption 2. *The TFP shifter function A_i satisfies*

$$\lim_{\eta \rightarrow 1} (1 - \eta)^{\chi_i} a_i(\eta) = -\gamma_i,$$

where $\chi_i > 0$ and $\gamma_i > 0$.

Assumption 1 is a special case with $\chi_i = 1$. Furthermore, Assumption 2 specifies the behavior of the TFP shifter only asymptotically, as $\eta \rightarrow 1$, while Assumption 1 specifies the full parametric form of the TFP shifter for all admissible values of η .

We start by proving a useful lemma.

Lemma 9. *Under Assumption 2, the following holds:*

$$\lim_{\eta \rightarrow 1} (1 - \eta)^{1+\chi_i} \frac{da_i}{d\eta} = -\gamma_i \chi_i.$$

Proof. Write $\tilde{a}_i(s) = -a_i(1 - s)$. Then $\tilde{a}_i(s)$ is positive, decreasing, convex around $s = 1$. For $h \in (0, 1)$, we have

$$\frac{\tilde{a}_i(s) - \tilde{a}_i(s(1 + h))}{sh} \leq -\frac{d\tilde{a}_i}{ds} \leq \frac{\tilde{a}_i(s(1 - h)) - \tilde{a}_i(s)}{sh}.$$

Multiplying by $s^{1+\chi_i}$ and using $\lim_{s \rightarrow 0} s^{\chi_i} \tilde{a}_i(s) = \gamma_i$, we see that the lower and upper bounds converge to, respectively, $\frac{\gamma_i(1 - (1+h)^{-\chi_i})}{h}$ and $\frac{\gamma_i((1-h)^{-\chi_i} - 1)}{h}$. In turn, as $h \rightarrow 0$, these two expressions converge to $\gamma_i \chi_i$. Therefore,

$$\lim_{s \rightarrow 0} s^{1+\chi_i} \frac{d\tilde{a}_i}{ds} = -\gamma_i \chi_i,$$

which is the result. □

We can now prove the tail result which extends Proposition 1.

Proposition 10. *Suppose that Assumption 2 holds. Without endogenous returns to scale, the distribution of Q_{il} in sector i is log-normal. With endogenous returns to scale, the right tail of the distribution of Q_{il}*

behaves as follows:

$$\log(\mathbb{P}(Q_{il} > q)) \sim -\frac{(1 + \chi_i)^2 \gamma_i^{\frac{2}{1+\chi_i}}}{2\sigma_i^2 \chi_i^{\frac{2\chi_i}{1+\chi_i}}} (\log q)^{\frac{2\chi_i}{1+\chi_i}}, \text{ as } q \rightarrow \infty.$$

Proof. Without endogenous returns to scale, the log of Q_{il} is given by (64). The only random term is ε_{il} and so Q_{il} is log-normal. We now turn to the case with endogenous returns to scale.

Step 1. We want to characterize the Complementary CDF (CCDF) $S_Q(q) = \mathbb{P}(Q > q)$ as $q \rightarrow \infty$. Denote $B_i = \log P_i - \log H_i$. To simplify the notation, we drop the subscripts i and l from now on.

Let g be the function that maps ε to $\log Q$ via (66),

$$g(x) = -\log(1 - \eta(x)) - \frac{da(\eta(x))}{d\eta} - B,$$

where $\eta(x)$ solves (65). By Lemma 4, g is strictly increasing and therefore invertible. We can write

$$S_Q(q) = \mathbb{P}(\log Q > \log q) = \mathbb{P}(g(\varepsilon) > \log q) = \mathbb{P}(\varepsilon > g^{-1}(\log q)).$$

The right tail of Q corresponds to the right tail of ε .

Step 2. Let $y = g(x)$. We need to characterize the asymptotic behavior of $x = g^{-1}(y)$ as $y \rightarrow \infty$. Letting $X = 1/(1 - \eta(x))$, Assumption 2 and Lemma 9 imply that the equation $y = g(x)$ can be written as $y \sim \gamma \chi X^{1+\chi}$ when $X \rightarrow \infty$. Furthermore, from (65), $x \sim \gamma(\chi + 1)X^\chi$. Therefore,

$$x \sim \frac{\gamma^{\frac{1}{1+\chi}} (\chi + 1)}{\chi^{\frac{\chi}{1+\chi}}} y^{\frac{\chi}{1+\chi}}. \quad (82)$$

Step 3. As in the proof of Proposition 1, we can show that

$$\log S_\varepsilon(x) \sim -\frac{x^2}{2\sigma^2}, \text{ as } x \rightarrow \infty.$$

Step 4. We now combine the results. Let $x_q = g^{-1}(\log q)$. From Step 1, $S_Q(q) = S_\varepsilon(x_q)$. From Step 3, for large q , and consequently large x_q , $\log S_Q(q) \sim -x_q^2/2\sigma^2$. We can now substitute the asymptotic form for x_q from Step 2. Let $y = \log q$. As $q \rightarrow \infty$, $y \rightarrow \infty$ and

$$x_q = g^{-1}(\log q) \sim \frac{\gamma^{\frac{1}{1+\chi}} (\chi + 1)}{\chi^{\frac{\chi}{1+\chi}}} (\log q)^{\frac{\chi}{1+\chi}}.$$

Therefore,

$$\log S_Q(q) \sim -\frac{1}{2\sigma^2} \frac{\gamma^{\frac{2}{1+\chi}} (\chi + 1)^2}{\chi^{\frac{2\chi}{1+\chi}}} (\log q)^{\frac{2\chi}{1+\chi}}, \text{ as } q \rightarrow \infty,$$

which is the result. $\square$

Proposition 10 shows that the thickness of the right tail of the sales distribution is governed by the shape of the TFP shifter, particularly its asymptotic behavior as $\eta \rightarrow 1$, summarized by χ_i . When $\chi_i = 1$, as in the baseline model, the right tail is Pareto. When $\chi_i > 1$, the productivity cost of adopting technologies with returns to scale close to one rises more steeply, and the right tail is thinner than Pareto but remains thicker than log-normal. Intuitively, as χ_i increases, highly productive firms choose returns to scale that approach one more slowly. This weakens the amplification generated by endogenous returns to scale and makes the sales distribution less heavy-tailed.

The shape of the TFP shifter also determines whether firm-level outcomes can be aggregated, in the sense that expected profits, sales, and input expenditures are finite. The following proposition shows that the model admits aggregation if $\chi_i > 1$ or if $\chi_i = 1$ and $\gamma_i > \sigma_i^2/2$. The latter condition is imposed in the baseline Assumption 1.

Proposition 11. *Suppose that Assumption 2 holds. Then $E_i \Pi_i(\varepsilon, P, W) < \infty$ for any given prices (P, W) if $\chi_i > 1$ or if $\chi_i = 1$ and $\gamma_i > \sigma_i^2/2$. The same conditions apply to sales and input expenditures.*

Proof. Whether $E_i \Pi_i(\varepsilon, P, W) < \infty$ depends on the right-tail behavior of Π_i . From (5) and the proof of Proposition 10 (see (82), in particular), we have

$$\log \Pi_i(\varepsilon) = \log H_i - \frac{da_i(\eta_{il}(\varepsilon))}{d\eta_{il}} \sim \frac{\chi_i}{\gamma_i^{\frac{1}{\chi_i}}(\chi_i + 1)^{\frac{1+\chi_i}{\chi_i}}} \varepsilon^{\frac{1+\chi_i}{\chi_i}}, \text{ as } \varepsilon \rightarrow \infty.$$

If $\chi_i > 1$, the exponent $(1 + \chi_i)/\chi_i < 2$, and so integrating Π_i with respect to the normal ε yields a finite value, $E_i \Pi_i < \infty$. At the same time, if $\chi_i < 1$, then $E_i \Pi_i = \infty$. If $\chi_i = 1$, write the product of $\Pi_i(\varepsilon)$ and the probability density function $f_i(\varepsilon)$ as

$$\log [\Pi_i(\varepsilon) f_i(\varepsilon)] \sim \frac{1}{4\gamma_i} \varepsilon^2 - \frac{(\varepsilon - \mu_i)^2}{2\sigma_i^2} \sim -\frac{1}{2\sigma_i^2} \varepsilon^2 \left(1 - \frac{\sigma_i^2}{2\gamma_i}\right).$$

Therefore, $E_i \Pi_i < \infty$ if $\sigma_i^2 < 2\gamma_i$ and $E_i \Pi_i = \infty$ if $\sigma_i^2 > 2\gamma_i$.⁶⁶

Turning to sales, we can write

$$\log P_i Q_{il} = \log \Pi_{il} - \log(1 - \eta_{il}).$$

But, from the proof of Proposition 10, we have $\log(1 - \eta_{il}) \sim -\frac{1}{\chi_i} \log \varepsilon$ and hence $\log P_i Q_{il} \sim \log \Pi_{il}$, and all the results derived above for profits also apply to sales. Finally, since input expenditures are fractions of sales, these results also apply to input expenditures. $\square$

⁶⁶If $\sigma_i^2 = 2\gamma_i$, whether expected profits are finite depends on lower-order terms in a , not specified in Assumption 2.

D.3 The impact of γ and κ on returns to scale

In this appendix, we characterize how entry costs κ and the cost of scalability γ affect returns to scale.

D.3.1 Entry cost

We examine the impact of entry costs on returns-to-scale decisions.

Lemma 10. *The impact of the entry cost κ_j on the effective returns to scale $\hat{\eta}_i$ is given by*

$$\frac{d\hat{\eta}_i}{d\log \kappa_j} = \Psi_i^{-1} \left[-\mathcal{K}_{ij} (1 - \hat{\eta}_j) - \mathbb{I}_{\{i=j\}} \right]. \quad (83)$$

In particular, $d\hat{\eta}_i/d\log \kappa_j \leq 0$ for $i \neq j$.

Proof. Applying the implicit function theorem to (76), we get

$$\frac{d\hat{\eta}}{d\log \kappa_j} = - \left[\frac{\partial \mathcal{F}}{\partial \hat{\eta}} \right]^{-1} \frac{\partial \mathcal{F}}{\partial \log \kappa_j}.$$

We have already computed the first term in the proof of Lemma 6, so consider the second one. We have

$$\frac{\partial \mathcal{F}_i}{\partial \log \kappa_j} = \alpha_i^\top \mathcal{L}(\hat{\eta}) \frac{\partial z(\hat{\eta})}{\partial \log \kappa_j} + \frac{\partial \log \kappa_i}{\partial \log \kappa_j} = -\alpha_i^\top \mathcal{L}(\hat{\eta}) \mathbf{1}_j (1 - \hat{\eta}_j) + \mathbb{1}(i = j).$$

Putting the pieces together we find the result. $\square$

An increase in the entry cost in sector j always reduces the effective returns to scale of any other sector $i \neq j$. The mechanism is similar to that of a shock to μ_j . Increasing κ_j decreases j 's productivity z_j , which increases the price of the input bundle of any sector that relies on j . Firms in those sectors then reduce their returns to scale to rely less on expensive intermediate inputs. At the same time, the effective returns to scale $\hat{\eta}_j$ of sector j itself typically increases with κ_j . This is because, when entry costs are large, there is more pressure to have fewer but larger firms, which requires large $\hat{\eta}_j$.

D.3.2 Cost of adjusting returns to scale

The productivity cost γ_i of adjusting returns to scale also affects firms' scalability decisions.

Lemma 11. *The impact of the productivity cost of higher returns to scale γ_j on the effective returns to scale $\hat{\eta}_i$ is given by*

$$\frac{d\hat{\eta}_i}{d\gamma_j} = \Psi_i^{-1} \left(\mathcal{K}_{ij} \frac{\partial z_j}{\partial \gamma_j} - \mathbb{I}_{\{i=j\}} \frac{\partial^2 z_i}{\partial \gamma_i \partial \hat{\eta}_i} \right) \quad (84)$$

where $\frac{\partial z_j}{\partial \gamma_j} = -\frac{1}{1-\hat{\eta}_j} - \frac{1}{2\gamma_j} \frac{\varphi_j}{1-\varphi_j} (1 - \hat{\eta}_j) < 0$ and $\frac{\partial^2 z_i}{\partial \gamma_i \partial \hat{\eta}_i} = -\frac{1}{(1-\hat{\eta}_i)^2} + \frac{1}{2\gamma_i} \frac{\varphi_i}{1-\varphi_i}$. In particular, $d\hat{\eta}_i/d\gamma_j \leq 0$ for $i \neq j$.

Proof. Applying the implicit function theorem to (76), we get

$$\frac{d\hat{\eta}}{d\gamma_j} = - \left[\frac{\partial \mathcal{F}}{\partial \hat{\eta}} \right]^{-1} \left[\frac{\partial \mathcal{F}}{\partial \gamma_j} \right].$$

We have already computed the first term in the proof of Lemma 6, so consider the second one. We have

$$\frac{\partial \mathcal{F}_i}{\partial \gamma_j} := \alpha_i^\top \mathcal{L}(\hat{\eta}) \frac{\partial z(\hat{\eta})}{\partial \gamma_j} + \frac{\partial^2 a_i(\hat{\eta}_i)}{\partial \gamma_j \partial \hat{\eta}_i} + \frac{1}{2} \frac{\partial}{\partial \gamma_j} \log(1 - \varphi_i).$$

If $i \neq j$,

$$\frac{\partial \mathcal{F}_i}{\partial \gamma_j} := -\mathcal{K}_{ij} \frac{\partial z_j}{\partial \gamma_j},$$

where

$$\frac{\partial z_j}{\partial \gamma_j} = -\frac{1}{1 - \hat{\eta}_j} - \frac{1}{2\gamma_j} \frac{\varphi_j}{1 - \varphi_j} (1 - \hat{\eta}_j) < 0.$$

For $i = j$, we have an extra term,

$$\frac{\partial \mathcal{F}_i}{\partial \gamma_i} := -\mathcal{K}_{ii} \frac{\partial z_i(\hat{\eta})}{\partial \gamma_i} - \frac{1}{(1 - \hat{\eta}_i)^2} + \frac{1}{2\gamma_i} \frac{\varphi_i}{1 - \varphi_i}.$$

□

Consider first the impact of a higher γ_j on the effective returns to scale of another sector $i \neq j$. Unsurprisingly, a higher productivity cost of adjusting returns to scale leads to a lower productivity in sector j . Through input-output linkages, that lower productivity increases the price of the intermediate input bundles of firms that rely, directly or indirectly, on j as an input ($\mathcal{L}_{ij} > 0$). Those firms, to limit the negative impact of higher inputs, lower their returns to scale. A similar impact is at work when considering the impact of a higher γ_j on j itself, but in addition, j is also affected more directly by the increase in γ_j . Indeed, a larger γ_j mechanically makes a high $\hat{\eta}_j$ more expensive, which amplifies the negative movement in $\hat{\eta}_j$. In general, these forces combine to create a stronger negative impact of γ_j on $\hat{\eta}_j$.

D.4 Wedges

In this appendix, we consider an economy with wedges. In the presence of wedges, the firm's problem (2) becomes

$$\Pi_{il} := \max_{\eta_{il}, L_{il}, X_{il}} (1 - \tau_i^S) P_i F_i(L_{il}, X_{il}, \eta_{il}) - (1 + \tau_i^L) W L_{il} - \sum_{j=1}^N (1 + \tau_{ij}^X) P_j X_{ij,l}. \quad (85)$$

Firms in sector i have to pay $(1 + \tau_{ij}^X) P_j$ for each unit of good j , $\tau_{ij}^X > -1$, and $(1 + \tau_i^L) W$ for each unit of labor, $\tau_i^L > -1$. Firms face an effective sales tax $\tau_i^S < 1$. Finally, we introduce a corporate tax rate τ_i^Π .

This tax does not directly affect the profit-maximization problem (85). However, it affects the free-entry condition (8):

$$E_i \left[(1 - \tau_i^\Pi) \Pi_i (\varepsilon_{il}, P^*, W^*) \right] = \kappa_i W^*.$$

As we can see, the profit tax effectively increases the entry cost.

Wedges $\{\tau^X, \tau^L, \tau^S, \tau^\Pi\}$ can capture a variety of economic factors, such as tariffs, transportation costs, taxes, markups, etc. Some of those wedges can be associated with loss of resources, while others only lead to resource redistribution. To capture this, we assume that a fraction of wedge income is rebated to the household, such that its budget constraint (7) becomes

$$\sum_{i=1}^N P_i C_i \leq W \bar{L} + \mathcal{T},$$

where

$$\mathcal{T} = \sum_{i=1}^N \theta_i^S \tau_i^S P_i Q_i + \sum_{i=1}^N \tau_i^L W L_i + \sum_{i=1}^N \sum_{j=1}^N \theta_{ij}^X \tau_{ij}^X P_j X_{ij} + \sum_{i=1}^N \tau_i^\Pi \Pi_i.$$

Here $\theta_i^S, \theta_{ij}^X \in [0, 1]$. Note that wedges $\{\tau^X, \tau^L, \tau^S, \tau^\Pi\}$ can be both positive or negative. For example, τ_{ij}^X is positive in case of transportation costs. If those are iceberg costs, nothing is rebated to the household, and $\theta_{ij}^X = 0$. Tariffs would also correspond to a positive τ_{ij}^X . Different from transportation costs, tariff income is likely partially rebated to the household, in which case θ_{ij}^X is positive. On the other hand, τ_{ij}^X would be negative in case of government subsidies. Such subsidies are financed by lump-sum taxation of the household, such that $\theta_{ij}^X = 1$.⁶⁷

The model can be analyzed analogously to our baseline model. In particular, we can derive that the equilibrium price vector is given by

$$\log \frac{P}{W} = -\mathcal{L}(\hat{\eta}) z(\hat{\eta}), \quad (86)$$

where, as in the baseline model, $\hat{\eta}$ is a vector of sales-weighted average returns to scale, $\mathcal{L}(\hat{\eta}) = (I - \text{diag}(\hat{\eta}) \alpha)^{-1}$, and

$$z_i(\hat{\eta}_i) = \mu_i - T_i + a_i(\hat{\eta}_i) + \frac{\sigma_i^2}{2} \frac{1}{1 - \hat{\eta}_i} + \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \varphi_i} \right) - (1 - \hat{\eta}_i) \log \frac{\kappa_i}{1 - \tau_i^\Pi}. \quad (87)$$

⁶⁷We restrict labor and profit wedge revenues to be fully rebated, $\theta_i^L = \theta_i^\Pi = 1$, to avoid introducing government demand. Allowing $\theta_i^L \neq 1$ or $\theta_i^\Pi \neq 1$ would mean that the government retains part of labor income or profits. For goods markets to clear, the retained revenue would have to be matched by explicit government spending or waste, which we do not model here. By contrast, allowing $\theta_{ij}^X \neq 1$ or $\theta_i^S \neq 1$ has a natural iceberg-cost interpretation in the resource constraint, as shown in equation (89) below.

The productivity shifter T_i is

$$T_i = T_i(\tau_i^L, \tau_i^S, \tau_i^X, \hat{\eta}_i) = \log \left[\frac{(1 + \tau_i^L)^{\hat{\eta}_i(1 - \sum_j \alpha_{ij})} \prod_{j=1}^N (1 + \tau_{ij}^X)^{\hat{\eta}_i \alpha_{ij}}}{1 - \tau_i^S} \right]. \quad (88)$$

Introducing wedges $\{\tau^X, \tau^L, \tau^S\}$ is, therefore, equivalent to a change in sectoral total factor productivities. An increase in wedges τ_i^L , τ_i^S or τ_{ij}^X reduces the effective productivity of sector i , resulting in a reduction in the returns to scale in all sectors. This result is analogous to the effect of a reduction in μ_i , described in Lemma 6. At the same time, an increase in the corporate tax τ_i^Π effectively increases the entry cost, and so its impact on returns to scale is analogous to that of $\log \kappa_i$, described in Lemma 10.

Proposition 12. *An increase in wedges $\{\tau^X, \tau^L, \tau^S\}$ reduces returns to scale in all sectors. An increase in the profit tax τ_i^Π reduces returns to scale in other sectors but can increase returns to scale in sector i .⁶⁸*

The market-clearing conditions (9) also change. Specifically, for good i , the resource constraint becomes

$$(1 - (1 - \theta_i^S) \tau_i^S) Q_i = C_i + \sum_{j=1}^N (1 + (1 - \theta_{ji}^X) \tau_{ji}^X) X_{ji}. \quad (89)$$

Then the Domar weight of sector i is

$$\tilde{\omega}_i = \frac{P_i Q_i}{PY} = \mathbf{1}_i (I - \text{diag}[(1 - \theta^S) \circ \tau^S] - \tilde{\alpha}^\top \text{diag}(\hat{\eta}))^{-1} \beta,$$

where $\circ$ denotes element-wise product of two vectors, and

$$\tilde{\alpha}_{ji} = \alpha_{ji} (1 - \tau_j^S) \left(\frac{1 + (1 - \theta_{ji}^X) \tau_{ji}^X}{1 + \tau_{ji}^X} \right).$$

Clearly, if $\tau_j^S \geq 0$ and $\tau_{ji}^X \geq 0$, then $\tilde{\alpha}_{ji} \leq \alpha_{ji}$.

Using these results, we can derive how wedges affect the expression for the aggregate output.

Proposition 13. *Equilibrium log GDP $y := \log Y$ is given by*

$$y(\hat{\eta}) = \underbrace{\beta^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta})}_{\text{Contribution of productivity}} + \underbrace{\log \bar{L}}_{\text{Labor endowment}} - \underbrace{\log \Gamma_\tau}_{\text{Wedges income}}, \quad (90)$$

⁶⁸We provide expressions for derivatives of returns to scale with respect to wedges in the proof of this proposition.

where

$$\Gamma_\tau = 1 - \sum_{i=1}^N \tilde{\omega}_i \left((1 - \tau_i^S) \hat{\eta}_i \left(\sum_{j=1}^N \alpha_{ij} \frac{\theta_{ij}^X \tau_{ij}^X}{1 + \tau_{ij}^X} + \left(1 - \sum_{j=1}^N \alpha_{ij} \right) \frac{\tau_i^L}{1 + \tau_i^L} \right) + \theta_i^S \tau_i^S + \tau_i^\Pi (1 - \tau_i^S) (1 - \hat{\eta}_i) \right).$$

As discussed above, some wedges can lead to a destruction of resources while others may lead to redistribution of resources. In the latter case, aggregate output needs to be adjusted for wedges income. This is the last term in expression (90). If all the wedges are nonnegative, then $\log \Gamma_\tau < 0$, which leads to a higher y .⁶⁹

The presence of wedges distorts the economy. Intuitively, firms do not internalize that part of the wedge income is rebated to the household, and their decisions are inefficient as a result. If none of the wedge income is rebated to the household, then $\log \Gamma_\tau = 0$, and the economy is efficient. In that case, firms correctly perceive wedges as resource-destructive. In the inefficient economy, the equilibrium returns to scale do not maximize GDP, and any marginal change in returns to scale can have a nontrivial impact on GDP. Specifically, a change in the underlying parameter χ leads to the following response of GDP:

$$\frac{dy}{d\chi} = \frac{\partial y}{\partial \chi} + \sum_{j=1}^N \frac{\partial y}{\partial \hat{\eta}_j} \frac{d\hat{\eta}_j}{d\chi}.$$

In general, the sign of the response of GDP to a marginal change in returns to scale, $\frac{\partial y}{\partial \hat{\eta}_j}$, depends on the sign of wedges. However, we can provide a sharp characterization in a few important special cases.

Proposition 14. *Suppose that there are no profit taxes, $\tau_i^\Pi = 0$, and all other wedges are positive, $\tau_{ij}^X > 0$, $\tau_i^L > 0$, and $\tau_i^S > 0$ for all i, j , and suppose that some of the wedge income is rebated to the household, $\log \Gamma_\tau < 0$. Then any marginal increase in the returns to scale leads to an increase in GDP, $\frac{\partial y}{\partial \hat{\eta}_j} > 0$.*

Consider first the case with no profit taxes. If other wedges are positive, the equilibrium returns to scale are too low (Proposition 12) as the firms do not internalize that part of the wedge income is rebated to the household. Then, any change in the parameter that leads to an increase in returns to scale is beneficial for GDP. For example, if the economy becomes more productive, as captured by a higher μ_j , the equilibrium returns to scale increase (Lemma 6).⁷⁰ Such a change has a positive impact on GDP because the equilibrium returns to scale were inefficiently low before the change.

Profit taxes affect the equilibrium returns to scale differently. As Proposition 12 suggests, an increase in the profit tax τ_i^Π is equivalent, from the firms' perspective, to an increase in the entry cost κ_i . Such an increase typically leads to a higher $\hat{\eta}_i$ (see our discussion following Lemma 10). Therefore, if profit taxes

⁶⁹Of course, in that case, sectoral productivities (87) are also lower than in the no-wedges economy.

⁷⁰If $\chi \in \{\mu_j, \sigma_j^2, \kappa_j, \gamma_j\}$, it is straightforward to show that $\frac{\partial y}{\partial \chi}$ and $\frac{d\hat{\eta}_j}{d\chi}$ are given by the same expressions as in the baseline model.

are rebated to the household, equilibrium returns to scale tend to be inefficiently high as firms incorrectly perceive entry costs as being too high. In that case, any change in the parameter that leads to a further increase in returns to scale is harmful for GDP. Expression (95) in the proof of Proposition 14 provides an exact expression for $\frac{\partial y}{\partial \hat{\eta}_j}$ in that case.

D.4.1 Proof of Proposition 12

Proposition 12. *An increase in wedges $\{\tau^X, \tau^L, \tau^S\}$ reduces returns to scale in all sectors. An increase in the profit tax τ_i^Π reduces returns to scale in other sectors but can increase returns to scale in sector i .*

Proof. Taking first-order conditions of (85) with respect to L_{il} and X_{il} , we can derive the following expression for log profit of firm l in sector i :

$$\begin{aligned} \log \Pi_{il} = & \log P_i + \log (1 - \tau_i^S) + \frac{a_i(\eta_{il}) + \eta_{il} \left(\log \frac{P_i}{W} - \sum_{j=1}^N \alpha_{ij} \log \frac{P_j}{W} \right)}{1 - \eta_{il}} \\ & + \frac{\varepsilon_{il} - \eta_{il} \left(\log \frac{(1 + \tau_i^L)^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N (1 + \tau_{ij}^X)^{\alpha_{ij}}}{(1 - \tau_i^S)} \right)}{1 - \eta_{il}}. \end{aligned}$$

Then the first-order condition with respect to η_{il} yields

$$\begin{aligned} \frac{d \log \Pi_{il}}{d \eta_{il}} = 0 \Leftrightarrow \\ \varepsilon_{il} - \log \left[\frac{(1 + \tau_i^L)^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N (1 + \tau_{ij}^X)^{\alpha_{ij}}}{(1 - \tau_i^S)} \right] + \log \frac{P_i}{W} - \sum_{j=1}^N \alpha_{ij} \log \frac{P_j}{W} + a_i(\eta_{il}) + (1 - \eta_{il}) \frac{da_i}{d \eta_{il}} = 0. \end{aligned} \quad (91)$$

Following the same steps as in the proof of Proposition 2, we can derive that the equilibrium price vector is given by (86), and the sales-weighted average of firm-level returns to scale $\hat{\eta}_i$ satisfies

$$\frac{1}{1 - \eta_{il}} = \frac{1 - \varphi_i}{1 - \hat{\eta}_i} + \frac{\varepsilon_{il} - \mu_i}{2\gamma_i}. \quad (92)$$

Plugging (86) and (92) into (91), we get the following equation for $\hat{\eta}$:

$$\begin{aligned} \mathcal{F}_i = & \frac{da_i(\hat{\eta}_i)}{d \hat{\eta}_i} + \frac{\sigma_i^2}{2} \frac{1}{(1 - \hat{\eta}_i)^2} + \alpha_i^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta}) + \frac{1}{2} \log (1 - \varphi_i) + \log \frac{\kappa_i}{1 - \tau_i^\Pi} \\ & - \log \left[(1 + \tau_i^L)^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N (1 + \tau_{ij}^X)^{\alpha_{ij}} \right] = 0. \end{aligned} \quad (93)$$

Denote by χ_i any of τ_{ij}^X , τ_i^L or τ_i^S . Then, by the implicit function theorem,

$$\frac{d\hat{\eta}}{d\chi_i} = - \left[\frac{\partial \mathcal{F}}{\partial \hat{\eta}} \right]^{-1} \left[\frac{\partial \mathcal{F}}{\partial \chi_i} \right].$$

As in the baseline model, we have

$$\frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_i} = (1 - \varphi_i) \frac{d^2 a_i}{d\hat{\eta}_i^2}$$

and $\frac{\partial \mathcal{F}_i}{\partial \hat{\eta}_j} = 0$ if $i \neq j$. Furthermore,

$$\frac{\partial \mathcal{F}_i}{\partial \chi_k} = \alpha_i^\top \mathcal{L}(\hat{\eta}) \frac{\partial z(\hat{\eta})}{\partial \chi_k} - \frac{\partial \log \left[(1 + \tau_i^L)^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N (1 + \tau_{ij}^X)^{\alpha_{ij}} \right]}{\partial \chi_k}.$$

From (87), it is clear that $\frac{\partial z_k}{\partial \chi_k} < 0$. Therefore, $\frac{d\hat{\eta}}{d\chi_i} \leq 0$. In particular, we have

$$\frac{d\hat{\eta}_i}{d\tau_j^S} = - \frac{1}{1 - \tau_j^S} \frac{d\hat{\eta}_i}{d \log(1 - \tau_j^S)} = - \frac{1}{1 - \tau_j^S} \left[(1 - \varphi_i) \frac{d^2 a_i}{d\hat{\eta}_i^2} \right]^{-1} \mathcal{K}_{ij}.$$

□

D.4.2 Proof of Proposition 13

Proposition 13. *Equilibrium log GDP $y := \log Y$ is given by*

$$y(\hat{\eta}) = \underbrace{\beta^\top \mathcal{L}(\hat{\eta}) z(\hat{\eta})}_{\text{Contribution of productivity}} + \underbrace{\log \bar{L}}_{\text{Labor endowment}} - \underbrace{\log \Gamma_\tau}_{\text{Wedges income}},$$

where

$$\Gamma_\tau = 1 - \sum_{i=1}^N \tilde{\omega}_i \left((1 - \tau_i^S) \hat{\eta}_i \left(\sum_{j=1}^N \alpha_{ij} \frac{\theta_{ij}^X \tau_{ij}^X}{1 + \tau_{ij}^X} + \left(1 - \sum_{j=1}^N \alpha_{ij} \right) \frac{\tau_i^L}{1 + \tau_i^L} \right) + \theta_i^S \tau_i^S + \tau_i^\Pi (1 - \tau_i^S) (1 - \hat{\eta}_i) \right).$$

Proof. From the household's budget constraint, we have

$$Y = W\bar{L} + \mathcal{T}, \tag{94}$$

where

$$\begin{aligned}
\mathcal{T} &= \sum_{i=1}^N \left(\sum_{j=1}^N \theta_{ij}^X \tau_{ij}^X P_j X_{ij} + \tau_i^L W L_i + \theta_i^S \tau_i^S P_i Q_i + \tau_i^\Pi \Pi_i \right) \\
&= \sum_{i=1}^N \left(\sum_{j=1}^N \hat{\eta}_i \alpha_{ij} \frac{\theta_{ij}^X \tau_{ij}^X}{1 + \tau_{ij}^X} (1 - \tau_i^S) P_i Q_i + \hat{\eta}_i \left(1 - \sum_{j=1}^N \alpha_{ij} \right) \frac{\tau_i^L}{1 + \tau_i^L} (1 - \tau_i^S) P_i Q_i \right. \\
&\quad \left. + \theta_i^S \tau_i^S P_i Q_i + \tau_i^\Pi (1 - \tau_i^S) (1 - \hat{\eta}_i) P_i Q_i \right) \\
&= Y \sum_{i=1}^N \tilde{\omega}_i \left((1 - \tau_i^S) \hat{\eta}_i \left(\sum_{j=1}^N \frac{\alpha_{ij} \theta_{ij}^X \tau_{ij}^X}{1 + \tau_{ij}^X} + \frac{(1 - \sum_{j=1}^N \alpha_{ij}) \tau_i^L}{1 + \tau_i^L} \right) + \theta_i^S \tau_i^S + \tau_i^\Pi (1 - \tau_i^S) (1 - \hat{\eta}_i) \right).
\end{aligned}$$

Plugging this into (94) gives the result. $\square$

D.4.3 Proof of Proposition 14

Proposition 14. *Suppose that there are no profit taxes, $\tau_i^\Pi = 0$, and all other wedges are positive, $\tau_{ij}^X > 0$, $\tau_i^L > 0$, and $\tau_i^S > 0$ for all i, j , and suppose that some of the wedge income is rebated to the household, $\log \Gamma_\tau < 0$. Then any marginal increase in the returns to scale leads to an increase in GDP, $\frac{\partial y}{\partial \hat{\eta}_j} > 0$.*

Proof. Differentiating y , given by (90), with respect to $\hat{\eta}_i$ and using the first-order condition (93), we get

$$\frac{\partial y}{\partial \hat{\eta}_i} = - \frac{\partial \log \Gamma_\tau}{\partial \hat{\eta}_i} = \frac{\text{Num}_i}{\Gamma_\tau},$$

where the numerator is

$$\begin{aligned}
\text{Num}_i &= \tilde{\omega}_i (1 - \tau_i^S) \left(\sum_{j=1}^N \alpha_{ij} \frac{\theta_{ij}^X \tau_{ij}^X}{1 + \tau_{ij}^X} + \left(1 - \sum_{j=1}^N \alpha_{ij} \right) \frac{\tau_i^L}{1 + \tau_i^L} - \tau_i^\Pi \right) \\
&\quad + \sum_{k=1}^N \frac{d\tilde{\omega}_k}{d\hat{\eta}_i} \left((1 - \tau_k^S) \hat{\eta}_k \left(\sum_{j=1}^N \alpha_{kj} \frac{\theta_{kj}^X \tau_{kj}^X}{1 + \tau_{kj}^X} + \left(1 - \sum_{j=1}^N \alpha_{kj} \right) \frac{\tau_k^L}{1 + \tau_k^L} \right) + \theta_k^S \tau_k^S + \tau_k^\Pi (1 - \tau_k^S) (1 - \hat{\eta}_k) \right).
\end{aligned}$$

The derivative of the Domar weights is given by

$$\frac{d\tilde{\omega}_k}{d\hat{\eta}_i} = \sum_{j=1}^N \tilde{\alpha}_{ij} (I - \text{diag}[(1 - \theta^S) \circ \tau^S] - \text{diag}(\hat{\eta}) \tilde{\alpha})_{jk}^{-1} \tilde{\omega}_i > 0.$$

Therefore, if taxes are positive ($\tau^X, \tau^L, \tau^S > 0$) but there is no profit tax ($\tau^\Pi = 0$), all terms in the numerator are positive (assuming some rebates $\theta > 0$), implying $\frac{\partial y}{\partial \hat{\eta}_i} > 0$. $\square$

In contrast, if $\tau^X = \tau^L = \tau^S = 0$ and $\tau^\Pi > 0$ with $\theta^\Pi = 1$, then $\tilde{\omega} = \omega$, and we get

$$\frac{\partial y}{\partial \hat{\eta}_i} = \omega_i \frac{-\tau_i^\Pi + \sum_{k=1}^N \left(\sum_{j=1}^N \alpha_{ij} \mathcal{L}_{jk} \right) \tau_k^\Pi (1 - \hat{\eta}_k)}{1 - \sum_{i=1}^N \omega_i \tau_i^\Pi (1 - \hat{\eta}_i)}. \quad (95)$$

As $\hat{\eta}_k \rightarrow 1$ for all k , the term $(1 - \hat{\eta}_k)$ vanishes, leaving only the negative term $-\omega_i \tau_i^\Pi$. Thus, $\frac{\partial y}{\partial \hat{\eta}_i}$ becomes negative.

D.5 Sales wedge correlated with productivity

In this appendix, we consider an economy in which firms face sales tax (38). The firm's problem (2) becomes

$$\Pi_{il} := \max_{\eta_{il}, L_{il}, X_{il}} (1 - \tau_{il}^S) P_i F_i(L_{il}, X_{il}, \eta_{il}) - W L_{il} - \sum_{j=1}^N P_j X_{ij,l},$$

where $F_i(L_{il}, X_{il}, \eta_{il})$ is given by (1). Clearly, this problem is equivalent to the one in the main text if we redefine the productivity as

$$\tilde{\varepsilon}_{il} = \varepsilon_{il} + \log(1 - \tau_{il}^S) = (1 - b_i)(\varepsilon_{il} - \mu_i) + \mu_i + \log(1 - \tau_i^S),$$

such that $\tilde{\varepsilon}_{il} \sim \text{iid } \mathcal{N}(\tilde{\mu}_i, \tilde{\sigma}_i^2)$, where

$$\tilde{\mu}_i = \mu_i + \log(1 - \tau_i^S) \quad \text{and} \quad \tilde{\sigma}_i = (1 - b_i) \sigma_i.$$

Similar to the baseline model, define

$$\hat{\eta}_i := \int_0^{M_i} \frac{\tilde{\omega}_{il}}{\tilde{\omega}_i} \eta_{il} dl, \quad (96)$$

with $\tilde{\omega}_{il} := (1 - \tau_{il}) \omega_{il}$ and $\tilde{\omega}_i := (1 - \hat{\tau}_i^S) \omega_i$, where

$$\hat{\tau}_i^S = \int_0^{M_i} \frac{\omega_{il}}{\omega_i} \tau_{il}^S dl. \quad (97)$$

As in the baseline model, we get

$$\frac{1}{1 - \hat{\eta}_i} = \frac{\tilde{\mu}_i + s_i}{2\gamma_i(1 - \tilde{\varphi}_i)},$$

where $s_i = \log P_i - \log H_i$. Integrating (97), we get

$$\frac{1}{1 - \hat{\tau}_i^S} = \frac{1}{1 - \tau_i^S} \left(1 + \frac{\tilde{\varphi}_i}{\frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i}} \frac{b_i}{1 - b_i} \right) \exp \left(\frac{b_i}{1 - b_i} \frac{4\tilde{\varphi}_i \gamma_i \frac{1 - \tilde{\varphi}_i}{1 - \hat{\eta}_i} + \tilde{\sigma}_i^2 \frac{b_i}{1 - b_i}}{2} \frac{1}{(1 - \tilde{\varphi}_i)} \right). \quad (98)$$

Then, following the same steps as in the main model, we can derive

$$\log W = \beta^\top (I - \text{diag}(\hat{\eta}) \alpha)^{-1} \tilde{z}(\hat{\eta}),$$

where

$$\tilde{z}_i(\hat{\eta}_i) = \tilde{\mu}_i + a_i(\hat{\eta}_i) + \frac{\tilde{\sigma}_i^2}{2} \frac{1}{1 - \hat{\eta}_i} + \frac{1}{2} (1 - \hat{\eta}_i) \log \left(\frac{1}{1 - \tilde{\varphi}_i} \right) - (1 - \hat{\eta}_i) \log \kappa_i$$

and $\tilde{\varphi}_i = \frac{\tilde{\sigma}_i^2}{2\gamma_i}$.

We assume that all tax proceeds are rebated to the household. Therefore, using the market clearing condition (9), we get

$$\omega_i := \frac{P_i Q_i}{PY} = \beta^\top (I - \text{diag}(1 - \hat{\tau}^S) \text{diag}(\hat{\eta}) \alpha)^{-1} \mathbf{1}_i.$$

The rebate amount is

$$\mathcal{T} = \sum_{i=1}^N \int_0^{M_i} \tau_{il}^S P_i Q_{il} dl = \sum_{i=1}^N \hat{\tau}_i^S \omega_i \bar{P} Y.$$

GDP is then

$$C = W \bar{L} + \mathcal{T} \Leftrightarrow \log C = \log W + \log \bar{L} - \log \left(1 - \sum_{i=1}^N \hat{\tau}_i^S \omega_i \right).$$

D.6 Dispersed returns-to-scale economy

In this appendix, we consider the dispersed returns-to-scale economy. Specifically, consider the initial economy (we will use subscripts b to mark any quantities in that economy). From (10), firm l in sector i chooses the following returns to scale:

$$\frac{1}{1 - \eta_{il}^b} = \frac{1}{2\gamma_i} (\varepsilon_{il}^b + s_i^b), \quad (99)$$

where $s_i^b = \log P_i^b - \log H_i^b$ in the initial economy. Furthermore, from (17), we know that

$$\frac{1}{1 - \hat{\eta}_i^b} = \frac{1}{2\gamma_i (1 - \varphi_i^b)} (\mu_i^b + s_i^b), \quad (100)$$

where $\varphi_i^b = \frac{(\sigma_i^b)^2}{2\gamma_i}$.

Suppose now that there is a change in the distribution of ε_{il}^b , such that the mean changes from μ_i^b to μ_i , and the standard deviation changes from σ_i^b to σ_i . Such a change can reflect an increase in μ for all sectors (Section 7.2) or removal of sales tax (Section 7.3). Then, productivity ε_{il}^b shifts to ε_{il} , where

$$\frac{\varepsilon_{il}^b - \mu_i^b}{\sigma_i^b} = \frac{\varepsilon_{il} - \mu_i}{\sigma_i}.$$

In the dispersed economy, firms can adjust all their choices except returns to scale. The free-entry condition (8)

$$\underbrace{\int_{-\infty}^{\infty} \exp \left(\log P_i + \frac{\varepsilon_{il} + a_i (\eta_{il}^b) + \eta_{il}^b s_i}{1 - \eta_{il}^b} \right) f_i(\varepsilon_{il}) d\varepsilon_{il}}_{=\Pi_{il}(\varepsilon_{il}, \eta_{il}^b)} = \kappa_i W,$$

where $s_i = \log P_i - \log H_i$, and η_{il}^b is given by (99). Taking this integral, we get

$$\exp \left(\frac{\left[(\mu_i^b + s_i^b) \left(1 - \varphi_i^b \frac{\sigma_i}{\sigma_i^b} \right) + \varphi_i^b (\mu_i + s_i) \right]^2 \frac{1}{1 - \varphi_i^b \left(2 \frac{\sigma_i}{\sigma_i^b} - 1 \right)} - (\mu_i^b + s_i^b)^2}{2 (\sigma_i^b)^2} \right) \times \quad (101)$$

$$\exp \left(\sum_{j=1}^N \alpha_{ij} \log \frac{P_j}{W} \right) \frac{1}{\sqrt{1 - \varphi_i^b \left(2 \frac{\sigma_i}{\sigma_i^b} - 1 \right)}} = \kappa_i.$$

Next, we can define $\hat{\eta}_i$ in the same way as usual, $\hat{\eta}_i = \int_0^{M_i} \frac{\omega_{il}}{\omega_i} \eta_{il}^b dl$, where again η_{il}^b is given by (99). Omitting tedious yet straightforward calculations, we get

$$\frac{1}{1 - \hat{\eta}_i} = \frac{1}{2\gamma_i \left(1 - \varphi_i^b \left(2 \frac{\sigma_i}{\sigma_i^b} - 1 \right) \right)} \left[(\mu_i^b + s_i^b) + \varphi_i^b \left(\mu_i + s_i - (\mu_i^b + s_i^b) \frac{\sigma_i}{\sigma_i^b} \right) \right]. \quad (102)$$

Combining (100), (101), and (102), we get

$$\log W = \beta^\top (I - \text{diag}(\hat{\eta})) \alpha)^{-1} z,$$

where

$$z_i = \mu_i - \left[\frac{1 - \varphi_i^b \left(2 \frac{\sigma_i}{\sigma_i^b} - 1 \right)}{1 - \hat{\eta}_i} - 2 \frac{1 - \varphi_i^b}{1 - \hat{\eta}_i^b} \left(1 - \varphi_i^b \frac{\sigma_i}{\sigma_i^b} \right) + (1 - \hat{\eta}_i) \left(\frac{1 - \varphi_i^b}{1 - \hat{\eta}_i^b} \right)^2 \right] \frac{\gamma_i}{\varphi_i^b}$$

$$- (1 - \hat{\eta}_i) \left[\frac{1}{2} \log \left(1 - \varphi_i^b \left(2 \frac{\sigma_i}{\sigma_i^b} - 1 \right) \right) + \log \kappa \right].$$

GDP is then $Y = W\bar{L}$.

E Dynamic extension and validation of returns-to-scale estimation

Our baseline model is static, whereas the returns-to-scale estimations in Section 6 and Appendix A.2 rely on panel variation. In this appendix, we extend the model to a dynamic setting. We show that the main results of the static model continue to hold in this setting. We further show that applying our estimation procedure to data simulated from the dynamic model accurately recovers the model's true returns to scale.

E.1 A dynamic model

We begin by presenting a dynamic extension of our model. Since many of the results of the static baseline model have direct analogs in the dynamic extension, we keep the discussion brief and do not repeat every derivation. Instead, we focus on the results needed to simulate the model and estimate returns to scale.

Time is discrete and infinite. In every period, each incumbent firm dies with probability $\delta \in (0, 1)$, independently of its characteristics, and the mass of new entrants is determined by free entry. A firm entering sector i pays κ_i^e units of labor. The discount factor is $\beta_d \in (0, 1)$, and there is no aggregate risk.

The log productivity of firm l in sector i in period t is

$$\varepsilon_{ilt} = \varepsilon_{il}^P + \varsigma_{ilt}, \text{ where } \varsigma_{ilt} = \rho_e \varsigma_{il,t-1} + e_{ilt}. \quad (103)$$

The permanent component $\varepsilon_{il}^P \sim \text{iid } \mathcal{N}(\mu_i, \sigma_{P,i}^2)$ is drawn once at entry and fixed for the firm's life. The transitory component ς_{ilt} has innovations $e_{ilt} \sim \text{iid } \mathcal{N}(0, \sigma_{e,i}^2 (1 - \rho_e^2))$, such that the stationary variance of ς_{ilt} is $\sigma_{e,i}^2$. Entrants draw ς_{ilt} from its stationary distribution.

Following Hsieh and Klenow (2009), we introduce firm-specific labor and intermediate-input wedges. As we explain below, the variation generated by these wedges allows us to identify the labor and intermediate-input elasticities. Specifically, firm l faces the effective wage $W \exp(\tau_{ilt}^L)$ and effective intermediate-input prices $P_j \exp(\tau_{ilt}^X)$, while its financial statements record expenditures at market prices W and P_j , respectively. The wedges can therefore be interpreted as firm-specific (shadow) taxes or subsidies on inputs, with the resulting revenue rebated lump-sum to the household.

The wedges follow

$$\tau_{ilt}^J = \rho_{J,i} \tau_{il,t-1}^J + \xi_{ilt}^J,$$

where $\xi_{ilt}^J \stackrel{\text{iid}}{\sim} N(0, (1 - \rho_{J,i}^2) \sigma_{J,i}^2)$ and $J \in \{L, X\}$.⁷¹ Entrants draw their wedges from the stationary distribution. Denoting the variable-input-bundle price as $H_i = W^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}}$ and letting $\bar{\alpha}_i := \sum_{j=1}^N \alpha_{ij}$, we can write the effective wedge on the variable-input bundle as $\bar{\tau}_{ilt} = (1 - \bar{\alpha}_i) \tau_{ilt}^L + \bar{\alpha}_i \tau_{ilt}^X$. Because

⁷¹We impose that the wedges are independent for simplicity, but our results do not require this assumption. The estimation procedure described below requires only that their innovations are not perfectly correlated.

the two wedges are independent, its stationary variance is

$$\sigma_{\tau,i}^2 = (1 - \bar{\alpha}_i)^2 \sigma_{L,i}^2 + \bar{\alpha}_i^2 \sigma_{X,i}^2.$$

Firm l chooses its returns to scale η_{il} at entry, after observing its permanent productivity ε_{il}^P but before observing the transitory productivity shocks and input wedges. We think of returns to scale as a slow-moving firm characteristic and, for tractability, assume that η_{il} remains fixed throughout the firm's life. A richer version of the model could allow returns to scale to adjust over time.⁷² Conditional on η_{il} , the firm's problem in each period is the same as in the baseline static model, evaluated at its current productivity and input prices adjusted for wedges.

The only intertemporal decision is the choice of returns to scale at entry. Because exit is independent of firm characteristics and entrants draw the transitory shocks and wedges from their stationary distributions, expected profits conditional on ε_{il}^P are the same in every period.⁷³ Therefore,

$$\max_{\eta_{il}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta_d^t (1 - \delta)^t \Pi_{ilt} \middle| \varepsilon_{il}^P \right] = \frac{1}{1 - \beta_d (1 - \delta)} \max_{\eta_{il}} \mathbb{E} [\Pi_{ilt} | \varepsilon_{il}^P]. \quad (104)$$

The technology choice therefore reduces to maximizing expected per-period profit. As in the baseline static model, this problem admits a closed-form solution.

Lemma 12. *Suppose that $\hat{\gamma}_i := \gamma_i - (\sigma_{e,i}^2 + \sigma_{\tau,i}^2) / 2 > 0$. At an interior solution, the optimal technology of firm l in sector i is given by*

$$\frac{1}{1 - \eta_{il}} = \frac{1}{2\hat{\gamma}_i} (\varepsilon_{il}^P + \log P_i - \log H_i - \sigma_{\tau,i}^2). \quad (105)$$

Proof. By (62), the log profit is given by

$$\log \Pi_{ilt} = \frac{\varepsilon_{il}^P + a_{ilt} - \bar{\tau}_{ilt} + \log P_i - \log H_i}{1 - \eta_{il}} + \bar{\tau}_{ilt} - \frac{\gamma_i}{(1 - \eta_{il})^2} + \log H_i,$$

where $\log H_i$ is adjusted for the effective wedge $\bar{\tau}_{ilt}$ and productivity is written as $\varepsilon_{ilt} = \varepsilon_{il}^P + a_{ilt}$.

⁷²Appendix E.6 considers such an extension, in which firms re-optimize their technology each period. In particular, we show there that the within-firm design of Section 6.3 recovers the true dynamics of returns to scale.

⁷³Conditional on ε_{il}^P , the shocks $(e_{ilt}, \tau_{ilt}^L, \tau_{ilt}^X)$ are drawn from their stationary laws at entry and evolve by their stationary transitions, so their joint distribution, and hence the distribution of Π_{ilt} , is the same at every age. Moreover, exit is independent of the firm's shocks, so conditioning on survival leaves the expectation unchanged; and prices are constant in the stationary equilibrium. Expected per-period profit $\mathbb{E} [\Pi_{ilt} | \varepsilon_{il}^P]$ is therefore a time-invariant function of $(\eta_{il}, \varepsilon_{il}^P)$. The net present value is that static expected profit times $\sum_{t=0}^{\infty} [\beta_d (1 - \delta)]^t$, regardless of how profits correlate across periods. As $\beta_d (1 - \delta) < 1$, the series converges.

Taking expectations over a_{ilt} and $\bar{\tau}_{ilt}$, we get

$$\log E [\Pi_{ilt} | \varepsilon_{il}^P] = \frac{\varepsilon_{il}^P + \log P_i - \log H_i - \sigma_{\bar{\tau},i}^2}{1 - \eta_{il}} - \frac{\hat{\gamma}_i}{(1 - \eta_{il})^2} + \log H_i + \frac{\sigma_{\bar{\tau},i}^2}{2},$$

where $\hat{\gamma}_i = \gamma_i - (\sigma_{e,i}^2 + \sigma_{\bar{\tau},i}^2)/2$. This objective is a strictly concave quadratic function of $1/(1 - \eta_{il})$ when $\hat{\gamma}_i > 0$. Setting its derivative with respect to $1/(1 - \eta_{il})$ to zero gives (105). $\square$

Equation (105) is the analogue of its baseline counterpart, equation (10), with ε_{il} replaced by the permanent productivity component ε_{il}^P , γ_i replaced by the effective curvature $\hat{\gamma}_i$, and an additional shift due to wedge variance. Three features of the η_{il} choice decision are worth emphasizing. First, the technology choice responds to permanent productivity but not to realizations of the transitory productivity shock or input wedges, which are not observed when η_{il} is chosen. Second, greater volatility in transitory productivity and input wedges raises the returns to scale chosen at entry by lowering $\hat{\gamma}_i$. Since the expected profit is convex in shocks to which the technology cannot respond, choosing a more scalable technology ex ante is more valuable in a more volatile environment. Finally, wedge volatility also generates the shift because input wedges directly affect the cost of the variable-input bundle.

The free-entry condition is analogous to its baseline counterpart (8),

$$\kappa_i W = E [\Pi_{ilt}], \quad (106)$$

where the entry cost is annualized, $\kappa_i = (1 - \beta_d (1 - \delta)) \kappa_i^e$.

One difference from the static model is that aggregate profits are not fully offset by entry costs because only entrants pay the entry cost. If M_i firms operate in sector i , stationarity requires a flow δM_i of new entrants each period. Aggregate net profit is therefore

$$\Pi_{\text{net}} := \sum_{i=1}^N (M_i E [\Pi_{ilt}] - \delta M_i \kappa_i^e W) = \frac{(1 - \beta_d)(1 - \delta)}{1 - \beta_d(1 - \delta)} \sum_{i=1}^N M_i E [\Pi_{ilt}] \geq 0,$$

where the second equality follows from the free-entry condition.

Household income therefore consists of labor income, net firm profits, and the net transfer generated by the input wedges:

$$\bar{P}Y = W\bar{L} + \Pi_{\text{net}} + \mathcal{T},$$

where

$$\mathcal{T} = \sum_{i=1}^N M_i E [\eta_{il} ((1 - \bar{\alpha}_i) (1 - \exp(-\tau_{ilt}^L)) + \bar{\alpha}_i (1 - \exp(-\tau_{ilt}^X))) P_i Q_{ilt}].$$

In sum, this dynamic setting can be analyzed along the same lines as our baseline static model. The only intertemporal decision is the choice of returns to scale at entry, and Lemma 12 shows that it takes a form analogous to the static optimality condition. Similarly, the free-entry condition (106) mirrors its static

counterpart, with the entry cost adjusted since firms expect to earn profits over multiple periods. We next derive the model-implied estimating equation and study the estimator's behavior in simulated panels.

E.2 Exact estimating equation and moment conditions

We now establish the firm-level estimating equation implied by the dynamic model and the validity of the associated population moment restrictions for our Blundell-Bond IV-GMM estimator described in Appendix A.2. The model delivers exact moment validity. Consistency of the pooled estimator additionally requires the within-cell (decile-sector unit) homogeneity and rank conditions stated below.

To match the estimation notation, let q_{ilt} denote log deflated sales and let l_{ilt} and m_{ilt} denote log labor and material inputs, constructed from the deflated wage bill and material expenditure. The first-order conditions for the two variable inputs imply

$$l_{ilt} = q_{ilt} + \log [\eta_{il} (1 - \bar{\alpha}_i)] - \tau_{ilt}^L, \quad m_{ilt} = q_{ilt} + \log [\eta_{il} \bar{\alpha}_i] - \tau_{ilt}^X, \quad (107)$$

up to sectoral price normalizations. The wedges thus move the inputs within a firm, independently of productivity. The following lemma gives the firm-level counterpart of the empirical specification.

Lemma 13. *In the dynamic firm-level environment described above, the sales and inputs of firm l in sector i satisfy, exactly, in every period,*

$$q_{ilt} = \beta_{il}^L l_{ilt} + \beta_{il}^M m_{ilt} + \kappa_{il} + \varsigma_{ilt}, \quad \varsigma_{ilt} = \rho_e \varsigma_{il,t-1} + e_{ilt}, \quad (108)$$

with $\beta_{il}^L = \eta_{il} (1 - \bar{\alpha}_i)$, $\beta_{il}^M = \eta_{il} \bar{\alpha}_i$, a firm fixed effect κ_{il} that collects the permanent type, the technology, and price constants, and an autoregressive error ς_{ilt} equal to the transitory productivity component of (103).

Proof. Define the technology-weighted input expenditure $B_{ilt} := \beta_{il}^L l_{ilt} + \beta_{il}^M m_{ilt}$. Equation (107) and $\beta_{il}^L + \beta_{il}^M = \eta_{il}$ imply

$$B_{ilt} = \eta_{il} q_{ilt} + c_{il} - \eta_{il} \bar{\tau}_{ilt},$$

where $c_{il} = \beta_{il}^L \log \beta_{il}^L + \beta_{il}^M \log \beta_{il}^M$ collects terms that are constant over the firm's life. Next, the profit-share identity (4) gives $q_{ilt} = \log \Pi_{ilt} + \log u_{il}$, with $u_{il} = 1/(1 - \eta_{il})$. Substituting realized log profit from the proof of Lemma 12 and dividing by u_{il} yields

$$(1 - \eta_{il}) q_{ilt} = \varepsilon_{il}^P + \varsigma_{ilt} - \bar{\tau}_{ilt} + (1 - \eta_{il}) \bar{\tau}_{ilt} + d_{il} = \varepsilon_{il}^P + \varsigma_{ilt} - \eta_{il} \bar{\tau}_{ilt} + d_{il},$$

where

$$d_{il} := \log P_i - \log H_i - \gamma_i u_{il} + (1 - \eta_{il}) (\log H_i + \log u_{il})$$

is time invariant. Finally, combining with $B_{ilt} = \eta_{il} q_{ilt} + c_{il} - \eta_{il} \bar{\tau}_{ilt}$ gives

$$q_{ilt} - B_{ilt} = (1 - \eta_{il}) q_{ilt} - c_{il} + \eta_{il} \bar{\tau}_{ilt} = \varsigma_{ilt} + \varepsilon_{il}^P + d_{il} - c_{il}.$$

The composite wedge therefore cancels exactly. Defining $\kappa_{il} := \varepsilon_{il}^P + d_{il} - c_{il}$ and substituting for B_{ilt} gives (108). $\square$

Lemma 13 maps the model directly into the two-input estimating equation. The two slope coefficients sum to the returns to scale, $\beta_{il}^L + \beta_{il}^M = \eta_{il}$, the parameter our procedure targets. The firm effect absorbs permanent firm characteristics, while transitory productivity follows an AR(1) process. In the model, sectoral prices are constant and the firm effect absorbs them; in the data they vary over time, so the empirical specification includes year effects, which absorb common price variation within each estimation cell. Quasi-differencing therefore yields (40) and its common-factor restrictions exactly, firm by firm. The next proposition states the corresponding population moment restrictions and identification conditions.

Proposition 15. *In the environment described above, the following holds:*

1. the difference moments (41) hold exactly, for any $\rho_e \in [0, 1)$;
2. the levels moments (42) hold exactly;
3. the common-factor restrictions imposed in the minimum-distance step of Appendix A.2 are exactly satisfied, and the estimated autoregressive root consistently estimates the shock persistence ρ_e ;
4. in a homogeneous cell, as long as wedges have nondegenerate input-specific variation, the population instrumented second-moment matrix has full column rank, the system GMM is consistent for $(\rho_e, \beta^L, \beta^M)$, with $\beta^L + \beta^M = \eta$.
5. as firms within a cell become technologically similar, the estimator consistently estimates their common returns to scale. Namely, suppose that there is a heterogeneity in firms' returns to scale within an estimation cell c , such that

$$\eta_{il} = \bar{\eta}_{ic} + \nu_{il}, \quad \bar{\eta}_{ic} := E_c[\eta_{il}], \quad E_c[\nu_{il}] = 0, \quad \nu_{ic}^2 := E_c[\nu_{il}^2],$$

where $\bar{\eta}_{ic}$ is the cell mean and ν_{ic}^2 is the within-cell variance of returns to scale. Under the uniform-rank and smoothness conditions (discussed in the proof), the estimated cell-specific returns to scale, $\hat{\eta}_{ic}$, differs from the cell mean only to the order of the within-cell variance,

$$\text{plim}_{N_{ic} \rightarrow \infty} \hat{\eta}_{ic} = \bar{\eta}_{ic} + O(\nu_{ic}^2).$$

Proof. The proof uses two distinct requirements. An instrument is valid if it is uncorrelated with the equation's current error. Identification additionally requires the instruments to predict the regressors in independent directions. Both sets of moments use the quasi-differenced representation (40), whose error consists of the transformed firm effect $\kappa_{il}^* := (1 - \rho_e) \kappa_{il}$ and the serially uncorrelated innovation e_{ilt} .

(i) First-differencing removes κ_{il}^* exactly, leaving $\Delta e_{ilt} = e_{ilt} - e_{il,t-1}$. A level of (q, l, m) at time $t - 2$ or earlier had been determined before e_{ilt} and $e_{il,t-1}$ were realized. By (103) and the independence of wedges

from productivity, it is therefore uncorrelated with both e_{ilt} and $e_{il,t-1}$, and hence with Δe_{ilt} . This is exactly the difference-moment condition in (i).

(ii) In the levels equation, an instrument $\Delta x_{il,t-1}$ must be uncorrelated with both components of the error. Because it uses changes dated $t-1$ and earlier, timing makes it uncorrelated with the new innovation e_{ilt} . Differencing also removes the permanent type and technology constants. Entrants draw their transitory states from the stationary distributions, so the expected change in each regressor is zero at every firm age, conditional on its permanent characteristics. It follows that $\Delta x_{il,t-1}$ is also uncorrelated with κ_{il}^* . This is the joint mean-stationarity condition of Blundell and Bond (2000). Here it follows from the model's stationary entry draws, so there is no initial-conditions violation. This proves (ii).

(iii) Equation (108) is the exact data-generating equation, and its only time-varying error component follows an AR(1). Substituting that AR(1) into (108) gives (40) and its common-factor restrictions. The autoregressive coefficient in that representation is therefore ρ_e , which proves (iii).

(iv) Valid moments alone do not identify the two input coefficients. If the wedges have no within-firm variation, (107) implies that labor and material expenditures each differ from sales only by a firm-specific constant. All three variables then have the same time variation. The instruments cannot determine how much of that common movement should be attributed to labor and how much to materials, so β^L and β^M are not separately identified; only ρ_e is identified. Separate identification therefore requires the population instrument-regressor matrix to have full column rank. As long as two input wedges are not perfectly correlated, (τ^L, τ^X) creates two independent directions of input variation, but the lagged instruments must also predict those directions. Under this rank condition and the orthogonality established in (i) and (ii), the population moments have a unique solution, and the sample GMM estimator converges to it as the number of firms grows. This proves (iv).

(v) We now distinguish the population effect of pooling heterogeneous firms from ordinary sampling error. Let y_{il} , X_{il} , and Z_{il} stack the transformed outcome, regressors, and instruments used by system GMM, and define

$$g_{ic}(\theta) := E_c [Z_{il}^\top (y_{il} - X_{il}\theta)], \quad Q_{ic}(\eta) := E_c [Z_{il}^\top X_{il} \mid \eta_{il} = \eta].$$

Thus, $g_{ic}(\theta)$ is the population GMM moment evaluated at a common coefficient vector θ , while $Q_{ic}(\eta)$ records the instrument-regressor moments for firms with returns to scale η . Maintain the orthogonality conditions from parts (i) and (ii) within cell c , so that each firm's true coefficient vector satisfies

$$E_c [Z_{il}^\top \{y_{il} - X_{il}\theta_i(\eta_{il})\}] = 0.$$

Write

$$\eta_{il} = \bar{\eta}_{ic} + \nu_{il}, \quad E_c [\nu_{il}] = 0, \quad \nu_{ic}^2 = E_c [\nu_{il}^2].$$

Because ρ_e and $\bar{\alpha}_i$ are common within the cell, every coefficient in the dynamic equation is either constant

(ρ_e) or proportional to η_{il} (the output elasticities). Specifically,

$$\theta_{il}^\top = c_i + b_i \eta_{il} = \begin{pmatrix} \rho_e & 0 & 0 & 0 & 0 \end{pmatrix} + \eta_{il} \begin{pmatrix} 0 & 1 - \bar{\alpha}_i & -\rho_e(1 - \bar{\alpha}_i) & \bar{\alpha}_i & -\rho_e \bar{\alpha}_i \end{pmatrix}.$$

It follows that

$$\theta_i(\eta_{il}) - \theta_i(\bar{\eta}_{ic}) = b_i(\eta_{il} - \bar{\eta}_{ic}) = b_i \nu_{il}.$$

We can now evaluate the population moment at the coefficients implied by the arithmetic cell mean:

$$\begin{aligned} g_{ic}(\theta_i(\bar{\eta}_{ic})) &= \mathbb{E}_c[Z_{il}^\top \{y_{il} - X_{il}\theta_i(\bar{\eta}_{ic})\}] \\ &= \underbrace{\mathbb{E}_c[Z_{il}^\top \{y_{il} - X_{il}\theta_i(\eta_{il})\}]}_{=0} + \mathbb{E}_c[Z_{il}^\top X_{il} b_i \nu_{il}] \\ &= \mathbb{E}_c[Q_{ic}(\eta_{il}) \nu_{il}] b_i. \end{aligned}$$

This expression makes the source of the population discrepancy explicit. If $Q_{ic}(\eta)$ were constant across firms, then

$$g_{ic}(\theta_i(\bar{\eta}_{ic})) = Q_{ic}(\bar{\eta}_{ic}) b_i \mathbb{E}_c[\nu_{il}] = 0.$$

Under the rank condition, the coefficients associated with the arithmetic cell mean would then be the common-coefficient population target exactly. In the model, however, $Q_{ic}(\eta)$ may vary with η because returns to scale affect the loadings of productivity and wedge shocks on sales and inputs. Smoothness is sufficient to bound the resulting discrepancy. To see this, add and subtract $Q_{ic}(\bar{\eta}_{ic})$:

$$\begin{aligned} g_{ic}(\theta_i(\bar{\eta}_{ic})) &= \mathbb{E}_c[\{Q_{ic}(\eta_{il}) - Q_{ic}(\bar{\eta}_{ic})\} \nu_{il}] b_i \\ &\quad + Q_{ic}(\bar{\eta}_{ic}) b_i \mathbb{E}_c[\nu_{il}] \\ &= \mathbb{E}_c[\{Q_{ic}(\eta_{il}) - Q_{ic}(\bar{\eta}_{ic})\} \nu_{il}] b_i. \end{aligned}$$

The second term vanishes because $\mathbb{E}_c[\nu_{il}] = 0$. This is the central cancellation: positive and negative deviations from the cell mean cancel to first order. Suppose $Q_{ic}(\eta)$ has a uniformly bounded derivative on the relevant compact subset of $\eta_{il} \in (0, 1)$. Then, for some finite L_{ic} ,

$$\|Q_{ic}(\eta_{il}) - Q_{ic}(\bar{\eta}_{ic})\| \leq L_{ic} |\nu_{il}|.$$

Therefore,

$$\|g_{ic}(\theta_i(\bar{\eta}_{ic}))\| \leq L_{ic} \|b_i\| \mathbb{E}_c[\nu_{il}^2] = C_{ic} \nu_{ic}^2,$$

where $C_{ic} := L_{ic} \|b_i\|$. Thus, the population moments evaluated at the arithmetic cell mean differ from zero only to the order of the within-cell variance. Finally, with fixed T , sample moments converge to their

population counterparts as $N_{ic} \rightarrow \infty$, implying

$$\text{plim}_{N_{ic} \rightarrow \infty} \hat{\eta}_{ic} = \bar{\eta}_{ic} + O(\nu_{ic}^2).$$

□

Part (v) justifies estimating one common coefficient for each sector-by-size-decile cell without requiring firms in the same cell to have identical returns to scale. Provided that the moment restrictions hold within the cell, the pooled population estimate differs from the arithmetic mean by only $O(\nu_{ic}^2)$, where ν_{ic} measures within-cell heterogeneity. With T fixed, sampling error also disappears as N_{ic} increases. Sorting firms into size deciles therefore replaces exact homogeneity with a local approximation. Namely, when within-cell dispersion is small, the pooled estimate is close to the cell's mean technology, and the approximation error tends to zero as that dispersion shrinks. The Monte Carlo exercise below uses the same rolling-window sector-decile grouping as the empirical analysis and examines how well this approximation holds in samples of comparable size.⁷⁴

E.3 Calibration of the Monte Carlo design

The Monte-Carlo experiment asks whether the estimator recovers returns to scale in finite samples generated by a quantitatively realistic version of the dynamic economy. We hold the sectoral parameters $(\gamma_i, \sigma_i^2, \hat{\eta}_i, \bar{\alpha}_i, \beta_i)$, the input-output matrix from Section 7, and the implied entry costs as in the baseline calibration. The decomposition into permanent and transitory productivity also preserves each sector's calibrated total dispersion: $\sigma_{P,i}^2 + \sigma_{e,i}^2 = \sigma_i^2$. The transitory productivity process is calibrated as follows. We set $\rho_e = 0.95$, within the range of annual firm-level estimates (Cooper and Haltiwanger, 2006; Foster et al., 2008; Syverson, 2011; Asker et al., 2014). We set the persistence parameters externally because productivity and returns to scale are recovered jointly in the data. We set $\sigma_{e,i} = 0.05$ to reproduce the observed within-firm volatility of sales growth.⁷⁵ This parameterization implies sales-growth volatility of 0.36, close to 0.375 in the data.

Wedge dynamics are disciplined directly by the Spanish data. Equation (107) implies that the input-share residuals $l_{ilt} - q_{ilt}$ and $m_{ilt} - q_{ilt}$ equal the negative wedges up to firm constants. We estimate each wedge's persistence and stationary variance, and the innovation correlation across wedges, by minimum distance on the autocovariances of the first-differenced residuals up to lag three, allowing an iid measurement-error component (Anderson and Hsiao, 1982). The resulting estimates, which we impose in every sector, are $\rho_{L,i} = 0.70$, $\rho_{X,i} = 0.59$, and $\text{corr}(\xi_{ilt}^L, \xi_{ilt}^X) = -0.18$. We assign the wedges the implied variance of the estimated AR(1) process, which gives stationary standard deviations of $\sigma_{L,i} = 0.16$ and $\sigma_{X,i} = 0.19$ in every

⁷⁴Although the Monte Carlo uses only labor and materials, the same argument accommodates capital under the corresponding timing and rank conditions, with returns to scale defined as $\beta^L + \beta^M + \beta^K$.

⁷⁵In the model, log sales load on the transitory productivity component with factor $1/(1 - \eta_{il})$ (see the proof of Lemma 13).

sector. The procedure removes firm effects, remains valid under serially independent measurement error, and avoids the fixed- T bias of the within estimator (Nickell, 1981). Where necessary, we make the minimum sector-specific adjustment needed for a well-defined equilibrium, so that expected operating profits at entry remain positive. Finally, we match the firm exit rate, $\delta = 0.10$.

Taken together, these choices combine the paper’s sectoral calibration with externally grounded productivity dynamics, wedge processes estimated from the Spanish data, and the observed exit rate. The Monte Carlo therefore evaluates finite-sample performance under a theoretically-disciplined and quantitatively realistic data-generating process.

E.4 Monte Carlo design and estimation

We simulate 10,000 firms in each sector for 32 years and retain the final 25 years. This means each estimation cell — one size decile of one sector over a seven-year window — contains roughly 1,000 firms, close to the average of about 750 firms per cell in the data.

We apply the exact system-GMM and minimum-distance estimator of Appendix A.2 to simulated sales and input series. First, for each seven-year window, we rank firms within sector by their mean log sales and form size deciles. We then run the system-GMM step, using the difference and levels moments (41) and (42). Lastly, the Söderbom (2009) minimum-distance step imposes the common-factor restrictions.

Sample construction otherwise mirrors the empirical pipeline, with two simulation-specific adjustments. First, Gaussian type draws can place a firm outside the interior technology range. We therefore exclude firms with $\eta_{il} \notin (0.01, 1)$. Second, we additionally trim the input-share residuals at their top and bottom 2 percent within each sector-year. For each sector-decile-year cell, we record the estimated returns to scale and its true counterpart, the unweighted mean of η_{il} across the cell’s firm-year observations. Lastly, we also compute the sectoral effective returns to scale $\hat{\eta}_i$ as the size-weighted firm-level returns to scale within each sector.

E.5 Results

Figure 22 shows that the estimator recovers returns to scale at all three levels used in the analysis. Panel (a) shows that the estimation recovers the size gradient accurately. The small gap reflects the finite-sample bias and the small within-cell heterogeneity in returns to scale, as argued by Proposition 15(v). Panel (b) compares estimated returns to scale with their benchmarks across all 1,860 sector-decile-year cells. The cell-level correlation is 0.982 and the mean absolute deviation is 0.021. Panel (c) compares estimated and true sectoral effective returns to scale. The estimate $\hat{\eta}_i$ is constructed by weighting the decile estimates within each sector by their sales shares. Across the 62 sectors, the estimated and true values align closely: the correlation is 0.984, the mean absolute deviation is 0.014, and the median estimation gap is +0.009.

Taken together, the Monte-Carlo exercise validates the estimator for both uses made of it in the paper. It recovers the size profile of firm-level returns to scale and closely tracks returns to scale at the sector-decile-year level. After aggregation, it also recovers sectoral effective returns to scale almost one for one. This last

result is central to the paper’s quantitative analysis: $\hat{\eta}_i$ is the key sectoral object used in the aggregation

E.6 Validation of the within-firm growth design

The Monte-Carlo exercise above validates the estimator for the cross-sectional and sectoral objects used in the paper. We now apply the same logic to the within-firm growth design of Section 6.3. In the dynamic model above, a firm only chooses its technology once, at entry, and never revises it. We therefore modify the data-generating process minimally, so that a firm can adjust its returns to scale during its life.

We make three changes to the model of Appendix E.1: the permanent productivity component is removed ($\sigma_{P,i} = 0$), each sector’s full dispersion is carried by the transitory process ($\sigma_{e,i} = \sigma_i$ with $\rho_e = 0.95$ as above), and firms re-optimize η_{it} each period under full information. The wedge processes, sector constants, sample sizes, and estimation protocol are unchanged. With these changes, the model now generates a true within-firm growth channel present in all 62 sectors.⁷⁶

Figure 23 reports the key result, replicating the four design variants in the robustness check in Figure 15. The simulated panels are put through the empirical procedure of Appendix A.6: firms are sorted into terciles of the predetermined growth measure G_{it} , and we estimate how firm growth affects a firm’s returns to scale via equation (46). On the baseline measure the true tercile spread is +0.005 and the estimator returns +0.008. When the trailing average in G_{it} is dated $t - 2$ through $t - 4$ instead of $t - 3$ through $t - 5$, the true spread is +0.009; system GMM returns +0.008. The third variant, which keeps only firms above the median size in their sector, is the most accurate of the three: its estimated spread is +0.0038 against a true spread of +0.0039. The fourth variant adds the firm’s permanent tercile, interacted with the inputs. Its estimated spread is +0.003 against an unchanged true spread of +0.005, so controlling for the permanent tercile absorbs part of the genuine within-firm variation and appears to be conservative. Across the four measures, the design closely recovers the sign and magnitude of the true within-firm growth profile.

⁷⁶We do not present a formal dynamic model here since it closely follows its analogue in Appendix E.1.

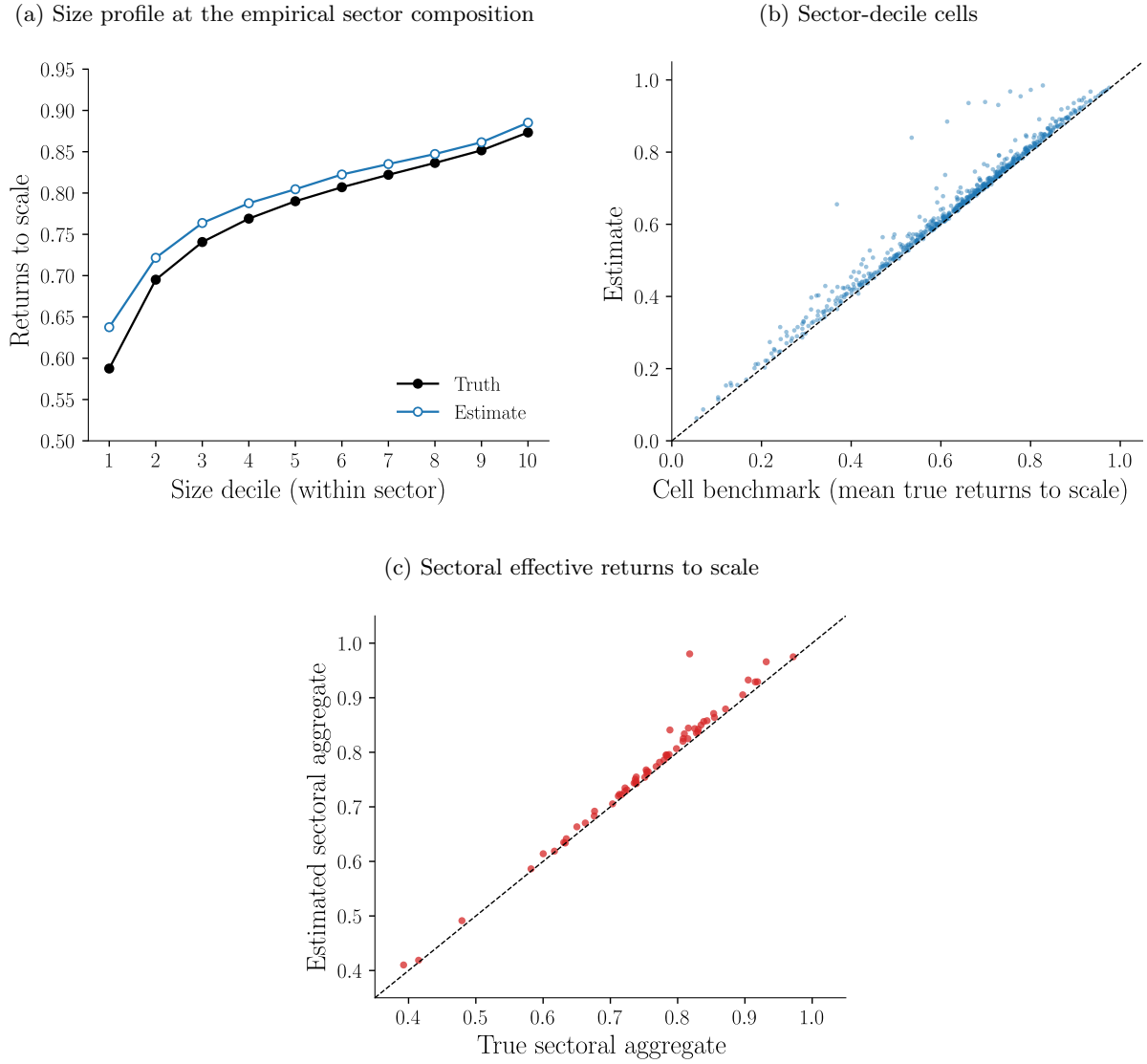


Figure 22: Monte Carlo validation of the returns-to-scale estimator.

Notes: Panel (a) plots true and estimated returns to scale by firm-size decile, aggregating sector-specific profiles using the empirical sector composition. Panel (b) compares estimated returns to scale with the corresponding benchmark for 620 sector-by-decile cells, averaged across three 7-year rolling estimation windows (centered on the 7th, 13th, and 19th years of the simulated panel); the dashed line denotes the 45° line. Panel (c) compares true and estimated effective returns to scale across the 62 sectors, computed by weighting decile-level values by their sales shares. The simulation trims the input-share residuals at their top and bottom 2 percent within each sector-year and contains 1,860 sector-by-decile-by-window cells.

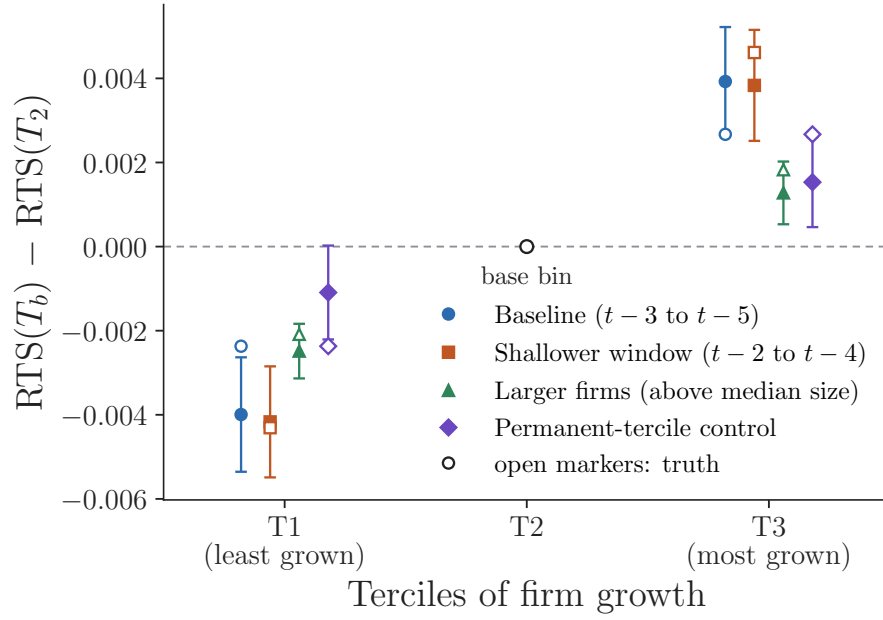


Figure 23: Within-firm growth designs estimated against their truths

Notes: The figure replicates the four growth-design variants of Figure 15 in the modified economy with no permanent types and per-period technology choice. Filled markers are system-GMM estimates with 90 percent confidence intervals from the cross-cell dispersion (62 sector cells); open markers are the true within-firm tercile contrasts on the same cells.

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